Project Title	Coastal Carbon Corridor: Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia
Project ID	1493
Version	1.0
Report ID	2525
Date of Issue	31-July-2019
Project Location	Aceh province and North Sumatra province, Indonesia
Project Proponent(s)	Livelihoods Fund Mr. Stéphane Perrier, Technical&Innovation Director 75009 Paris, France sperrier@livelihoods-venture.com YAGASU ACEH Mr. Bambang Suprayogi Medan 20154 Indonesia Email: bambang@yagasu.or.id
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Validation/Verification Body	Forest Solution Martin Opitz, MSc. Forestry phone: +49 177 3024475 skype: opitz_m email: m.opitz@forest-solution.com
GHG Accounting/ Crediting Period	14-06-2011 – 13-06-2031
Monitoring Period of this Report	01-10-2015 – 31-07-2019 for VCS 14-06-2011 – 31-07-2019 for CCB
History of CCB Status	Seeking Verification under CCB
Gold Level Criteria	Seeking Climate, Community and Biodiversity Gold Level Criteria for following reasons: 1) The Carbon Corridor Project contributes to commit local communities to protect and restore mangrove ecosystem through its role in preventing erosion, sedimentation and inhibiting the effects of sea storm tsunamis and flooding, 2) The project seeks to benefit local communities improving their quality of life and resilience by supporting sustainable businesses to be developed in the project lands, 3) East Coast of North Sumatra is a site identified as Important Bird and Biodiversity Area and Key Biodiversity Area. The site presents significant populations of globally threatened species (at least 4) and significant congregations of one or more bird species a certain time in their lifecycle or seasonal migration.

COASTAL CARBON CORRIDOR: MANGROVE RESTORATION AND COASTAL GREENBELT PROTECTION IN THE EAST COAST OF ACEH AND NORTH SUMATRA PROVINCE, INDONESIA

Document prepared by YAGASU ACEH and AGRESTA S. COOP.

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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
1) Increase in perception and recognition on mangrove restoration and protection	A total number of 299 actions were implemented from 2011 to 2019 attended by 14,104 people (villagers, students, and teachers)	4.3.1	The project activities are expected to improve the perception and recognition on mangrove restoration and protection of 19,697 people
2) Increase the capacity of coastal ecosystem for carbon sequestration	354,250 t CO_2-e	3.2.4	2,494,121 t CO_2 removed
3) Reduce the natural disaster risk	From 2011 to 2019, a total of 25,000 ha of coastal mangrove have been protected by local communities and 5,278 ha of mangrove have been restored with native species	3.3.1	Maintenance of the protection of 25,000 ha of coastal mangrove as well as the 5,278 ha of restored mangrove
4) Improvement in sustainable local community livelihood	Development of 634 sustainable businesses associated to mangrove: aquaculture (70), organic batik (34), eco-tourism (11), food processing (202) and others (317) involving 6,077 people (3,646 men and 2,431 women)	4.3.1	The maintenance and/or increasement of sustainable businesses developed associated to mangrove
5) Increase biodiversity conservation	The coastal mangrove is being protected and restored which ensures the recovery of ecosystem and the maintenance and enhancements of populations of species supported by them. To achieve protection goal: The Community Mangrove Patrolling Unit has been created, through 154 patrols which protection actions cover 25,000 ha of coastal mangrove protected and 17,388.77 ha of villages, involving 1,224 people. To achieve restoration goal: a total of 5,278 ha	5.3.1 and 5.4.1	The protection and restoration of mangrove forest is expected to enhance the population of this species.

Outcome or Impact	Achievements during the Monitoring Period	Section Reference	Achievements during the Project Lifetime
	has been restored since the beginning of the project.		

1.2 Standardized Benefit Metrics

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
GHG emission reductions & removals	Net estimated emission removals in the project area, measured against the without-project scenario	354,250	3.2.4	2,494,121
	Net estimated emission reductions in the project area, measured against the without-project scenario			
Forest[1] cover	For REDD[2] projects: Number of hectares of reduced forest loss in the project area measured against the without-project scenario	Not applicable		Not applicable
	For ARR[3] projects: Number of hectares of forest cover increased in the project area measured against the without-project scenario	5,278.71	3.1.2	5,278.71
Improved land management	Number of hectares of existing production forest land in which IFM[4] practices have occurred as a result of the project's activities, measured against the without-project scenario	Not applicable		Not applicable
	Number of hectares of non-forest land in which improved land management has occurred as a result of the project's activities, measured against the without-project scenario	Non applicable		Not applicable
Trai ning	Total number of community members who have improved skills and/or	13,197	4.3.1	19,697

[1] Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)
[2] Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)
[3] Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)
[4] Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	knowledge resulting from training provided as part of project activities			
	Number of female community members who have improved skills and/or knowledge resulting from training provided as part of project activities of project activities	4,094	4.3.1	6,141
Employment	Total number of people employed in of project activities,[5] expressed as number of full-time employees[6]	144	4.3.1	202
	Number of women employed in project activities, expressed as number of full-time employees	86	4.3.1	120
Livelihoods	Total number of people with improved livelihoods[7] or income generated as a result of project activities	6,077	4.3.1	9,004
	Number of women with improved livelihoods or income generated as a result of project activities	2,431	4.3.1	3,646
Health	Total number of people for whom health services were improved as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable
	Number of women for whom health services were improved as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable

[5] Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.
[6] Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])
[7] Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
Education	Total number of people for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable
	Number of women and girls for whom access to, or quality of, education was improved as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable
Water	Total number of people who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable
	Number of women who experienced increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applicable		Not applicable
Well-being	Total number of community members whose well-being[8] was improved as a result of project activities	6,077	4.3.1	8,507
	Number of women whose well-being was improved as a result of project activities	2,431	4.3.1	3,640
Biodiversity conservation	Change in the number of hectares significantly better managed by the project for biodiversity conservation,[9] measured against the without-project scenario	25,000 ha	5.3.1	25,000 ha

[8] Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Health, Education, Water, etc.), but could also include other benefits such as empowerment of community groups, strengthened legal rights to resources, conservation of access to areas of cultural significance, etc.
[9] Biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation.

Category	Metric	Achievements during Monitoring Period	Section Reference	Achievements during the Project Lifetime
	Number of globally Critically Endangered or Endangered species[10] benefiting from reduced threats as a result of project activities,[11] measured against the without-project scenario	4 at least	5.4.1	4 at least

[10] Per IUCN's Red List of Threatened Species
[11] In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 GENERAL

2.1 Project Description

2.1.1 Implementation Description

The projects have created direct and indirect impacts to the local communities such as increasing perception to the mangrove conservation, creating new skills and employment from social business development in the villages, establishing village poly initiatives and increasing income in the family. In addition, the impacts to the biodiversity include stable condition of coastal ecosystem due to mangrove restoration and protection, returning of various fauna that include flagship species into mangrove ecosystem, increasing quality of coastal green-belt for natural disaster prevention and increasing population of commercial fauna biodiversity that support income generation for coastal communities.

The long-term target is to ensure mangrove ecosystem in the project sites contributing to the Sumatra's future ecological, social and economic development.

Currently the project has restored 5,000 ha of degraded mangrove, 278.71 ha have been included as new instances in this monitoring period. This will increase the carbon sequestration capacity of the project by the end of the project lifetime.

Besides, the project activities included additional mangrove restoration, protection of 25,000 ha mangroves, scientific research, Income Generating Activity (IGA) models, awareness program, community resilience and policy development.

As demonstrated in section 3.3 of the VCS Project Description, leakage due displacement of agricultural activities in the areas of this project has been accounted as zero. The non-permanence risk rating is assessed as documented in the AFOLU non-permanence risk report.

The total GHG emission removals generated in this monitoring period is 354,250 t CO_2-e.

A change in the project proponent have been made in this monitoring period. Livelihoods Fund was the project proponent in the first place. However, Livelihoods recognize Yagasu as the implementors and developers on the community and biodiversity activities, raising awareness, getting more funding for these activities that allows reach the CCB certification. Thereby, Yagasu is considered as project proponent, in addition to Livelihoods Fund. A project contract between Yagasu and Livelihoods does clearly state that Livelihoods Fund is the owner of carbon credits.

2.1.2 Project Category and Activity Type

The Coastal Carbon Corridor is a project categorized as an Agriculture, Forestry and Other Land Use (AFOLU) project under the Afforestation, Reforestation and Revegetation (ARR) project category.

As indicated in section 3.1.11 of VCS AFOLU requirements, when soil organic carbon pool in project scenario is not deemed below de minimis, all ARR projects shall comply also the WRC requirements (Wetlands Restoration and Conservation). Soil Organic Carbon pool is an important part of the total amount of the carbon sequestration of this project; therefore, this project will comply not only ARR requirements but also WRC. It is important to observe, though, that the present project do not consider GHG emissions reductions and therefore do not fall under the description of WRC project in section 4.2.19 of AFOLU requirements (version 3.4).

This is a grouped project.

2.1.3 Project Proponent(s)

Organization name	Livelihoods Fund
Contact person	Mr. Stéphane Perrier,
Title	Technical & Innovation Director
Address	48-50 rue de la Victoria, 75009, Paris, France
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Organization name	YAGASU ACEH
Role in the project	Project development & implementation / CCB data & documentation
Contact person	Bambang Suprayogi
Title	CEO of Yagasu
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Email	bambang@yagasu.org

2.1.4 Other Entities Involved in the Project

There are no other entities involved in the project.

2.1.5 Project Start Date (G1.9)

The start date of the project activity is 14-06-2011, which is the date of the first project activity on ground of planting campaign. For the current grouped project, the crediting period will be 20 years, from June 14, 2011 to June 13, 2031.

Biodiversity and Community assessment will be carried out throughout the lifetime of the project for each verification period according the monitoring plan described in Section CM4 and B4.

2.1.6 Project Crediting Period (G1.9)

Project crediting period: 14-06-2011 – 13-06-31.

Total numbers of years: 20 years renewable.

2.1.7 Project Location

The project sites stretch along 497 km Ujung Pancu village in Aceh province to Pante Labu in North Sumatra province, crossing 13 districts of Aceh Besar, Banda Aceh, Pindie, Pidie Jaya, Bireun, Lhoksemawe, Aceh Utara, Aceh Timur, Langsa, Aceh Tamiang, Langkat, Medan and Deli Serdang. The mangrove width covers are from as 20 m to more than 5 km from the shore. Along this width, the areas are dissected by many coastal streams.

- *Zone I & II in Aceh Besar, Banda Aceh, Aceh Utara, Pidie, Pidie Jaya and Bireun*. This area follows the northern tip of Sumatra island. Around 40 km shoreline was severely damaged by the 2004 Asian Tsunami, with entire communities lost. Prior to the tsunami, the area had been converted either for fishpond or nearby coastal agriculture.
- *Zone III in Aceh Timur and Langsa.* As the same with other zones, this area was historically under aquaculture and now is abandoned. It consists of a mosaic of shoreline; interior and island areas. Ongoing pressures include the conversion to oil palm. Priority is given to rehabilitate Pulau Sembilan, an island system with abandoned ponds that are also close to accreting shorelines.
- *Zone IV in Tamiang and Langkat*. This area consists of numerous potential plots close to the Karang Gading and Langkat Timur Laut Wildlife Reserve. The entire habitat has been degraded since the 1960s and 1970s; the Reserve has been recovered during the past 25 years, but the nearby unprotected areas remain under pressure. The area is available for reforestation in the interior of abandoned ponds and along estuary embankments.
- *Zone V in Deli Serdang*. This area forms a long accretion zones along about 20 km of beaches and estuaries. Previous attempts at reforestation have been successful, establishing thick mangroves forests since 2007 through plantation augmented by natural regeneration.

Figure 1. Map of Project Zonation. (Source: Own elaboration. AGRESTA S. COOP.)

The project region has up to 456,896 of potential hectares for plantation and mangrove restoration and protection. Among 9,600 ha restored mangroves by Yagasu, the first instance of project verification has carried out over an area of 5,000 hectares. Besides, the second instance of the project have been carried out over an area of 278.71 hectares.

Figure 2. Outer boundary for the project (Source: Project Description of the project)

2.1.8 Title and Reference of Methodology

The project applies the following methodologies:

AR-AM0014: Afforestation and reforestation of degraded mangrove habitats (Version 3.02).
ARR methodological tools:

- "Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities" (Version 01)
- Estimation of non-CO_2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0)
- "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities" (Version 03.1)
- "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities" (Version 04.2)
- "Estimation of the increase in GHG emissions attributable to displacement of pre-project agricultural activities in A/R CDM project activity" (Version 02.0)

Other methodological ARR CDM tools which are applied include,

- Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in ARR CDM project activities (Version 01.0.0)

- Calculation of the number of sample plots for measurements within ARR CDM project activities (Version 2.1.0)

2.1.9 Other Programs (G5.9)

This grouped project is not currently subject to an emission trading program and other binding limits. The project only seeks carbon credits under the VCS program, and has not received other forms of environmental credits participating under other GHG program and has not been registered under any emissions trading program.

2.1.10 Sustainable Development

This Coastal Carbon Corridor program addresses the Indonesia Biodiversity Strategy and Action Plan (IBSAP) for implementing national development in accordance with CBD that strongly link to the RAN-GRK guidance. More specifically, this initiative is in full alignment with the IBSAP's vision (V) on biodiversity management ("Indonesian biodiversity preservation and development that contributes to national competitiveness and a fair and sustainable use of resources to improve the welfare of current and future generations") and the following Mission Statements (MS):

- MS1: "To improve ownership and utilization of Indonesian biodiversity";
- MS2: "To use biodiversity as source of improving welfare & sustainability of life for the Indonesian people";
- MS3: "To manage biodiversity with full of responsibility for the sustainability of life of the world".

This program also contributes towards the Sustainable Development Goals (SDG):

- SDG01: Economic opportunities will improve the resilience of beneficiaries and increase their net income;
- SDG02: The aquaculture will become a source of sustainable seafood, protein and B-group vitamins in local markets, and thus help reduce food insecurity and malnutrition prevalence locally;
- SDG05: 3 women-led interventions on organic "batik", mangrove foods, and shrimp paste will bring additional income and empower women economically. Overall male/female ratio of beneficiaries: 40/60.
- SDG13&14: The mangrove restoration and protection will help to restore depleted fish stocks in surrounding marine ecosystems and to sequestrate over 10m tCO_2e over 20 years. Additionally, this intervention will contribute towards SDG04, SDG12, SDG15 and SDG17.

The program components already matched with the basic principles of provincial's economic development: local resources and environmental base, community base and value added and sustainable market base. The scheme of added value of those community's key commodities is to establish an integrated model of social business.

The program intervention enhances economic returns of biodiversity by capitalizing on ecosystem services. Small scale models of IGAs and village policy development have been piloted across 28 villages with incurrent Coastal Carbon Corridor. The Coastal Carbon Corridor support will be instrumental to scale up the most successful ones across over 50 villages. Furthermore, the project will bring additional processing, packaging, branding, certification, licensing, and marketing capacity to Sumatra for the commercialization of project outputs and the development of potential exports. This will expand trade and increase access to market of coastal communities.

Indonesia is party of the UNFCCC (United Nations Framework Convention on Climate Change) since August 1994. The country has ratified the UNFCCC (1995), the Kyoto Protocol (2004), has signed the Paris Agreement (Paris 2015, COP21). The Republic of Indonesia participates voluntarily in the Clean Development Mechanism (CDM).

2.2 Project Implementation Status

2.2.1 Implementation Schedule (G1.9)

Currently the project has restored 5,000 ha of degraded mangrove, 278.71 ha have been included as new instances in this monitoring period This will increase the carbon sequestration capacity of the project by the end of the project lifetime.

Besides, the project activities included additional mangrove restoration, protection of 25,000 ha mangroves, scientific research, Income Generating Activity (IGA) models, awareness program, community resilience and policy development.

As demonstrated in section 3.3 of the VCS Project Description, leakage due displacement of agricultural activities in the areas of this project has been accounted as zero. The non-permanence risk rating is assessed as documented in the AFOLU non-permanence risk report.

The total GHG emission removals generated in this monitoring period is 354,250 t CO2-e.
A change in the project proponent have been made in this monitoring period. Livelihoods Fund was the project proponent in the first place. However, Livelihoods recognize Yagasu as the implementors and developers on the community and biodiversity activities, raising awareness, getting more funding for these activities that allows reach the CCB certification. Thereby, Yagasu is considered as project proponent, in addition to Livelihoods Fund. A project contract between Yagasu and Livelihoods does clearly state that Livelihoods Fund is the owner of carbon credits.

Table 1. Milestones in the project development

Date	Milestone in the project's development and implementation
2011-2015	Reforestation (planting new mangroves)
December 2015	Project registration in VCS database (VCS validation)
	VCS Monitoring and auditing
	Project VCU issuance (1st VCS verification)
2016-2019	Project maintenance (including replanting) and field monitoring
08-05-2019 to 15-05-2019	First site visit for 2nd VCS verification and CCBS validation and verification
May-June	VCS + CCB Monitoring Addition second instances (278,71 ha)
31-July-2019	CCBS PDD Draft (CCB validation) + VCS and CCB MR Draft (2nd VCS verification, 1st CCB verification)
14-September-2019	VVB site visit for CCB validation and 1st verification and VCS 2nd verification

Table 2 Implementation schedule planned for the project

Activities	Project lifetime (years)				
	2011-2015	2015-2019	2019-2023	2023-2027	2027-2031
Reforestation	X	X			
Monitoring	X	X	X	X	X
Capacity building restoration, protection and, mangrove-based business program	X	X	X	X	X

2.2.2 Methodology Deviations

No methodology deviations were made during this monitoring period.

2.2.3 Minor Changes to Project Description (*Rules* 3.5.6)

No community or biodiversity changes to project design have occurred during monitoring period compared with the validated project description.

2.2.4 Project Description Deviations (*Rules* 3.5.7 – 3.5.10)

A change in the project proponent have been made in this monitoring period. Livelihoods Fund was the project proponent in the first place. However, Livelihoods recognize Yagasu as the implementors and developers on the community and biodiversity activities, raising awareness and getting more funding for these activities that allows reach the CCB certification. Thereby, Yagasu is considered as project proponent, in addition to Livelihoods Fund. A project contract between Yagasu and Livelihoods does clearly state that Livelihoods Fund is the owner of carbon credits.

According to the rule 3.5.7, a change in the project proponent is considered as a project description deviation, however, there has not been a change of the project proponent responsible for the implementation, but an addition of a new project proponent. This change from the validated PD occurred during this monitoring period. This deviation is determined that do not impact the applicability of methodology, additionality or the appropriateness of the baseline scenario. The project remains in compliance with the applied methodology.

2.2.5 Grouped Projects

This project is a grouped project. This project has included a new instance of 278.71 ha. These areas meet the eligibility criteria and scalability limits to be included as new instance.

1) New Project Areas and Communities (G1.13):

A total of 278,71 ha has been included during this monitoring period as second instance.

2) Removed Project Areas and Communities (G1.13)

Not applicable.

3) Eligibility Criteria for Grouped Projects (G1.14)

The second project activity instance included in this monitoring period (278.71 ha) meets the eligibility criteria. The eligibility criteria for inclusion of new project activity instances are demonstrated in accordance with the paragraph 3.4.9 of the VCS Standard (Version 3.4).

4) Scalability Limits for Grouped Projects (G1.15)

New instances included in this monitoring period (528.71 ha) is aligned with the scalability limits since all the plots included are abandoned fish ponds and degraded coastal and river mudflats. Besides, each owner is responsible for performing management activities and ensuring the sustainability of the plantations.

5) Risk Mitigation for Grouped Projects (G1.15)

Yagasu conducted 1.417 village meetings attended by 23,573 people, 360 districts meetings attended by 18,000 people to build local commitment and village policy endorsement. Yagasu signed 10 years MoUs with 138 village governments and 198 community groups for mangrove protection and community development; and signed 5 years MoUs with 6 government institutions at provincial level.

Yagasu facilitated local communities to enact 8 Village Land-use Plan (VLP), 6 Mangrove Protected Area (MPA) and 10 Villages Regulation (VR) for mangrove forest protection and local economic development. These initiatives include: (a) producing VLP through participative mapping that cover the village areas for settlements, economic activities, disaster protection, disaster evacuation routes, and other purposes; (b) setting up MPA for coastal disaster risk reduction and climate change adaptation; (c) producing VR for implementation of VLP and MPA; and (d) integrating VLP and MPA into District Land-use Plan.

6) Project Zone Map (G1.13)

Not applicable.

7) Changes to Management (G4.1)

No changes to the management structure, roles and/or responsibilities have resulted from new entities joining the project.

2.2.6 Risks to the Project (G1.10)

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
MANGROVE RESTORATION AND PROTECTION		
Land tenure and ownership issues related to mangrove restoration and protection	Conflicts between communities and village government on the land borders and selling lands to outsider people Conversion of mangrove lands into intensive pond, oil palm plantation, settlement and other purposes Land use change of planted mangrove areas for other commercial purposes	1,417 village meetings attended by 23,573 people conducted to build consensus and agreement on series of field activities, 360 districts meetings attended by 18,000 people to build local commitment and village policy endorsement. Individual approaches based on the agreement signed in village government and local community group MoUs Socialize the signed MoUs with village governments and local community groups. Yagasu signed 10 years MoUs with 138 village governments and 198 community groups for mangrove protection and community development; and signed 5 years MoUs with 6 government institutions at provincial level. A total of 122 meeting for implement and enforce the agreed Village Land-use Plan (VLP), Mangrove Protected Area (MPA) and Village Regulation were conducted and attended by 2,400 people.
Poor understanding of the ecological requirements of mangroves	Excessively high planting densities	A total of 299 awareness actions about climate change, mangrove conservation and benefits of mangrove for human life attended by 14,104 people were conducted.
Mortality of planted mangroves due to natural disturbances and human activities	Less density or loss of planting plots	Replant the dead trees Change the “problem” plots with the new one
Mortality due to pests and diseases	Loss of planted mangroves	Regular biological control, use of organic pesticides

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Unclear legal status between community lands and "*appointed*" state forests that are shown as only indicative (general) forest sites and no legal formal status	Conflict between community groups with central government due to unclear legal status of communal lands	Conduct intensive consultations with technical depatments of GOI to get achievable consensus among the people in the villages and governments
Deforestation of mangroves Lack of government supports and low responses on law enforcement if there is any special case of mangrove disturbance	Less density or loss of planting plots Direct confrontation with the illegal loggers and poachers	Five maps of threatened areas and key species occurance and populations have been produced Increase commitment and integrity among law enforcers (*BBKSDA* and *Dinas Kehutanan*) is sufficient to uphold effectiveness of protection initiatives Report any specific findings related to the need of law-enforcement will be reported directly to the District Forestry Department and local police for further actions
Increase risks of natural disasters (sea-rise level, tsunami and strong wind)	Flooding in community lands due to permanent high tide Loss of land due to coastal erosion Loss of human life, village economic activities and other facilities Displacement of coastal communities Loss of flora and fauna	Scale-up mangrove restoration and protection for disaster prevention and local economic development. A total of 299 awareness actions about climate change, mangrove conservation and benefits of mangrove for human life attended by 14,104 people were conducted to villagers, school children, decision makers, legislatives and wider public.
SCIENTIFIC RESEARCH		
Less use of research findings	Very little or almost no environmental and social risk of research activities	Utilize and maintain research equipment properly Encourage local stakeholders and partners to be receptive to scientific inputs

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
		Encourage government patners to commits using scientific inputs on their management and policy development
SOCIAL BUSINESS DEVELOPMENT		
Limited scope of project cover	The project can't cover all the beneficiaries in the villages; not enough opportunities to participate which makes villagers less interested in the project	Find other fundings to scale-up all aspects of social business in the villages Develop business entity that is able coordinate the scatered business into an aggregated structure to link to markets Create more business and new job opportunities
Increase of intensive aquaculture practices	Cut planted and existing mangroves for open aquaculture system Alter and change the composition of water and soil of fish ponds due to supplementary feeding	A total of 231 trainings on Income Generation Activities were carried out. Support equipment and capital to run organic aquaculture using silvo-fishery system
Coastal construction and settlement	Loss of coastal green-belt	Approach individual land owner based on the principle of mutual respect, trust and benefits; A total of 8 VLP, 6 MPA and 10 VR have been established to support long -term mangrove carbon conservation. Consult district and provincial technical departments in enforcing their regulations to protect and save the coastal green-belt.
COMMUNITY RESILIENCE		
Less commitment of non-participated people on the project implementation	Abandonment of project activities	Yagasu signed 10 years MoUs with 138 village governments and 198 community groups for mangrove protection and community development; and signed 5 years MoUs with 6 government institutions at provincial level.

Identify Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Less understanding of wider public to the climate change issues and mangrove conservation	Less participation in supporting project activities and co-investment to project leverages	Yagasu has produced awareness materials (leaflets, posters, project presentation, up-loaded videos and photos of project activities) Ensure that local stakeholders and village leaders advocate and participate in awareness program implementation Conduct attitude survey at the beginning and end of project period
PROJECT MANAGEMENT		
Day-to-day management operations and communications	Turn-over in the project team and local stakeholders Very little or almost no management risk and project monitoring & audit	Produce contract documents of project staff, village stakeholder liaison and consultants (term of reference, contract period and job description). A total of 715 people is employed directly and indirectly in the village businesses. Increase commitment and integrity of project team (part-time and full-time staff and consultants) work proffesionally and on time basis; capacity building and exposure visits of the staff as needed Periodic review and project progress meetings and workshops

2.2.7 Benefit Permanence (G1.11)

The main measures taken and needed to maintain and enhance the climate, community and biodiversity benefits beyond the project lifetime are:

- Species used in the restoration actions are native species adapted to the ecological conditions in the project sites
- The carbon conservation of restored mangroves will be managed by strong commitment of local community groups who receive benefits from the long-term program of non-planting mangroves, village policy initiatives and scale-up of social business;
- The sustainability of the project is ensured with the active participation of local communities and local stakeholders; policy supports from village governments; endorsement of district, provincial and national governments; and the support funding from various donors and social business development in the villages
- Management teams of the project activities include individuals with significant experience in all skills necessary to perform the activities related to the planting of the species and monitoring of the planted trees, community development, social business development, and also experience in staff management, human resources, training, among others.
- Project is protected by legally commitment of 10-year MoUs with local community groups, 10-year MoUs with village governments and endorsement of GOI at district, provincial and national level that continue management practices that protect the credited carbon stocks over the lifetime project.
- The financial profitability projected will develop into a strong community's commitment and confident in investors under the principle of mutual respect, mutual trust and mutual benefits.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

Similar to the PDD-carbon VCS document, this PDD-CCB will also be available in VCS database, which is a public database where anyone can find and download the project documents (Project description, Monitoring Report).

2.3.2 Informational Meetings with Stakeholders (G3.1)

334 sessions for village governments attended by 5,010 people and 466 sessions for community groups attended by 10,718 people have been carried out in order to explain the main goals and expected impacts of the project, obtain community perceptions and key aspects in improving the design of the project and establish and sign Memorandum of Understanding (MoUs) between Yagasu and village government, and Yagasu and community groups.

Yagasu has organized different meetings that aims to get public engagement with the project. The topics of these meetings were:

- Restorations Program, the sessions on this topic have been about socialization of restoration program, discuss with community groups leaders about planting system organization, preparation for nursery, discuss with communities regarding plantation monitoring, training in plantation monitoring

- Village meeting to conduct MoU, the aim of these meetings is to sign the Memorandum of Understanding with the villages, a document of agreement between villages and Yagasu to establish mutual respect. There exist MoUs of Mangrove restoration program and plantation with community group
- Village meetings to conduct VLP, MPA and Perdes, to establish and disseminate the progress and results of VLP, MPA and village regulation program.
- Economic development activity, meetings with communities regarding economic development, training on organic batik using natural coloring and mangrove food processing, ecotourism development and how to use the loan for these purposes.

There have been also meetings at the district level to support establish and disseminate VLP, MPA and Perdes program with Forestry and Agronomy District Government of different districts.
The project ensures the maximum transparency in managing its activities by promoting implementation of the disclosure policy through provision of, but not limited to:

- Information which has direct relevance to the beneficiaries and donor such as details of activities in each respective location
- All quarter procurement plans and schedules and updates will be reported
- All bidding documents will be reported in accordance with the procurement guidelines
- All short list consultants both individual or organizations will be reported in accordance with the consultant guidelines
- All information regarding contracts will be reported
- External audit and financial mid-term report will be reported

2.3.3 Community Costs, Risks, and Benefits (G3.2)

The community consultation process ensures a free, prior and informed consent through informational meetings to establish MoUs. During the meetings the possible impacts that the project might have on individual or collective actors in terms of economic, social, climate and biodiversity aspects are considered.

2.3.4 Information to Stakeholder on Verification Process (G3.3)

All participants were informed about the VCS validation and registry process, as well as, CCB validations and verification process while pointing out that an external auditor will visit the project area to interact with stakeholders, evaluate the project information and issue a report of the evaluation.

Referring to the project owners, Livelihoods Fund (project proponent) has maintained constant and direct communication with Yagasu (project proponent) in order to give the guidelines and clarity aspects related to the validation and verification process.

2.3.5 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

The auditor will be assisted by a local person in order to communicate with stakeholders and communities. This person does not belong to Yagasu staff or Livelihoods Fund staff, to guarantee they can communicate freely, and the process is transparent and independent.

2.3.6 Stakeholder Consultation (G3.4)

During the period 2011 to 2019, a total of 360 meetings have taken place with local communities, institutions (private sectors, universities, NGOs) and governments, attended by 18,000 people. Yagasu team members have been working in their respective areas for many years and the organization’s approaches and practices are well known to local communities. Early consultations provided important inputs and laid the foundation for continuing consultations during the project implementation. Community and local government demand for the services provided by this project were confirmed through meetings during the preparation missions.

The project assists the communities in obtaining official recognition of such agreements through the local and provincial government agencies. This consultation process and the resulting agreements are intended to increase involvement from local communities. Regulations agreed to between local communities and relevant government agencies, and developed with the project's appropriate technical inputs give legal and enforceable status to the long-term restoration and community development program. Likely, the coastal green belt protection strategy introduced through the project increase livelihood opportunities for local communities, by institutionalizing communal or individual land ownership. Should any issue arise, the problem is handled through the institutionalized Local Stakeholders Forum.

2.3.7 Continued Consultation and Adaptive Management (G3.4)

Yagasu have regular and continuous communication with heads of communities and the rest of stakeholders to address possible suggestions and complaints, training activities, dissemination of monitoring reports, etc.

At the same time, Yagasu have regular and continuous communication with Livelihoods Fund.

2.3.8 Stakeholder Consultation Channels (G3.5)

Participatory processes have been undertaken directly with villagers of local communities involved in the project activities through community groups meetings (466 sessions), village meetings (334 sessions) and district meetings (360 sessions), providing adequate information. Village meetings held can be seen in Section 7.2 Community monitoring data, as well as in CCB PDD Appendix I.

2.3.9 Stakeholder Participation in Decision-Making and Implementation (G3.6)

As described in Section 2.3.6, during the period 2011 to 2019, a total of 360 meetings have taken place with local communities, institutions (private sectors, universities, NGOs) and governments, attended by 18,000 people. These consultations provided important inputs and laid the foundation for continuing consultations during the project implementation.
Local consultation allows stakeholders to influence the project design through their perceptions of the project activities. Nevertheless, the design and implementation of the project have not been modified, as the comments received do not affect them.

Besides, a total of 442 governance communications (village meetings, district meetings and workshops) have taken place since the start of the project, attended by 8,833 people, where 3,092 were women.

2.3.10 Anti-Discrimination Assurance (G3.7)

Livelihoods Fund and Yagasu have excellent reputation and are not involved in any form of discrimination or sexual harassment with respect to the project.

The community involvement in the project has been inclusive, according to individual capabilities and independent of gender, cultural identity and religion.

2.3.11 Grievances (G3.8)

Grievance redressal is one of the most critical components in to provide a forum to the internal and external stakeholders to voice their concerns, queries and issues with the Project. Some examples grievance received and the ways that Yagasu provided to solved using project's grievance redress procedure:

- Stakeholder submitted grievance to Yagasu due to lack of ability of community in collecting propagules, nursery and planting techniques. The local community expressed their concerns through field staff. After received this grievance, Yagasu had conducted a training in collecting propagules, nursery and planting techniques. The training was conducted before the planting activity started.
- Local groups of Desa Percut, Desa Tanjung Rejo, Desa Rugemuk had sent their grievance to Yagasu about the abrasion and erosion occurred in their village, especially in coastline area. Yagasu had sent field staff to investigate the problem by survey and field assessment. Abrasion is a dynamic sediment moving which makes Yagasu conducted series of research on how to prevent a large abrasion by planting mangrove and established waves breaker.
- After introduced green product development from mangrove, local community still found difficulties to sell their products. In this case, Yagasu helped the community by developing village cooperatives, conducted training on business skills, processing, licensing, branding, marketing and product promotion and exhibition to link the community with buyer.

2.3.12 Worker Training (G3.9)

Yagasu execute training activities to the communities regarding restoration program, aiming the conduction of MoUs, VLP, MPA, and PerDes and Economic Development Activity.

- Village meetings for community development and Income Generation Activities (IGA) preparation: A total of 289 meetings have been carried out in 12 different districts, attended by 2,767 people, where 1,445 were women.
- Village meetings for VLP, MPA and VR: A total of 122 meetings have been carried out in 4 different districts, attended by 2,440 people, 976 women.
- Village meetings for preparing nursery and planting mangroves: A total number of 206 meetings, in 12 districts, attended by 2,678 people, 1,073 women.
- Awareness actions about climate change issues, mangrove conservation and benefits of mangrove for human life: A total of 299 actions were implemented, attended by 14,104 people, villagers (25%), teachers (17%), students (45%), others (13%).
- Capacity building sessions for local stakeholders: A total of 249 trainings have been carried out attended by 13,197 participants, villagers (26%), village administrators (13%), university students (28%), teachers (11%), government administrators (11%) and NGOs (9%).
- Capacity building sessions on IGAs: A total of 231 trainings have been carried out, attended by 3,922 people, where 1,569 were women, and 60% were young people.
- Capacity building for women groups: A total of 43 women were trained for organic batik business, 77 women were trained for food processing (syrup, dodol, jam), 40 women were trained for shrimp paste production and 30 women were trained for dried fish/oyster processing.

Once built, local capacity is not lost, since people are implementing actions on mangrove nursery (297 people), planting mangroves (351 people), replanting dead trees (225 people), mangrove patrolling (11 people),

Currently there are 634 social businesses ran by local communities on aquaculture (70 businesses), organic batik (34 businesses), eco-tourism (11 businesses), food processing (202 businesses) and other (317 businesses). These businesses involve 6,077 people, which 2,431 are women. All businesses are present in all 12 districts, excepting eco-tourism businesses which have been developed in 7 districts (Banda Aceh, Aceh Timur, Langsa, Aceh Tamiang, Langkat, Deli Serdang and Medan). Women manage and participate in the following businesses: organic batik, ecotourism and food processing. They don’t participate in aquaculture businesses.

A total of 9,790 people was employed on nursery works, planting mangroves, policy development and IGAs, where 4,094 were women.

See Section 7.2. Community monitoring data, as well as CCB PDD Appendix I.

2.3.13 Dissemination of Summary Project Documents (G3.1)

As mentioned before, the program is being conducted based on full participation of all components in multi-stakeholder collaborative management. Each stakeholder is actively involved in the project activities which guarantees the dissemination of the project implementation status.

2.3.14 Community Employment Opportunities (G3.10)

Yagasu verify that the person to be hired meets the hiring profile established for the vacant position, without any discrimination of age, sex, marital status, ethnicity, social status or religion, political ideas and/or sexual orientation. It is not allowed to employ under-age young people, 15 years for Indonesia, 13 if the work does not interrupt the physic, mental and social development of the underaged, and during a maximum of 3 hours per day.

Data explained above, in Section 2.3.12, show that people from the communities have been given an opportunity to fill all work positions, there are 6,077 people running different businesses associated to mangroves, 4,094 of whom are women.

Capacity building have been made for different people, to be employed in project activities, as well as to be the managers of their own sustainable businesses to be develop.

2.3.15 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

Indonesia is a member of the International Labor Organization since 1950 and have worked to protect the welfare and rights of workers. The following list cover worker's rights:

- Law on Poverty Management (Law No. 13/2012)
- Presidential Decree No. 23/2011 on the National Action Plan for Human Rights in Indonesia for 2011-2014
- Law No. 21 of 2000 Concerning Trade Union/Labour Union
- Registration of workers organizations (PER-05/MEN/1998).
- Law on Social Organisations (No. 8 of 1985).
- Law on Elimination of Human Trafficking Crimes (Law No. 21/2007)
- Law on Immigration (Law No. 6/2011)
- Law on Child Protection (No. 23/2002)
- Law No. 4/1979 on Child Welfare
- The Elimination of Racial and Ethnic Discrimination (Law No. 40/2008
- Law of the Republic of Indonesia on the Elimination of all Forms of Racial Discrimination (No. 29/1999)
- Decree of the Minister of Manpower and Transmigration of the Republic of Indonesia No. 184 of 2013 establishing a Task-force on Equal Employment Opportunity.
- Regulation of the Minister of Labour No. 6 of 2016 establishing Religious Holiday Allowances for Employees / Workers in the Company.
- Decree on National Holiday and Collective Leave in 2004
- Government Regulation of the Republic of Indonesia No. 50 of 2012 concerning the Implementation of Occupational Safety and Health Management Systems
- Social Welfare Law (No. 11/2009).
- Regulation of the Minister of Manpower No. PER-02/MEN/1984 on insurance against sickness, pregnancy and confinement for employees and their relatives.

- Law on the Protection and Empowerment of Fishers, Fishing Resources and Salt Farmers (No. 7/2016)

The workforce has followed international rules on employment and has provided with information about rights and obligations as workers.

2.3.16 Occupational Safety Assessment (G3.12)

The project has taken into account the risks that may arise with the project activity on the workforce involved, namely by providing labor security facilities and infrastructure. These facilities include the required speedboats and buoys (boyas). Planting these mangroves does not have a risk of workplace accidents for workers and it can be said that the risk is zero.

2.4 Management Capacity

2.4.1 Required Technical Skills (G4.2)

Livelihoods Fund

The Livelihoods funds are supported by private companies who believe in working and learning together with project developers to effect change. They are convinced that smallholder farmers are the key stakeholders that need support in achieving food security while preserving ecosystems on a global scale. The Livelihoods funds promote sustainable farming practices that can increase food production while preserving our natural resources.

The staff have significant ground and hands-on experience in skills of mobilizing communities, technical expertise in agriculture and forestry; and management skills of handling large scale projects. In addition, they come with the competence and experience of working with the mainstream markets to work for the rural, remote and poor communities.

Yagasu

Bambang Suprayogi and the Yagasu technical program coordinators are responsible for the technical implementation of the whole project. In the main office the data are processed, analyzed and archived following SOP and good practice guidelines.

Yagasu field staff are trained in all necessary activities to perform the forest inventory, tracking with GPS, socio-economic monitoring, biodiversity monitoring and forest establishment monitoring (survival rate, nursery monitoring, etc.).

More than 20 field officers of Yagasu are responsible for the different districts of the project areas ranging East to West of Northern coast of Sumatra. Generally, they are part of the communities and are well acquainted with the specific conditions within the different planting areas. The mangrove inventory and monitoring surveys are conducted by them in the field as part of the carbon survey teams. All results are directly brought to the main office of Yagasu in Medan

Yagasu has worked on post-tsunami coastal rehabilitation, reconstruction and economic restoration, and manage Mangrove Education Center (MEC) in Muara Angke – Jakarta. These actions created significant positive impacts among groups of local people in some villages in Aceh and North Sumatra. Up to end 2011 Yagasu has planted around 6,4 million mangroves trees with a survival rate around 75%. The donors supporting the mangrove rehabilitation programs are UNESCO-Jakarta, Wetlands International, Conservation International, KEHATI Foundation, Atlas Logistique-France, HELP-Germany, Keidanren-Japan, Islamic Relief, Muslim Aid, Rahmat Foundation, Waspada-local media, Newmont Company, BRR Aceh-Nias, District

Forestry Department and Indonesian Environmental Department and Planete Urgence.

In addition, Yagasu have also managed its own Carbon and Biodiversity Research Center in Percut for permanent plots of carbon and biodiversity study.

Local community groups

Workers involved in the project are people from the local area that have been trained through capacity building sessions to be able of develop the project activities and sustainable businesses associated to mangroves.

North Sumatra University in Medan

The North Sumatra University of Medan collaborate in research and scientific review. Brings knowledge and support the correct implementation of the project activities.

2.4.2 Management Team Experience (G4.2)

See Section 2.4.1.

2.4.3 Project Management Partnerships/Team Development (G4.2)

To develop the activities regarding the CCB validation/verification and VCS verification, Yagasu count with an expert in social assessment, an expert in biodiversity and also a professor expert in carbon quantification.

The community groups that develop the plantation activities have received an adequate capacity building from Yagasu that ensures the good practices. See Section 7.2. Community monitoring data, as well as CCB PDD Appendix I.

2.4.4 Financial Health of Implementing Organization(s) (G4.3)

There are no reports that demonstrate that the Project participants (Livelihoods Fund and Yagasu) are involved or complicit in any form of corruption.

Livelihoods Fund is able to provide adequate financial support to new instances included in this monitoring period.

2.4.5 Avoidance of Corruption and Other Unethical Behavior (G4.3)

See Section 2.4.4.

2.4.6 Commercially Sensitive Information (*Rules* 3.5.13 – 3.5.14)

There is no any commercially sensitive information excluded from the public version of the project description.

2.5 Legal Status and Property Rights

2.5.1 Recognition of Property Rights (G5.1)

The ownership of project sites is 92% as private lands owned by the local communities and 8% communal lands that are managed by the village governments. The abandoned private lands usually have arisen after the intensive shrimp collapse.

Under MoUs with local community groups and village governments, the local community groups plant the mangroves in community and village lands. In many plots, the mangroves are planted as part of silvo-fishery system (70% mangrove vegetation and 30% open water for fish/shrimp/crab farming).

The communities have agreed that the plantation mangrove is carried out for the benefit of the local communities who have recognized the benefits generated out of such plantations, including mangroves as a bio-shield, increased biodiversity, to protect embankments against climate change consequences, increased productive activities like fishing, crab production, organic silvo-fishery, mangrove foods, mangrove inks for batik, eco-tourism, etc. and increased new income generating options, with the only exception being the carbon credits generated by project.

The communities have agreed that the property rights on the carbon credits generated by this restoration are exclusively allocated to the proponent of the project. Under this agreement, the beneficiary community is committed not to assert any property rights over the carbon credits generated and/or to be generated. The sharing of VCUs between project stakeholders is governed by a series of agreements that extend over all activities in Aceh and North Sumatra. A copy of the community agreement is available as supporting documentation.

Due to the careful selection process of the project areas access/use rights are clearly defined. Disputes over land tenure or ownership do not exist.

2.5.2 Free, Prior and Informed Consent (G5.2)

According to the previously exposed it can be demonstrated that:

- The project is being developed at the properties of the land-owners and that the project does not invades other communitarian property or any government property or any private property.
- The participation process guarantees free, prior and informed consent obtained from those whose property rights are affected by the project as well as an appropriate compensation has been allocated to these parties.

Livelihoods Fund has the right over the Certified Emission Reductions generated and to be generated by the project activities taking place in these properties.

2.5.3 Property Right Protection (G5.3)

The project is not complicit of involuntary relocation. The environmental authorities have knowledge about the project before the project start date and gave approval for the implementation of the plantation for all instances.

Project areas are protected through the legally binding agreements between Yagasu and the communities as well as between Livelihoods and Yagasu.

2.5.4 Identification of Illegal Activity (G5.4)

The illegal activities occurring inside the project zone are logging for wood and charcoal, intensive aquaculture and land conversion (oil palm). A total of 241 illegal findings have been identified since the beginning of the project (77% illegal logging, 10% poaching of fauna and 13% land encroachment). See in Section 7.2. Community monitoring data, Indicator C.HC.7.

Yagasu has implemented a village regulation program to develop specific regulation (Perdas) containing policy and technical guidelines regarding these activities. The establishment and implementation of this regulation have been developed through 122 village meetings with local communities' participation.

Regarding green belt protection a Mangrove Patrolling Unit has been created in order to prevent, detect and control illegal activities. Currently there are 154 patrols covering a total area of 17,388.77 ha in the villages and 24,841.10 ha of coastal mangrove. These patrols involve 1,244 people.

2.5.5 Ongoing Disputes (G5.5)

There are no ongoing or unresolved disputes or conflicts regarding to rights to lands in the project zone. However, sometimes land-owners change their minds and want to sell the lands, or convert the mangroves into oil palm. Measures taken to resolve and prevent this kind of issues are the establishment of agreements with owners prior to any activity. In this sense, a total of 178 MoUs have been signed between Yagasu and local community groups, 126 MoUs have been signed between Yagasu and village governments, and 6 MoUs have been signed at provincial level.

2.5.6 National and Local Laws (G5.6)

Regulatory framework

Indonesia is party of the UNFCCC (United Nations Framework Convention on Climate Change) since August 1994. The country has ratified the UNFCCC (1995), the Kyoto Protocol (2004), has signed the Paris Agreement (Paris 2015, COP21).

The Republic of Indonesia participates voluntarily in the Clean Development Mechanism (CDM).

The current grouped project is in compliance with this regulatory framework, as afforestation is one of the mechanisms considered in the AFOLU scope, by which GHG emissions are expected to be reduced.

National legislation

- Law No. 48 Year 1991
- Biodiversity Strategy and Action Plan (IBSAP), 2003
- Law No. 5 Year 1960, Land Management
- Law No. 5 Year 1967, Basic Provisions of Forestry
- Law No. 9 Year 1985, Fishery
- Law No. 5 Year 1990, Conservation of Biological Nature Resource and Its Ecosystem
- Law No. 9 Year 1990, Tourism
- Law No. 12 Year 1992, Crop Cultivation System
- Law No. 16 Year 1992, Fauna, Fish and Flora Quarantine
- Law No. 24 Year 1992, Land Use Planning
- Law No. 5 Year 1994 on Legalization of United Nations Convention concerning Biological Diversity
- Law No. 23 Year 1997, Environment Management
- Law No. 22 Year 1998, Autonomy of Local Government
- Law No. 41 Year 1999, Forestry
- Law No. 32 Year 2004, Indonesian Decentralization Law
- Law No. 27 Year 2007, Management of Coastal Areas and Small Islands
- Government Regulation No. 13 Year 1994, Fauna Hunting
- Government Regulation No. 28 Year 1985, Forest Protection

- Government Regulation No. 29 Year 1986, Environmental Impact Assessment
- Government Regulation No. 20 Year 1990, Water Pollution Control
- Government Regulation No. 35 Year 1991, Rivers
- Government Regulation No. 45 Year 1992, Autonomy of District Government
- Government Regulation No. 31 Year 2004, Fishery
- Government Regulation No. 68 Year 1998, Areas of Natural Conservation and Preservation
- Presidency Decree No. 32 Year 1990, Management of Protected Areas
- Regulation of Minister of Forestry No. 37/Menhut-II/2007, Community Forest Management
- Decree of Minister of Forestry No. 239/Menhut-V/2007, National Mangrove Task Force 2007

3 CLIMATE

3.1 Monitoring GHG Emission Reductions and Removals

3.1.1 Data and Parameters Available at Validation

Data and parameters available at validation of the grouped project are described in the project description document. The following list includes only data and parameters used in the second verification of the first and second project activity instances.

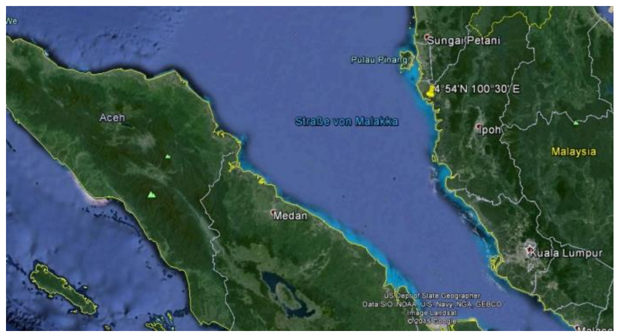
Data / Parameter	$\Delta C_{BSL,t}$
Data unit	t CO2-e
Description	Baseline net GHG removals by sinks in year t
Source of data	*Calculated*
Value applied:	0
Justification of choice of data or description of measurement methods and procedures applied	The project area is stratified into zero baseline and dwarf mangrove baseline stratum considering as shrubs according to the AR CDM tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities”. Based on this, no removals by sinks are assumed
Purpose of Data	Calculation of ex-ante and ex-post baseline emissions
Comments	-

Data / Parameter	CF_{TREE}
Data unit	t C (t d.m.)-1
Description	Carbon fraction of tree biomass
Source of data	Default value of AR CDM tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities” is applied.
Value applied:	0.47
Justification of choice of data or description of measurement methods and procedures applied	According to AR-TOOL14, the above default value (0.47) should be used unless transparent and verifiable information can be provided to justify a different value.
Purpose of Data	Determination of project emission/removals
Comments	-

Data / Parameter	R_j
Data unit	Dimensionless
Description	Root-shoot ratio for tree species *j*
Source of data	Application of the following equation from the AR-TOOL14: R_j=exp[-1.085+0.9256*ln(AGB)]/AGB Where AGB is the above-ground tree biomass per hectare (in t d.m. ha^{-1})
Value applied:	See database
Justification of choice of data or description of measurement methods and procedures applied	According to AR-TOOL14, root-shoot ratio for tree species shall be calculated with the formula: Rj=exp[-1.085+0.9256*ln(AGB)]/AGB
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	$f_j\ (x_{1,I}, x_{2,I}, x_{3,I}, ...)$
Data unit	t d.m (tone of dry matter).
Description	Above-ground biomass of the tree returned by the allometric equation for species *j* relating the measurements of tree *i* to the above-ground biomass of the tree.
Source of data	For ex-post: Two equations are used for this verification: 1. Specific equation for Rhizophora spp. from Ong et al. (2004)[12] 2. Common allometric equation for other mangroves from *Komiyama et al., 2005*[13].
Value applied:	For ex-post: Rhizophora spp.: *Rhizophora apiculata* $W_{top} = 0.235\mathrm{DBH}^{2.42}$ $r^2 = 0.98$, $n = 57$, $D_{max} = 28$ cm, Ong et al. (2004) Other mangrove species: Common equation $W_{top} = 0.251\rho D^{2.46}$ $r^2 = 0.98$, $n = 104$, $D_{max} = 49$ cm, Komiyama et al. (2005) Where: W_{TOP} = Above-ground biomass; kgd.m

[12]http://www.sciencedirect.com/science/article/pii/S0378112703003906 (also provided as supporting documentation)
[13]Komiyama, A.; Poungparn,S. & Kato, S. (2005) Common Allometric equations for estimation the tree weight of mangroves. Journal of tropical Ecology 21: 471-477. Cambridge University Press (available as supporting documentation), also in http://www.cifor.org/publications/pdf_files/WPapers/WP86CIFOR.pdf

	D = Diameter (at breast height); cm ρ = wood density; g/cm^3
Justification of choice of data or description of measurement methods and procedures applied	Ex-post allometric equations are appropriate as determined by the A/R Methodological Tool “Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in A/R CDM project activities”. Since the majority of trees in the project are R. apiculata, this species specific equation is considered. It is also applied to the other Rhizophora species (Stylosa and Mucronata), being all stilt mangroves. This equation represents the best option for this first verification since it was derived (Ong et al. 2004) in Malaysia, just opposite of the project area in Sumatra (see location on the map below).  Further, it is one of few equations which also included stilt roots into the aboveground biomass. Finally it is the only equation found which includes also smaller diameters (the study ranged from 1.1 – 28 cm). The common equation developed by Komiyama excludes stilt roots of mangroves and represents one of the most widely used equations, see supporting documentation.
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	$dSOC_t$
Data unit	t C ha^{-1} yr^{-1}
Description	The rate of change in SOC stocks within the project boundary, in year t.
Source of data	
Value applied:	3.32
Justification of choice of data or description of measurement methods and procedures applied	Value calculated based on Murdiyarso et al 2015[14] publication.
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	$EF_{CH4,i}$
Data unit	g CH_4 (kg dry matter burnt)$^{-1}$
Description	Emission factor for CH_4 in stratum i
Source of data	CDM A/R Methodological Tool “Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity” (Version 04.0.0)
Value applied:	A default value of 6.8 from the above CDM A/R Methodological Tool is used.
Justification of choice of data or description of measurement methods and procedures applied	A default value is used since values are not available from sources numbered a., b., and c. described below. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference: a. Regional/national inventories e.g. national forest inventory, national GHG inventory; b. Inventory from neighbouring countries with similar conditions; c. Globally available data applicable to the project site or to the region/country where the site is located; d. Default values
Purpose of Data	Calculation of ex-post project emissions
Comments	-

[14] Murdiyarso, D et al (2015). The potential of Indonesian mangrove forests for global climate change mitigation. Nature climate change. Available as supporting documentation

Data / Parameter	$EF_{N2O,i}$
Data unit	g N_2O(kg dry matter burnt)$^{-1}$
Description	Emission factor for N_2O in stratum i
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	A default value of 0.2 from the above CDM A/R Methodological Tool is used.
Justification of choice of data or description of measurement methods and procedures applied	A default value is used since values are not available from sources numbered a., b., and c. described below. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference: a. Regional/national inventories e.g. national forest inventory, national GHG inventory; b. Inventory from neighbouring countries with similar conditions; c. Globally available data applicable to the project site or to the region/country where the site is located; d. Default values;
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	GWP_{CH4}
Data unit	Dimensionless
Description	Global warming potential for CH_4
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	21
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	GWP_{N2O}
Data unit	Dimensionless
Description	Global warming potential for N_2O
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	310
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of ex-post project emissions
Comments	-

<table>
<tr><td>Data / Parameter</td><td>$COMF_i$</td></tr>
<tr><td>Data unit</td><td>Dimensionless</td></tr>
<tr><td>Description</td><td>Combustion factor for stratum i</td></tr>
<tr><td>Source of data</td><td>CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)</td></tr>
<tr><td>Value applied:</td><td>The corresponding default according to the age is used, unless transparent and verifiable information can be provided to justify a different value:
<table>
<tr><th>Mean age (years)</th><th>Default value</th></tr>
<tr><td>3-5</td><td>0.46</td></tr>
<tr><td>6-10</td><td>0.67</td></tr>
<tr><td>11-17</td><td>0.50</td></tr>
<tr><td>18 and above</td><td>0.32</td></tr>
</table></td></tr>
<tr><td>Justification of choice of data or description of measurement methods and procedures applied</td><td>A default value is used since values are not available from sources numbered a., b., and c. described below. Default emission factor for tropical forest from the CDM A/R tool. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference:
a. Project-specific calculation, regional/national inventories e.g. national forest inventory, national GHG inventory;
b. Inventory from neighbouring countries with similar conditions;</td></tr>
</table>

	c. Globally available data applicable to the project site or to the region/country where the site is located d. Default values
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	*E*
Data unit	t d.m. (or t d.m. ha^{-1})
Description	Acceptable margin of error (i.e. one-half the confidence interval) in estimation of biomass stock within the project boundary
Source of data	AR-TOOL14
Value applied:	10% of the mean value of biomass stock
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	t_{val}
Data unit	Dimensionless
Description	Two-sided Student's t-value at infinite degrees of freedom for the required confidence level
Source of data	AR-TOOL14
Value applied:	According to the student's t-distribution table, 1.654 for confidence level 90% and infinite degrees of freedom
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of ex-post project emissions
Comments	The value used is for the 90% confidence level for determination of biomass stock prescribed in the A/R CDM project activities since a different confidence level is not prescribed in the methodology.

Data / Parameter	DF_{DW}
Data unit	Percent
Description	Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass
Source of data	A/R Methodological tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities
Value applied:	6%
Justification of choice of data or description of measurement methods and procedures applied	Default value depending on the biome, elevation and precipitation of the area. The tool recommended this value unless transparent and verifiable information can be provided to justify a different value.
Purpose of Data	Calculation of ex-post project emissions
Comments	-

3.1.2 Data and Parameters Monitored

Data / Parameter	A_i
Data unit	*Ha*
Description	Area of tree biomass stratum i
Source of data	*GIS and GPS*
Description of measurement methods and procedures to be applied	Once the criteria for ex-post stratification, has been stratified and Ai has been calculated using GIS.
Frequency of monitoring/recording	At least for the first monitoring but can be updated in every verification.
Value applied:	

Stratum ID	Description	A_i (ha) *First project activity instances*	A_i (ha) second project activity *2nd instances*
1	Stratum 1, dense: highest crown cover (>60%) and lower mortality. Planting lines are not visible in the high-resolution image available from Google Earth since the trees have reached the tangency of canopies. This stratum probably is related to oldest plantations made in 2011-2013.	6,13.74	151.82
2	Stratum 2, medium: medium crown cover (10-60%) and lower mortality. Planting lines are visible in the high-resolution image available from Google Earth. This stratum is related to plantations made in 2013-2015.	2,425.44	110.28
3	Stratum 3, scattered: low crown cover and high mortality. No plants or poor growth where it is not possible to differentiate the planting lines in the high-resolution image available from Google Earth. This stratum is related to	1,961.09	16,61

		plantations made in 2015 or after[15].		
	TOTAL		5,000.27	278.71
Monitoring equipment	*GIS and other spatial information software*			
QA/QC procedures to be applied	N/A			
Purpose of data	Calculation of project emissions/removals			
Calculation method	Use of GIS tool			
Comments	-			

Data / Parameter	n_i
Data unit	*Dimensionless*
Description	Number of sample plots in stratum i
Source of data	*Calculated*
Description of measurement methods and procedures to be applied	*N/A*
Frequency of monitoring/recording	Ni is calculated for each monitoring event, at least every five years.
Value applied:	***The following values are estimated from the first and second project activity instance:***
Monitoring equipment	*N/A*
QA/QC procedures to be applied	*N/A*
Purpose of data	*Calculation of project emissions/removals*

Stratum	n_i
1	77
2	96
3	69
	242

[15] Pending new plantation plot called recalled plots

Calculation method	The calculation method is described in the tool“ Calculation of the number of sample plots for measurements within A/R CDM project activities”(version 02.1.0)[16]
Comments	-

Data / Parameter	w_i
Data unit	*Dimensionless*
Description	Relative weight of the area of stratum i, the area of the stratum i divided by the project area.
Source of data	*Calculated as the area of the stratum i divided by the project area.*
Description of measurement methods and procedures to be applied	*N/A*
Frequency of monitoring/recording	Calculated for each monitoring event, at least every five years
Value applied:	*The following values are estimated from the first and second project activity instances:* *Stratum* — w_i *1* — 0.14 *2* — 0.48 *3* — 0.37
Monitoring equipment	*N/A*
QA/QC procedures to be applied	*N/A*
Purpose of data	*Calculation of project emissions/removals*
Calculation method	Area of the stratum i divided by the project area
Comments	-

Data / Parameter	s_i
Data unit	t d.m. (or td.m. ha^{-1})
Description	Estimated standard deviation of biomass stock in stratum i
Source of data	*Inventory or default value*

[16]https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf

Description of measurement methods and procedures to be applied	*N/A*
Frequency of monitoring/recording	s_i is calculated for each monitoring event, at least every five years
Value applied:	*The following values are estimated from the first and second project activity instance:* *Stratum* / s_i: *1* – 35.77 *2* – 14.75 *3* – 19.34
Monitoring equipment	*N/A*
QA/QC procedures to be applied	*N/A*
Purpose of data	*Calculation of project emissions/removals*
Calculation method	Excel or tool available to calculate standard deviation
Comments	-

Data / Parameter	$A_{PLOT,i}$
Data unit	*Ha*
Description	Size of sample plot in stratum i
Source of data	Calculated
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Every three years since the year of the initial verification
Value applied:	*0.009 (5m radius)*
Monitoring equipment	*N/A*
QA/QC procedures to be applied	As this value is calculated based on the expected planting density, during the sampling process it needs to be confirmed that the mean number of trees per sample plot is more than 15.
Purpose of data	*Calculation of project emissions/removals*
Calculation method	$A_{PLOT,i}$ will be calculated depending on the expected density (trees/ha) in each stratum, with the objective of having around 15 trees per sample plot.

Comments	-

Data / Parameter	$A_{BURN,i,t}$
Data unit	*Ha*
Description	Area burnt in stratum i
Source of data	Field measurement, remote sensing measurement or any other spatial information available
Description of measurement methods and procedures to be applied	The area is delineated on the ground using GPS or georeferenced remote sensing data or from any other spatial information available.
Frequency of monitoring/recording	The area burnt is measured whenever forest fire has occurred.
Value applied:	*N/A, wildfires did not occur in any mangrove forests in the areas in this monitoring period*
Monitoring equipment	GPS
QA/QC procedures to be applied	Field teams are trained in the correct use of GPS. They are also made fully aware of all procedures and the importance of collecting data as accurately as possible.
Purpose of data	*Calculation of project emissions/removals*
Calculation method	N/A
Comments	*Applicable only if wild fires occurs.*

Data / Parameter	X_i
Data unit	*Variable*
Description	Variables measured per tree for the calculation of above-ground biomass applying an allometric equation: DBH, height, D_{30},
Source of data	*Measured*
Description of measurement methods and procedures to be applied	*DBH and* D_{30}readings are taken from a caliper/ diameter tape at 1.3 m along the stem (for DBH) or at 30 cm from the ground for D_{30}. Height – readings are taken on calibrated pole placed along the longitudinal axis of the tree.
Frequency of monitoring/recording	Measured every monitoring event, at least every five years
Value applied:	*n.a.*
Monitoring equipment	*DBH and* D_{30}, a caliper/ diameter tape is used.

	Height, a calibrated pole is used.
QA/QC procedures to be applied	Field teams are trained in all inventory procedures including correct measurement – as described in the Pilot Inventory, Sampling & Monitoring Plan. Field-team members are fully aware of all procedures and the importance of collecting data as accurately as possible. Finally, a quality check is undertaken as part of inventory to identify and correct errors if any.
Purpose of data	*Calculation of project emissions/removals*
Calculation method	n.a.
Comments	*D_{30},only for the first verification*

Data / Parameter	T
Data unit	*Year*
Description	Time period elapsed between two successive estimations of carbon stock in a carbon pool
Source of data	Recorded time
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	*Calculation of project emission removals*
Calculation method	N/A
Comments	If the two successive estimations of carbon stock in a carbon pool are carried out at different points of time in year t_2 and t_1, (e.g. in the month of April in year t_1 and in the month of September in year t_2), then a fractional value will be assigned to T

3.1.3 Monitoring Plan

3.1.3.1 Organizational structure

Institutionally, a permanent YAGASU Carbon Survey Team has been set up consisting of several field teams consisting of 2-3 field officers and under the supervision of the project coordinator (Mr. Bambang Suprayogi) and carbon monitoring coordinator (Mr. Rangga Bayu). These teams will undertake all surveys in the project to ensure consistency in measurements and will implement the carbon monitoring inventory of permanent sampling points (PSPs). The field measurement is supported with a project based android data collection system (YAGASU survey App) which ensures standardized data collection with QA/QC functions to allow minimizing errors already during this stage.

Further, YAGASU and Livelihoods have set up a stringent verification system with external tree audits including annual boundary verification and revision, if required.

The organizational structure of the monitoring is divided into two layers. The first layer is represented by the YAGASU field staff which are trained in all necessary activities to perform the forest inventory, boundary tracking with GPS, socio-economic monitoring and forest establishment monitoring (survival rate, nursery monitoring, etc.). More than 20 field officers of YAGASU are responsible for the different districts of the project areas ranging east to west of northern coast of Sumatra. Generally, they are part of the communities and are well acquainted with the specific conditions within the different planting areas. The forest inventory and monitoring surveys are conducted by them in the field (as part of the carbon survey teams). All results are directly brought to the main office of YAGASU in Medan; the second layer of the project. Bambang Suprayogi and the YAGASU technical program coordinators are responsible for the technical implementation of the whole project. In the main office the data are processed, analyzed and archived following standard operation procedures and good practice guidelines. To guarantee high level of certainty of the results, the YAGASU technical program coordinators will periodically crosscheck the data in the field as an independent survey. It is envisaged to at least verify 10% of the data after each inventory or survey conducted.

The project implementation is based on the local presence of YAGASU staff in project area. The main role of the field officers is to manage the reforestation/restoration activity in close cooperation with YAGASU technical program coordinators.

- Selecting randomly and verify the GPS location of at least 10% of the plots planted during a particular planting season.
- Comparing the trees planted with the trees recorded in the planting plans
- Assessing the degree (in %) of the survival of the mangrove seedlings and preparation of a report with the findings considering a minimum precision of 10% at the 90% confidence level. Replanting of mangroves is only necessary if the optimal tree density of 3,000 trees per ha cannot be achieved due to very high natural mortality. This has to be decided on a site-by-site basis due to the varying local tidal and ecological conditions.
- Area verification. Project parcels will be verified using GPS in the field as well as through Google Earth imagery analysis.

The Livelihoods' mangrove restoration grouped project in The East Coast of Aceh and North Sumatra Province is part of the portfolio of the Livelihoods Fund and for the monitoring process it has followed the Monitoring Livelihoods Projects– Operating Procedures Series. These procedures consist in several guidance documents for different monitoring phases:

- Mapping and stratifying
- Inventory design
- Field work planning
- Field work

This document complements those general procedures with the description of the specific characteristics of the monitoring process of the Sumatra project, based on the abovementioned guidance and adapted to the project, and with specific procedures for data recording and quality assessment & quality checks.

3.1.3.2 Stratifying

Summary

For this second verification, new project stratification has been performed. In the first monitoring of the project the area was stratified in four strata considering based on growth and survival rate. However, this stratification is not consistent with the state of the vegetation during the second verification, as can be seen from the high-resolution Google Earth images. Furthermore, in the first stratification each planted parcel was considered in one project strata and it has been demonstrated that there is a gradient of ecological conditions (mainly salinity) inside every planted parcel depending on the distance to the closest water stream. This issue has led to a high heterogeneity inside each plantation polygon having as consequence a high sampling uncertainty. For these reasons and in order to reduce the sampling error, it has been decided to carry out a new stratification based on a supervised classification of Sentinel-2 satellite images. The latest Google Earth high-resolution images have been used to generate a points sample for training the classifier.

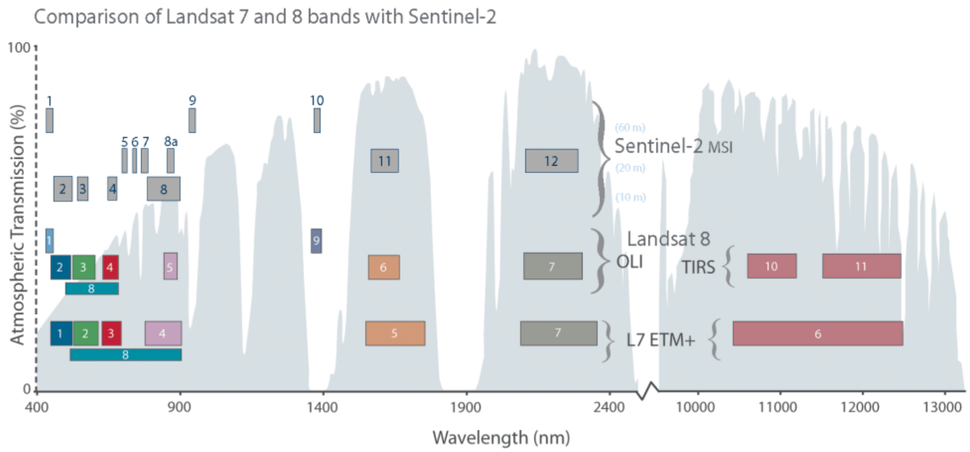
Remote sensing data

The different soil and salinity conditions mean that the spatial variability of aerial biomass is high even within the same plantation plot. In order to improve the forest inventory estimates, stratification has been carried out based on the automatic classification of Sentinel-2A satellite images.

Sentinel-2 mission of the European Space Agency (ESA) consists of two twin satellites: Sentinel 2A, launched on 23 June 2015 and Sentinel-2B, launched on 7 March 2017. The launch of these satellites is part of ESA's Copernicus Programme. Sentinel-2 represents a high-resolution multispectral mission designed to monitor the Earth's surface by analyzing the vegetation cover. Its temporal resolution of each of the satellites is 10 days, 5 days considering both satellites. It

consists of 13 spectral bands, four bands at 10 m, six bands at 20 m and three bands at 60 m.

Figure 3 Identification of Sentinel-2 bands with respect to Landsat 7 and 8 bands.



As can be seen in the figure the Sentinel-2 bands are like the Landsat 7 and 8 bands, except for the thermal band of the spectrum. In addition to extending the spectral and temporal resolution, Sentinel-2 offers bands at different spatial resolutions of 10, 20 and 60 m. The size of each Sentinel-2 image is 100 x 100 Km2.

Sentinel-2 images with a processing level of 1C have been used for the creation of the free cloud mosaic. This level of processing provides geometrically corrected images with reflectance values on the roof of the atmosphere.

Proposed stratification

Three strata that have been considered are listed below:

- Stratum 1, dense: highest crown cover (>60%) and lower mortality. Planting lines are not visible in the high-resolution image available from Google Earth since the trees have reached the tangency of canopies. This stratum probably is related to oldest plantations made in 2011-2013.
- Stratum 2, medium: medium crown cover (10-60%) and lower mortality. Planting lines are visible in the high-resolution image available from Google Earth. This stratum is related to plantations made in 2013-2015.
- Stratum 3, scattered: low crown cover and high mortality. No plants or poor growth where it is not possible to differentiate the planting lines in the high-resolution image available from Google Earth. This stratum is related to plantations made in 2015 or after17.

Second instance

In this verification period a new instance is added. The stratum per instance is described in table below:

[17] Pending new plantation plot called recalled plots

	Areas per stratum in first Project activity instance (ha)	Areas per stratum in first second project activity instance (ha)
Stratum 1	613,74	151,82
Stratum 2	2425,44	110,28
Stratum 3	1961,09	16,61
	5000,27	**278,71**

The second instance includes plantation areas that were planted between 2011 and 2017 but that were not included in the first verification period due to economic reasons. The year of the plantation of both instances are included below:

	Planted areas of first project activity instance (ha)	Planted areas of Second project activity instance (ha)
2011	193,145	7,77
2012	860,09	52,15
2013	2006,985	12,78
2014	1639,9	170,54
2015	299,86	1,35
2016		26,45
2017		8,24
	4999,98	**279,28**

Image classification

Random forest (Breiman 2001, henceforth RF) classification models have been applied in the stratification process. RF is part of the automatic learning methods and has become a widespread technique in vegetation mapping and prediction of forest variables.

RF is a machine learning technic based on the ensemble of many decision trees. For each individual decision tree, data is randomly segregated in two data sets for training and validation. The procedure used in the random generation of decision allows low correlation between the individual decision trees, ensuring robustness in the RF results. Several advantages have been observed comparing RF with other machine learning alternatives, such as the capacity for working with numerous predictor variables, the lack of overfitting problems, and robustness with respect to noise in the data.

The training sample has been generated using high-resolution image available in Google Earth.

Sentinel-2 spectral bands and vegetation indices derived from these bands have been used as predictive covariables in the classification. As it can be seen in figure 1, Sentinel-2 adds information on the wavelengths between red and infrared, an area of the electromagnetic spectrum where vegetation has a very different spectral behavior with respect to other land cover. The following table shows the predictor variables used in the classification.

Table 3. Variables used for stratification

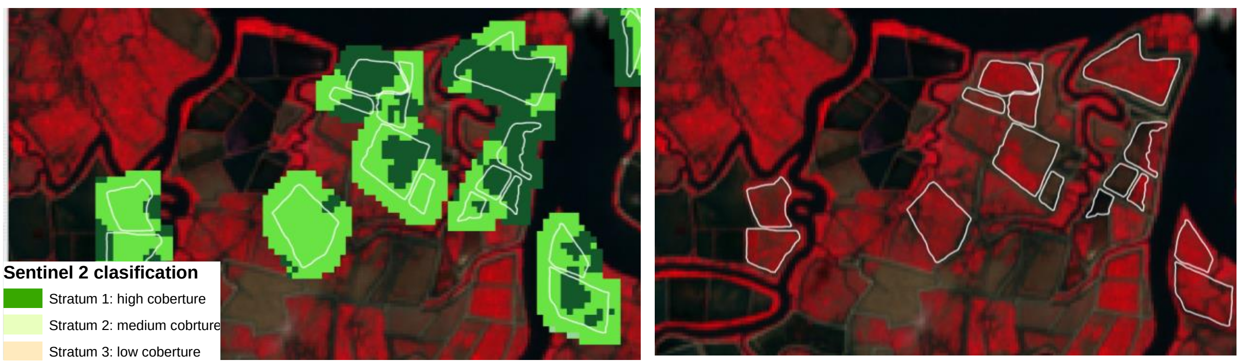
Variable types	Number of covariable	Covariable	Sentinel-2 bands
Spectrals	1	Blue	B2
	2	Green	B3
	3	Red	B4
	4	Red-edge	B5
	5	Red-edge	B6
	6	Red-edge	B7
	7	Near infrared (NIR)	B8
	8	Near infrared (NIR)	B8A
	9	Near infrared (SWIR-1)	B11
	10	Near infrared (SWIR-2)	B12

Variable types	Number of covariable	Covariable	Definition	Sentinel-2 bands
Vegetation index	11	NDVI1 (Rouse et al. (1973))	(NIR - R)/(NIR+ R)	(B7-B4)/(B7 + B4)
	12	NDVI2 (Rouse et al. (1973))	(NIR - R)/(NIR+ R)	(B8-B4)/(B8 + B4)
	13	NDI45 (Delegido et al. (2011	(NIR - R)/(NIR+ R)	(B5-B4)/(B5 + B4)
	14	SAVI	(1+L)*(NIR - R)/(NIR+ R+L)	(1+L)*(B7-B4)/(B7+B4+L)
	15	TCARI (Haboudane et al. (2002))	3*[(RE-R)-0.2*(RE-G)*(RE/R)]	3*[(B5-B4)-0.2*(B5-B3)*(B5/B4)]
	16	OSAVI	(1.16)*(NIR-R)/(NIR+R+0.16)	(1.16)*(B7-B4)/(B7+B4+0.16)
	17	MTCI (Dash and Curran (2004))I	(NIR-RE)/(RE-R)	(B6-B5)/(B5-B4)
	18	MCARI (Daughtry et al. (2000))	[(RE-R)-0.2(RE-G)]*(RE-R)	[(B5-B4) - 0.2(B5-B3)]*(B5-B4)
	19	GNDVI	(NIR-G)/(NIR + G)	(B7-B3)/(B7 + B3)

Variable types	Number of covariable	Covariable	Definition	Sentinel-2 bands
		(Gitelson et al. (1996))		
	20	PSSRa (Blackburn (1998))	NIR/R	B7/B4
	21	S2REP (Frampton et al, 2013)	705 +35*((((NIR + R)/2-RE1)/(RE2-RE1))	705 +35*((((B7 + B4)/2)-B5)/(B6-B5))
	22	IRECI (Frampton et al, 2013)	(NIR-R)/(RE1/RE2)	(B7-B4)/(B5/B6)

The "salt and pepper" effect of the classification has been refined by using a majority filter with a 3x3 window. The result has been converted to shape format in order to facilitate further work. Figure 2 shows a detail of the final classification.

Figure 4 Example of the classification on the Sentinel-2 image



3.1.3.3 Inventory design

Sampling design

Due to the heterogeneity of biomass carbon stocks in the project area, the total number of necessary sample plots to get a sampling error lower than 10% is too high. As the measurement of a high number of sample plots was not feasible the project proponent decided to estimate biomass carbon stocks with a higher uncertainty than 10% and apply an uncertainty discount to the monitoring results following Appendix 2 of AR-TOOL14.

A number of 40 or 60 plots per stratum should be the minimum stablished. So the intensity of monitoring has being calculated, resulting in the following squared grids: stratum 1: 300 x 300 m; stratum 2: 400 x 400 m and stratum 3: 800 x 800 m. Those points that are placed within the project area will be the sampling pots and will be measured within a 5,64 meters radium. Therefore, it was decided to measure 230 sample plots based on the expert criteria and in the previous experience: 77 in stratum 1, 84 in stratum 2 and 69 in stratum 3.

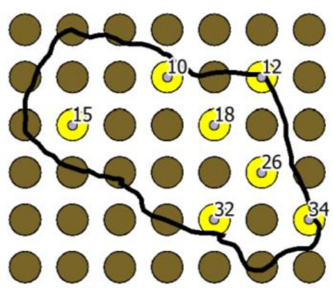
Table 4 Stratification

Stratum ID	Description	A_i (ha)	Number of sample plots
1	Low crown cover and high mortality (< 10%)	764,83	77
2	Higher crown cover (10-60%) and lower mortality	2.534,45	96
3	Highest crown cover (60-100%)	1.975,70	69
Total		**5274,98**	**242**

The stratum area per plot (strata area/number of plots per stratum) has been calculated following next steps:

1. In order to describe the stratum variability it is necessary to establish at least 40-60 sample plots in each stratum. It has been defined the following grid in order to place the permanent sample plot:
   - Stratum 1: 300 meters of grid size
   - Stratum 2 : 500 meters of grid size
   - Stratum 3: 500 meters of grid size
2. To locate the measurement plots within the planting areas a rectangular grid is used that is overlaid on the map of the planting area. This should be done directly with a GIS software

Figure 5 Random localization of sample plots within the selected planting areas.



ID	lon	lat
10	33502.57	2032810.87
12	33532.57	2032810.87
15	33472.57	2032795.87
18	33517.57	2032795.87
26	33532.57	2032780.87
32	33517.57	2032765.87
34	33547.57	2032765.87

Sampling procedure

Both the field work planning and the implementation of the field work are described in a specific section below.

The following procedure shall be followed for the collection, storage and transfer of data:

- Field data will be collected for each of the sampling plots following the template designed for the project (see Annex I). The form will be filled in the field both in paper and in digital through the application for mobile terminals CyberTracker.
- Team leaders shall download field forms from mobile terminals not more than once a week. The project will use a cloud storage service where team leaders will also upload mobile terminal downloads on a weekly basis.

- Each download will be named using a name that describes its content as follows: (team leader's name) _(Date with format aa/mm/dd).

The sampling procedure has a quality control and quality assessment described below in a specific section.

3.1.3.4 Data collection

A detailed description of the YAGASU carbon inventory procedures can be found in the SOPs (supporting documentation) including standard operating procedures on work safety, field measurement planning and organization, navigating in the field, and tree measurement procedures, etc.

In the field, several parameters are assessed/measured. The parameters relevant for the carbon monitoring are the following:

- Species name
- Origin: Classification of the tree according to the origin of the tree: Pre-existing tree, Planted tree or Natural regeneration.
- Size: Classification of trees and saplings.
- Diameter: Diameter at breast height (1.3 m above root collar) is measured if available, otherwise the diameter D30 is taken, 30 cm above the root collar.
- Total Height: Total tree height of the tree or sapling.

For navigation and data entry of tree and plot variables, a smartphone based inventory application was specifically designed for this project.

Figure 6. Field team measuring the sample plot

3.1.3.5 Data processing and sampling error estimation

Once the field inventory has been completed, the calculations are carried out considering the following steps:

- Record parcel data after field evaluation in the inventory database.
- Estimate the carbon content per tree, per plot, per hectare, and average carbon content per ha for each sampling plot separately for each project stratum.
- Estimate the carbon standard deviation [t C ha-1] for all plots.
- Estimate the relative error of the carbon variables using the equations of the Tool: Calculation of the number of sample plots for measurements within A/R CDM project activities (Version 02.1.0).

If the errors obtained in this phase are within the error margins required for the project, it will not be necessary to increase the sampling intensity, so the sampling will be considered as definitive.

If the error obtained in the sampling does not comply with the requirements, there are two alternatives:

- Recalculate the number of plots based on the results obtained using the tool Methodological Tool: Calculation of the number of sample plots for measurements within A/R CDM project activities (Version 02.1.0) and increase the number of inventory plots to the specified number. The location and sampling of these new plots will be carried out.
- Comply with the discounts in Appendix 2 of the tool Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities. Version 04.2. (AR-TOOL 14) (AR-TOOL 14).

3.1.3.6 Field work planning

The field work planning guidance describes all the necessary steps, to be followed during the work planning in pilot inventory and inventory field. In this section it is only included the specific aspects for monitoring field work implemented by Yagasu.

<u>Organizing sample plot information:</u>

One of the first steps in locating sample plots is to organize the information. Once the stratification and sampling design has been carried out (see previous sections) it is important to prepare the information from the sampling plots for the location of the project. Basically the following is necessary:

- Upload the coordinates of the sampling plots, including their IDs, to the GPS.
- Check that the GPS configuration is consistent with the GIS information.

- Check that GPS settings are coherent with GIS information (e.g. Reference system WGS 84).
- Print table of sample plots including ID and coordinates X, Y.
- Print maps to facilitate the location of sampling plots and identification of project plots (e.g. A4 with Google Earth image, project boundaries and sampling plots). This is important to avoid errors by placing parcels outside project boundaries.

Equipment and materials

The following list covers the necessary equipment for one survey team:

Table 5 Necessary equipment

Devices/ Materials	Check/Comment
Smartphone with Yagasu app	Check batteries and data app is ok. Check the app works properly.
GPS Receiver/ Field GIS e.g. Garmin or other	Check batteries and whether all necessary data are uploaded on the receiver, in particular the points of the measurement plots
Compass	
Caliper (>8 m)	To measure plot size
Caliper/ diameter tape in mm	To measure tree diameters.
Overview map / Roadmap / Topographic map	
Field forms + pencils	Depending on number of plots
Clipboard	For fixing and filling out field forms
Replacement batteries	Fresh batteries for GPS, UAVs, or others
Clothing (shoes, raincoat, etc. in outdoor outfit)	
Backpack for devices, usually it is waterproof	
First aid kit	

Calibration and testing of equipment

Calibration and test measurement of equipment is planned to be done regularly as it is specified following:

- Weekly, it is calibrated the caliper and the measuring tapes. GPS will be checked using other GPS to detect possible errors.

Orientation and training on standard field procedures

The coordinator of field survey has done a thorough training to inventory teams, specifying the following aspects:

- Different areas of the project: Number and localization of plots and parcels. Each team knows the id plot by the description id field.
- Management of different devices.
- Management of maps.
- Orientation and access to the plots.
- Characteristics of strata and plantations.
- Standard field procedures, focusing in following aspects:
  - Tree variables to be measured and procedures to reduce errors in measurements.
  - Procedure to fill the file forms/app.
  - Importance of systematic procedures and homogeneity between different teams.
  - Importance in transcription/download/send data.

Organizing monitoring work:

To organize field work GIS information is required. Before starting the field work the following questions have to be answered:

- Where is the new working area (use of the total project area base map)
- Which project sites can be covered from this location?
- How is the terrain (maps and/or satellite-images)?
- Where are good starting points to start the navigation to the first plot (grid map for each site 1:10.000)?
- What is the best track from plot to plot (grid map 1:10.000)?

3.1.3.7 Monitoring of net anthropogenic GHG removals

Ex post estimation of the baseline net GHG removals by sinks

The baseline net GHG removals by sinks are estimated following AR-TOOL14, as indicated in the PD, therefore they have not been monitored.

Ex post estimation of the actual net GHG removals by sinks

For the monitoring of GHG removals by sinks the project proponent has put in place within Yagasu a unit dedicated to the monitoring processes. This unit benefits from the expertise of a forest carbon specialist who is in charge of the design and calculations of the monitoring inventory.

The monitoring of the GHG removals has been based on the stratification of the project and the sampling design.

Ex post estimation of the actual net GHG removals by sinks has been done as explained in the following paragraphs.

- Changes in carbon stocks in tree and shrub biomass: once the project area is stratified and the sampling design is accomplished, ex-post changes in carbon stocks in tree and shrub biomass have been estimated according to AR-TOOL14. Specific allometric equations have been developed for the project. Based on those equations the following variables have been measured in all trees inside each sample plot (see further details in the SOP):
    - Ht = Total height; cm
    - Dmean = Mean crown cover diameter; cm
- Changes in carbon stocks in deadwood have not been monitored because they have been calculated using the conservative default-factor based method of the CDM tool AR-TOOL12.
- Changes in carbon stocks in soil organic carbon pool have been estimated using the default methods proposed in the selected AR CDM methodology (AR-AM00014), therefore no monitoring has been done of this carbon pool.
- GHG emissions within the project boundary: the only source of GHG emission considered in the selected methodology is burning of biomass attributable to the project activity and it is calculated using the A/R CDM Methodological tool: "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity, (Version 04.0.0)". In this grouped project there will not be fire for site preparation or fire to clear the land of harvest residue prior to replanting of the land or other forest management. The only potential source of GHG emission is forest fires that are monitored during the verification period.

Ex post estimation of leakage

In the first project activity instance displacement of pre-project agricultural activities does not occur. No specific data needs to be monitored for the assessment of this source of leakage.

Ex post estimation of the net anthropogenic GHG removals

The net anthropogenic greenhouse gas removals by sinks have been calculated based on the selected methodology as the actual net greenhouse gas removals by sinks minus the baseline net greenhouse gas removals by sinks minus leakage as appropriate.

Quality control and quality assurance (QA/QC) procedures

A rigid quality control and quality assurance (QA/QC) method is implemented to ensure: (i) reliability of collection of field and field measurements, (ii) verification of the methods used to collect field data, and (iii) verification of data maintenance and archiving.

Procedures to ensure reliable field measurements

Collecting reliable field measurement data is an important step to ensure the quality of field measurements and therefore is an important step in the quality assurance plan. In this monitoring field measurements have been done in some trees for the QC/QA of the field information. Those responsible for the measurement work have been trained in all aspects of the field data collection and data analyses. A Standard Operating Procedures (SOPs) have been developed including all steps of the field measurements, and this document has been shared with field teams. These SOPs describe in detail all steps to be taken of the field measurements (including standard field forms) and contain provisions for documentation for verification purposes so that future field personnel can check past results and repeat the measurements in a consistent way. To secure the quality of the standard operating procedures, they have been elaborated by third party experts, in close collaboration with the implementing partner. To ensure the collection and maintenance of reliable field data:

- Field-team members are fully aware of all procedures and of the importance of collecting data as accurately as possible; and
- Field teams install test plots if needed in the field and measure all pertinent components using the SOPs to estimate measurement errors;
- The document will list all names of the field team and the project leader will certify that the team is trained;
- New staff adequately trained.

Procedures to verify field data collection

To verify that plots have been installed and the measurements taken correctly, it is good practice to re-measure independently every 10 plots and to compare the measurements. The following quality targets should be achieved for the re-measurements, compared to the original measurements:

- Missed or extra trees → no error within the plot
- Tree species or groups → no error
- DBH of tree measurements →< ± 0.5 cm or 3 % whichever is greater
- Height measurements →< ± 10/ and -20%

Field data collected at this stage will be compared with the original data. Any errors found will be corrected and recorded. Any errors discovered will be expressed as a percentage of all plots that have been re-checked to provide an estimate of the measurement error.

Data maintenance and storage

Because of the relatively long-term nature of these project activities, data archiving (maintenance and storage) will be an important component of the work. Data archiving should take several forms and copies of all data should be provided to each project participant.

Copies (electronic) of all field data, data analyses, and models; estimates of the changes in carbon stocks and corresponding calculations and models used; any GIS products; and copies of the measuring and monitoring reports should all be stored in a dedicated and safe place, preferably offsite.

Given the time frame over which the project activity will take place and the pace of production of updated versions of software and new hardware for storing data, it is recommended that the electronic copies of the data and report be updated periodically or converted to a format that could be accessed by any future software application. Copies of all raw data, reports of analysis and supporting spreadsheets will be stored in a dedicated long-term electronic archive for at least 2 years following the end of the last crediting period.

Dissemination of Monitoring Plan and Results (CL4.2)

This MR will be available in Verra database, which is a public database where anyone can find and download the project documents (Project description, Monitoring Report).

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Under the applicability conditions of the applied methodology AR-AM0014 "Afforestation and reforestation of degraded mangrove habitats" (Version 03.0), it is expected that the baseline carbon stocks in litter and soil organic carbon pools will not show a permanent net increase.

The baseline net GHG removals by sinks are therefore calculated using Equation 1 of the methodology:

$$\Delta C_{BSL,t} = \Delta C_{TREE-BSL,t} + \Delta C_{SHRUB-BSL,t} + \Delta C_{DW-BSL,t}$$

Where:

$\Delta C_{BSL,t}$	= Baseline net GHG removals by sinks in year t; t CO_2-e
$\Delta C_{TREE\_BSL,t}$	= Change in carbon stock in baseline tree biomass within the project boundary in year t; t CO_2-e
$\Delta C_{SHRUB\_BSL,,t}$	= Change in carbon stock in baseline shrub biomass within the project boundary in year t; t CO_2-e
$\Delta C_{DW\_BSL,t}$	= Change in carbon stock in baseline dead wood biomass within the project boundary, in year t; t CO_2-e

As indicated in Section 3.1 of the PD, one baseline stratum was defined for the complete project area. The following land covers classes were:

- Blank areas: 97.59%
- Dwarf mangrove shrubs: 0.99%
- Pre-existing trees: 1.42%

Shrub baseline biomass

Shrub carbon stock changes in this stratum have been estimated as zero as explained in PD. Baseline carbon stocks in shrubs are calculated following the equation of the AR-TOOL14 "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities" (Version 4.1):

$$C_{SHRUB;T} = \frac{44}{12} * CF_s * (1 + R_s \sum_i A_{SHRUB}, * b_{SHRUB,i})$$

$$b_{SHRUB,t} = BDR_{SF} * b_{forest} * cc_{SHRUB}$$

Where:

$C_{SHRUB,t}$ = *Carbon stock in shrubs within the project boundary in the baseline; t CO2e*
CF_S = *Carbon fraction of shrub biomass, t C (t d.m.)-1*
R_s = *Root-shot ratio for shrubs; dimensionless*
$A_{SHRUB,i}$ = *Area of shrub biomass estimation stratum i, ha*
$b_{SHRUB,i}$ = *Shrub biomass per hectare in shrub biomass estimation stratum i, t d.m.ha-1*
BDR_{SF} = *Ratio of shrub biomass per hectare in land having a shrub crown cover of 1.0 and the default above-ground biomass content per hectare in forest in the region where the project is located.*
b_{FOREST} = *Default above-ground biomass content in forest in the region/country where the A/R CDM project activity is located, t d.m.ha-1*
CC_{SHRUB} = *Crown cover of shrubs in shrub biomass estimation stratum i at the time of estimation, expressed as a fraction*

Crown cover (CC_{SHRUB}):

Following the shrub classification (see PD), 1% of the project areas are covered with shrubs.

b_{FOREST}:
For the calculation of this second monitoring period, the default value for Indonesia presented IPCC table 3.A.1.4. is applied.

b_{FOREST}= 136.0 t d.m. ha^{-1}

For the following parameters, default values taken from the tool AR-TOOL14 are applied:

Parameter	Denotation	Value
Carbon fraction of shrub biomass	CF_s	0.47
Root-shot ratio for shrubs	R_s	0.4
Ratio of shrub biomass per hectare in land having a shrub crown cover of 1.0	BDR_{SF}	0.1

Pre-existing tree baseline biomass

The estimation of carbon stock in pre-project tree biomass was done following equation 20 and 21 of the methodological tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities (Version 04.1):

$$C_{TREE\_BSL} = \sum_{i=1} C_{TREE\_BSL,i},$$

$$C_{TREE\_BLSi} = \frac{44}{12} \times CF_{TREE} \times b_{FOREST} \times (1 + R_{TREE}) \times CC_{TREE\_BLSi} \times A_i$$

Where:

C_{TREE_BSL} = Carbon stock in pre-project tree biomass; t CO_2-e

$C_{TREE\_BSL,i}$ = Carbon stock in pre-project tree biomass in stratum i; t CO_2-e

CF_{TREE} = Carbon fraction of tree biomass (t C (t.d.m.)$^{-1}$

b_{forest} = Mean above-ground biomass in forest in the region or country where the A/R CDM project is located; t d.m.ha^{-1}

R_{TREE} = Root-shoot ratio for trees in the baseline; dimensionless

$CC_{TREE\_BSL,i}$ = Crown cover of trees in baseline stratum i, at the start of the A/R CDM project activity, expressed as a fraction; dimensionless

A_i = Area of baseline stratum i, delineated on the basis of tree crown cover at the start of the A/R CDM project activity; ha

Crown cover (CC_{tree})

According to the crown cover analysis, the crown cover of pre-existing trees in baseline represents 1.42% of the area in the project area.

b_{FOREST}:

The same value as for the shrubs baseline was applied.

b_{FOREST} = 136.0 t d.m. ha^{-1}

For the following parameters, default values taken from the tool AR-TOOL14 are applied:

Parameter	Denotation	Value
Carbon fraction of tree biomass	CF_T	0.47
Root-shot ratio for tree	R_T	0.25

Dead wood of pre-existing trees in the baseline

According to the methodological tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in AR CDM project activities (Version 03.1), when there are pre-existent trees in the baseline, dead wood carbon shall be accounted as baseline carbon.

The baseline carbon in dead wood is calculated using the conservative default-factor based method included in AR-TOOL12. The following equation is applied:

$C_{DW,i,t} = C_{TREE,i,t} * DF_{DW}$

Where:

$C_{DW,i,t}$ = Carbon stock in dead wood in stratum *i* at a given point of time in year t; t CO_2-e;

$C_{TREE,i,t}$ = Carbon stock in trees biomass in stratum *i* at a point of time in year t; t CO_2-e;

DF_{DW} = Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass; per cent.

As stated in the PD, the conservative default factor for dead wood biomass for this project according to the corresponding biome, elevation and annual precipitation is 6% of carbon stock of pre-existing tree biomass. Hence, this value is applied.

As result of all calculations, the pre-existing baseline carbon estimation for the first project activity instances is summarized in the table below:

Table 6 Calculation of pre-existing bassline carbon stocks in trees and shrubs. First project Activity instances

Year	Project year	Areas planted/ restored per year (ha)	$C_{TREE,t1}$ Pre-existing biomass of trees (tCO_2e)	C_{DW_BSL} Dead wood Pre-existing trees (tCO_2e)	C_{SHRUB_BSL} Pre-existing shrub biomass (tCO_2e)	C_{BSL} Total biomass baseline (t CO_2e)
2011	1	193.15	803.7	48.2	62.4	914.3
2012	2	860.09	3,578.7	214.7	278.1	4,071.5
2013	3	2,006.99	8,350.8	501.1	648.8	9,500.7
2014	4	1,639.90	6,823.4	409.4	530.2	7,763.0
2015	5	299.86	1,247.7	74.9	96.9	1,419.5
Total		4,999.98	20,804.4	1,248.3	1,616.4	23,669.1

Since the second project activity instances has been added in this verification period, the pre-existing baseline carbon estimation for the second project activity instance has been calculated. Here below it is summarized the pre-existing baseline carbon stock for the first and second activity instances.

Table 7 Calculation of pre-existing bassline carbon stocks in trees and shrubs. First and second project activity instances

Year	Project year	Areas planted/ restored per year (ha)	$C_{TREE,t1}$ Pre-existing biomass of trees (tCO_2e)	C_{DW_BSL} Dead wood Pre-existing trees (tCO_2e)	C_{SHRUB_BSL} Pre-existing shrub biomass (tCO_2e)	C_{BSL} Total biomass baseline (t CO_2e)
2011	1	200.92	836.0	50.16	65.0	951
2012	2	912.24	3795.7	227.74	294.9	4318
2013	3	2019.77	8404.0	504.24	653.0	9561
2014	4	1810.44	7533.0	451.98	585.3	8570
2015	5	301.21	1253.3	75.20	97.4	1426
2016	6	26.45	110.1	6.60	8.6	125
2017	7	8.24	34.3	2.06	2.7	39
Total		5279.26	21966.43	1317.99	1706.73	24991.15

3.2.2 Project Emissions

The actual net GHG removals by sinks have been calculated using equation 2 of the methodology: AR-AM0014: Afforestation and reforestation of degraded mangrove habitats (Version 03.0), as described below.

$$\Delta C_{ACTUAL,t} = \Delta C_{P,t} – GHG_{E,t}$$

Where:

$\Delta C_{ACTUAL,t}$ = Annual actual net greenhouse gas removals by sinks at time *t*; t CO_2-e yr^{-1}

$\Delta C_{P,t}$	= Change in carbon stocks in project, occurring in the selected carbon pools, at time *t*; t CO_2-e yr^{-1}
$GHG_{E,t}$	= Increase of non-CO_2GHG emissions within the project boundary as a result of the implementation of the A/R CDM project activity, in year t, t CO_2-e

Change in the carbon stocks in project have been calculated using equation 3 of the methodology:

$$\Delta C_{P,t} = \Delta C_{TREE\_PROJ,t} + \Delta C_{SHRUB\_PROJ,t} + \Delta C_{SOC\_PROJ,t}$$

Where:

$\Delta C_{P,t}$	= Change in carbon stocks in project, occurring in the selected carbon pools, at time *t*; t CO_2-e yr^{-1}
$\Delta C_{TREE\_PROJ,t}$	= Change in carbon stock in tree biomass in project in year t, as estimated in AR-TOOL14; t CO_2-e yr^{-1}
$\Delta C_{SHRUB\_PROJ,t}$	= Change in carbon stock in shrub biomass in project in year t, as estimated in AR-TOOL14; t CO_2-e yr^{-1}
$\Delta C_{SOC\_PROJ,t}$	= Change in carbon stock in the soil organic carbon (SOC) pool within the project boundary, as estimated in AR-AM0014, in year t; t CO_2-e yr^{-1}

Estimation of carbon stock changes in trees and shrubs

Carbon stock changes of trees and shrubs are estimated applying the AR-Tool 'A/R Methodological tool: Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (Version 04.2)18.

Estimation of carbon stock change in trees

The following equations of the methodology have been applied:

$$C_{TREE,t} = \frac{44}{12} * CF_{TREE} * B_{TREE,t}$$

$$B_{TREE,\,t} = b_{TREE,\,t} * A$$

$$b_{TREE,t} = \sum_{i=1}^{M} w_i \text{ x } b_{TREE,t,i}$$

Where:

$C_{TREE,\,t}$	= Carbon stock in tree biomass within the project boundary at a point in time in year *t*; t CO2-e.
CF_{TREE}	= Carbon fraction of tree biomass; t C (t d.m.)$^{-1}$.
$B_{TREE,\,t}$	= Total tree biomass within the project boundary at a given point in time in year *t*; t d.m.
$b_{TREE,\,t}$	= Mean tree biomass per hectare within the project boundary at a given point in time in year *t*; t d.m. ha^{-1}

[18]https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-14-v4.2.pdf

A	= Project area; ha
w_i	= Ratio of the area of stratum *i* to the sum of areas of tree biomass estimation strata (A_i/A), dimensionless
$b_{TREE,t,i}$	= Mean tree biomass per hectare in stratum *i* at a given point in time in year *t*; t d.m. ha^{-1}

Determination of tree biomass excluding saplings

A significant number of mangrove trees in this project during the second monitoring period are in a transitional condition between saplings and trees. During the carbon monitoring the field teams had to define for each mangrove tree if it is classified as sapling or tree. For this first verification, in order to remain a conservative carbon analysis, saplings will not be accounted in the carbon monitoring, but due this reason, the number of trees per ha it is expected to increase in following verifications.

The following allometric equations have been selected and applied for trees respectively for this monitoring period:

Table 8. Allometric equations selected

Applied to	***Equation***	R^2	***N (sample trees used to derive the equation)***	***Source***
Rhizophora spp.	$AGB = 0.235 * D^{2.42}$	0.98	57	Ong et al. (2004)[19]
Other mangrove trees	$AGB = 0.251* \rho*(D)^{2.46}$	0.98	104	Komiyama et al.(2005)[20]

Where:

AGB	= Above ground biomass (kg)
P	= Wood density, g/cm^3
D	=Diameter at Breast Height or Diameter at 30cm from the ground, cm

Wood density,ρ

Specific values for several species of planted genus were applied in order to get a higher accuracy in the calculations. Zanne et al (2009) 21presented a table with different examples of densities of mangroves, with a classification based on four priorities:

1. *Specie from South-East Asia (tropical)*
2. *Specie from South-East Asia*

[19]http://www.sciencedirect.com/science/article/pii/S0378112703003906 (also provided as supporting documentation)
[20]Komiyama, A.; Poungparn,S. & Kato, S. (2005) Common Allometric equations for estimation the tree weight of mangroves. Journal of tropical Ecology 21: 471-477. Cambridge University Press (available as supporting documentation), also in http://www.cifor.org/publications/pdf_files/WPapers/WP86CIFOR.pdf
[21]Zanne et al (2009) Global wood density database. http://datadryad.org/resource/doi:10.5061/dryad.234/1

3. *Genus from South-East Asia (tropical)*
4. *Genus from South-East Asia*

Taking this list as basis, wood density of the species of the project has been calculated. When the species matched, the corresponding value was directly taken. In the cases of matching genus, wood density would be calculated as an average value of the individuals of the genus. A site average of all the values was taken if species could not be matched.

Table 9 Wood density of project relevant mangrove species (Zanne et al (2011))

Scientific name	*Wood density (gcm^{-3})*
Avicennia spp.	0.61
Rhizophora Apiculata	0.85
Rhizophora Mucronata	0.82
Rhizophora Stylosa	0.84
Other	0.68

Root-to-shoot ratio, Rj:

The default equation from Appendix 1 of the Tool AR-TOOL14 is applied:

$$Rj: \frac{e^{(-1.085+0.9256\times lnb)}}{b}$$

Where,

- b is the AGB tree biomass in t d.m. ha^{-1}
- CF_{TREE}:

A default value of 0.47for CFTREE is applied – taken from AR CDM tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities”.

Carbon stock change in shrubs

The project area will be reforested with mangrove trees only. Due to the small size of overall vegetation in the project area, shrub vegetation is conservatively not estimated in this first verification. It may be included in subsequent verifications if it becomes significant.

Carbon stock change in dead wood

The change in carbon stock in dead wood is calculated using the conservative default-factor based method included in AR-TOOL12. The following equation is applied:

$$C_{DW,i,t} = C_{TREE,i,t} * DF_{DW}$$

Where:

$C_{DW,i,t}$	= Carbon stock in dead wood in stratum *i* at a given point of time in year t; t CO_2-e;
$C_{TREE,i,t}$	= Carbon stock in trees biomass in stratum *i* at a point of time in year t; t CO_2-e;
DF_{DW}	= Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass; per cent.

As stated in the PD, the conservative default factor for dead wood biomass for this project as influenced by the biome, elevation and annual precipitation is 6% of carbon stock in tree biomass. Hence, this value is applied.

The actual net GHG removals by sink have been calculated based on the forest inventory data. The change in the carbon stocks in project were calculated using equation 3 of the methodology as described in the preceding sections. The calculation is available in the Excel file "YAGASU ex-post carbon calculations_2nd verification period".

Table 10. Carbon stock in tree biomass per stratum (Ctree)

Stratum	Area (ha)	w_i	$b_{TREE.t2}$ t d.m. ha^{-1}	$B_{TREE.t2}$ t d.m.	$C_{TREE.t2}$ t CO_2-e
1	764.83	0.14	10.8	56.820.2	97.920.1
2	2,534.45	0.48			
3	1,975.70	0.37			
TOTAL	**5274.98**	1			

Calculation of uncertainty

This was done following the guidance of Appendix 2 of the A/R Methodological Tool. The values are presented in the table below:

Table 11. Uncertainty per stratum

Stratum	si t d.m. ha-1	ni	uc %	uc mean t CO2-e	Uc Discount t CO2-e
1	35.77	77	19.6	19188.3	**9594.2**
2	14.75	96			
3	19.34	69			
Total		242			

Total ΔCTREE_PROJ,t for this monitoring period **88,325.98 tCO2-e.**

Estimation of changes in soil organic carbon stocks

Changes in carbon stocks in the SOC pool is calculated as indicate in the Methodology AR-AM0014:

$$\Delta SOC_{PROJ,t} = \frac{44}{12} * \sum_{t=1}^{t} A_{PLANT,t} * dSOC_t * 1year$$

Where:

$\Delta SOC_{PROJ,t}$ = Change in SOC stock within the project boundary, in year t; t CO_2-e

$A_{PLANT,t}$ = Area planted in year t, ha

$dSOC_t$ = The rate of change in SOC stocks within the project boundary, in year t, $tCha^{-1}yr^{-1}$.

Calculation of dSOC has been done based on the publication Murdiyarso et al (2015)[22] , where the potential of Indonesian mangrove forest has been analyzed, providing evidence that mangroves in Indonesia are among the ecosystems with the highest carbon accumulation in the world and that soil carbon accounts for close to 80% of all carbon pools in these systems. The analysis of this study estimates soil carbon in mangroves until 2 m of soil depth. In order to be conservative, calculations for this project only considers 50 cm. In addition, in order to account for allochthones soil carbon buried at the project sites, the default approach of the recently accepted VCS Methodology VM0033 Methodology for Tidal Wetland and Seagrass Restoration is applied[23]. With this approach a deduction factor for allochthones soil carbon for each of the soil layers up to 50 cm measured in the Murdiyarso et al. study was derived.

Based on this study, dSOC was estimated as 3.32 t C/ha/year.

As indicated in the IPCC supplement for wetlands, where activity results in patchy or patches of biomass, the emission factor recommended as default value can only be applied when the mangrove cover is at least 10% of the overall area. All areas in the project with less than this mangrove cover were excluded of the project, so all the area under plantation for the second monitoring period can be included in the SOC calculations with this default value. This methodology was applied to the areas of this project.

Table 12 Derivation of eligible project areas for SOC accounting

Year of inclusion of project area	Area under plantation (ha)
2011 (6 months)	7.77
2012	52.15
2013	12.78
2014	170.54
2015	1.35
2016	26.45
2017	5274.98
2018	5274.98
2019 (7 months)	5274.98
Total	**5.274.98**

[22] Murdiyaso et al. (2015). The potential of Indonesian mangrove forest. Nature Climate Change. Available as supporting documentation.

[23] http://www.v-c-s.org/sites/v-c-s.org/files/VM0033%20Tidal%20Wetland%20and%20Seagrass%20Restoration%20v1.0%2020%20NOV%202015.pdf. In particular, section 8.1.4.3 of this new methodology and equations 32, 33, 34, 37, 77.

For this monitoring period, a 3.83-year period is used to calculate the actual soil organic carbon stock changes of the project. In this verification period are included for first time, the second project activity instances as are shown in table below.

Table 13 Calculation of SOC change for this monitoring period

Year	Project year	Annual carbon stock change (tCO2e)	Cumulative carbon stock change (tCO2e)
2011 (6 months)	1	47.3	47.3
2012	2	411.7	458.9
2013	3	806.5	1,265.4
2014	4	1,921.4	3,186.8
2015	5	18,043.5	21,230.3
2016	6	63,950.9	85,181.2
2017	7	64,161.8	149,343.0
2018	8	64,212.0	213,555.0
2019 (7 months)	9	37,382.4	250,937.4
Total		**250,937.44**	

Change in carbon stocks of trees

According to paragraph 16 of the tool AR-TOOL14A/R Methodological tool 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (Version 04.1) at the first verification CTREE, t1is set equal to the carbon stock in the pre-project tree biomass, i.e. CTREE,t1= CTREE_BSL,t1. Based on this, the following table shows the change in carbon stock during the period between three points of time t1 (pre-existing tree and shrub biomass), t2 and t3. The Excel sheet Yagasu ex-post carbon calculation_2019 details the full calculation of emission reductions for this monitoring period and is available upon request.

Table 14 Change in carbon stocks during the period between three points of time t1 (pre-existing tree and shrub biomass), t2 y t3

Year of inclusion of project areas	Project area included in year t (ha)	$C_{TREE,t1} = C_{TREE\_BSL,t}$ (tCO2e)	$C_{DW\_BSL;\ t1}$ (tCO2e)	$C_{SHRUB\_BSL,t1}$ (tCO2e)	$C_{TREE\_PROJ\_DISCOUNT;\ t2}$ (tCO2e)	$C_{DW\_PROJ;t2}$ (tCO2e)	$C_{TREE\_PROJ\_DISCOUNT;\ t3}$ (tCO2e)	$C_{DW\_PROJ;t3}$ (tCO2e)	**Change in carbon stock C_{PROJ} between two points of time t3 and t2 (tCO2e)**
2011	200.92	836.0	50.2	65.0	1153.4	69.2	3361.5	201.7	1389.4
2012	912.24	3795.7	227.7	294.9	5137.4	308.2	15262.5	915.7	6414.1
2013	2019.77	8404.0	504.2	653.0	11970.7	718.2	33792.2	2027.5	13569.6
2014	1810.44	7533.0	452.0	585.3	7994.6	479.7	30290.0	1817.4	15062.8
2015	301.21	1253.3	75.2	97.4	8.0	0.5	5039.5	302.4	3907.5
2016	26.45	110.1	6.6	8.6			442.5	26.6	343.9
2017	8.24	34.3	2.1	2.7			137.9	8.3	107.1
Total	**5279.26**	**21966.4**	**1318.0**	**1706.7**	**26264.1**	**1575.8**	**88326.0**	**5299.6**	**40794.5**

Next, the actual net greenhouse gas removals by sinks for the different years of the monitoring period are summarized below, separately for trees (including dead wood) and soil organic carbon as well as totals.

In addition the cumulative months of project implementation are shown, starting with 6 months in 2011 and ending with 7 months in 2019 until the end of this monitoring period 30-07-2019

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Table 15. Actual net greenhouse gas removals by sinks for the different years of the monitoring period

Year	Plantations areas of First Project activity instance (ha)	Plantations areas of second Project activity instance (ha)	Period included in this verification- 1st instance (months)	Period included in this second verification- 2nd instance (months)
2011	193.15	7.77	0	6
2012	860.09	52.15	0	6
2013	2,006.99	12.78	0	6
2014	1,639.90	170.54	0	6
2015	299.86	1.35	0	6
2016		26.45	12	6
2017		8.24	12	6
2018			12	12
2019			6.99	6.99
	4999.98	**279.28**		

*Table 16 A*ctual net greenhouse gas removals by sinks for the different years of the monitoring period

	Actual tree net GHG removals by sinks, in year t; (tCO_2e)	**Actual DW net GHG removals by sinks, in year t; (tCO2e)**	**Actual SOC net GHG removals by sinks, in year t; (tCO2e)**	**Actual total net GHG removals by sinks, in year t; (tCO2e)**
	$\Delta_{PLA,t}$	$\Delta_{DW,t}$	$\Delta_{SOC\_PROJ,t}$	$\Delta_{PROJ,t}$
Project year	**t CO_2-e**	**t CO_2-e**	**t CO_2-e**	**t CO_2-e**
2015	9,506.59	606.80	21,230.31	31,343.70
2016	38,346.79	2,447.67	63,950.88	104,745.34
2017	38,346.79	2,447.67	64,161.85	104,956.30
2018	38,346.79	2,447.67	64,211.96	105,006.42
2019	22,324.45	1,424.97	37,382.45	61,131.87
TOTAL	**38,346.79**	**2,447.67**	**250,937.44**	**407,183.63**

Other project emissions

GHG project emission from burning of biomass does not occur in any of the mangrove systems in the project area.

3.2.3 Leakage

As demonstrated in section 3.3 of the Project Description, leakage due displacement of agricultural activities in the areas of this project can be accounted as zero.

Table 17 GHG emissions due to leakage, in year t

Project year	LK_t t CO_2-e
2015	0
2016	0
2017	0
2018	0
2019	0
Total	0

3.2.4 Net GHG Emission Reductions and Removals

The net anthropogenic GHG emission reductions and removals are calculated using equation 6 of the methodology AR-AM0014:

$$\Delta C_{AR-CDM,t} = \Delta C_{\mathrm{ACTUAL,t}} - \Delta C_{\mathrm{BSL,t}} - \mathrm{LK_t}$$

Where:

$\Delta C_{\mathrm{AR-CDM,t}}$ = Net anthropogenic GHG removals by sinks, in year t; t CO_2-e
$\Delta C_{\mathrm{ACTUAL,t}}$ = Actual net GHG removals by sinks, in year t; t CO_2-e
$\Delta C_{\mathrm{BSL,t}}$ = Baseline net GHG removals by sinks, in year t; t CO_2-e
$\mathrm{LK_t}$ = GHG emissions due to leakage, in year t; t CO_2-e

The GHG emission reductions and removals for the first project activity instance are shown below.

Table 18 Total Yagasu project GHG emission reductions for this monitoring period

Year	$\Delta C_{\mathrm{BSL,t}}$ (t CO_2-e)	$\Delta C_{\mathrm{ACTUAL,t}}$ (t CO_2-e)	$\mathrm{LK_t}$ (t CO_2-e)	$\Delta C_{\mathrm{AR-CDM,t}}$ (t CO_2-e)
2015	0	31,343.70	0	31,343.70
2016	0	104,745.34	0	104,745.34
2017	0	104,956.30	0	104,956.30
2018	0	105,006.42	0	105,006.42
2019	0	61,131.87	0	61,131.87
Total	**0**	**407,183.63**	**0**	**407,183.63**

Therefore, the total GHG emission reductions of the first and second project activity instances, for the second monitoring period from 01-10-2015 to 31-07-2019 are 407,183 t CO2-e. The non-permanence risk rating is 13% –as assessed and documented in the AFOLU non-permanence risk report (provided as a separate document). Therefore, the total number of buffer credits that need to be deposited into the AFOLU pooled buffer account is 52,934 t CO2-e (13*%). The number of buffer credits per vintage year of this monitoring period is:*

Table 19 No of buffer credits per vintage year

Year	Buffer credits in year t; (tCO2e)
2015	4075
2016	13617
2017	13644
2018	13651
2019	7947
Total	52,933.87

The number of GHG credits eligible to be issued as VCUs for the first project instances of this monitoring period is 354,250 t CO2-e.

3.3 Optional Criterion: Climate Change Adaptation Benefits

3.3.1 Activities and/or processes implemented for Adaptation (GL1.3)

Based on the evidence showed in CCB Project description Section 3.1.1., the project zone is likely to suffer various and different effects of climate change over the next 20 years, deriving in impacts including:

- Decrease in the quality of the coastal environment and the small islands due to coastal erosion, the intrusion of seawater and pollution.
- Reduction on fishing productivity due to damage to mangrove and coral reef ecosystems and changes in fishing areas, outcrops, fish migration, etc.
- Damage to aquaculture land due to flooding caused by seawater and floods caused by sea level rise
- Damage to homes and possible loss of life due to extreme events like tropical storms and high waves.
- Forest degradation, damage to mangroves and aquatic biodiversity loss.
- Loss of rural productivity and infrastructure from forest/non-timber forest products
- Livestock deaths
- Reduce food security, quantity and quality of potable drinking water.
- Increase number of human deaths and injuries
- Increase risk of cardiovascular and respiratory diseases.

Preventing mangrove loss would be an effective climate change adaptation and mitigation strategy (Murduyarso *et al*, 2015).

The measures needed and taken to assist communities to adapt to the probable impacts of climate change are increases in income levels, education and technical skills, improvements in public food distribution, disaster preparedness and management, and health care system through sustainable and equitable development, to reduce the vulnerability to climate change.

Measures taken include:

- Protection of 25,000 ha of coastal mangrove ecosystem through protection actions carried out by Community Mangrove Patrolling Unit: 154 patrols covering 25,000 ha of mangrove and 17,388.77 ha of villages.
- Restoration of 5,278 ha of mangrove, 984 people implementing the knowledge for long-term program activities (mangrove nursery, planting mangroves, replanting trees and mangrove patrolling)
- Silvo-fishery management capacity building that ensures the permanence of a sustainable ecosystem: 249 trainings for local stakeholders, attended by 13,197 people (26% villagers), 70 profitable and sustainable aquaculture businesses managed by local communities
- Diversification of economic businesses by introducing sustainable options: 231 trainings on Income Generation Activities (aquaculture, organic batik, eco-tourism, food processing and other businesses) attended by 3,922 people and 634 profitable and sustainable businesses ran by local communities (70 aquaculture businesses, 34 organic batik businesses, 11 ecotourism businesses, 202 food processing businesses and other 317 businesses).
- Capacity building to local people regarding the importance of maintaining and protecting the mangroves: 229 awareness actions about climate change issues, mangrove conservation and benefits of mangrove for human life attended by 14,104 people (25% villagers, 17% teachers, 45% students, 13% other participants)

4 COMMUNITY

4.1 Net Positive Community Impacts

4.1.1 Community Impacts (CM2.1)

Project gives some supports related to mangrove protection such as:

- Socialization about positive impact of planting mangrove to fisheries product and environment
- Distribution of mangrove seedlings
- Giving some financial support for planting cost
- Giving some financial support for managing mangrove after planting

Women has an important role in fisheries activity. Based on this characteristic, during mangrove planting, the project asked women to participate in establishing mangrove seedling, planting and managing trees, as well as students in elementary school, senior and junior high school, which are also involved in the planting activity.

Capacity building have strengthened the knowledge through training for local people in the villages such as:

How to plant and manage mangrove tree
How to develop and rises income through mangrove product like cake, syrup from mangrove fruit etc.
Improving technique on fish pond production.
Empowering local community through establishing cooperation related to marketing product.

Tables below shows the changes in well-being.

Community Group	Fishermen
Impact	- Increase in perception, recognition of the value of mangroves, - Improvement in skills and knowledge in mangroves management, - Improvement if the relations of trust and mutual support, - Improvement in access to markets, ecotourism facilities developed, employment, - Increase in cash income, - Increase/stabilization of quality for local people livelihoods. - Stabilization of water flooding
Type of Benefit/Cost/Risk	Actual and direct benefit
Change in Well-being	Fishermen are developing and managing a sustainable and stable fishery business associated to mangroves fishponds, increasing their incomes and thereby, improving the life quality.

Community Group	Women
Impact	- Increase in perception, recognition of the value of mangroves, - Improvement in skills and knowledge in mangroves management, - Improvement if the relations of trust and mutual support, - Improvement in access to markets, ecotourism facilities developed, employment, - Increase in cash income, - Increase/stabilization of quality for local people livelihoods. - Increase gender equity Stabilization of water flooding
Type of Benefit/Cost/Risk	Actual and direct benefit
Change in Well-being	Women participate in planting activities and they are managing and developing a sustainable and stable batik and food business associated to mangroves, increasing the incomes of their families, having an independent mean of livelihood, and thereby, improving the life quality.

Community Group	Youth
Impact	- Increase in perception, recognition of the value of mangroves, - Improvement in skills and knowledge in mangroves management, - Improvement if the relations of trust and mutual support, - Improvement in access to markets, ecotourism facilities developed, employment, - Increase in cash income, - Increase/stabilization of quality for local people livelihoods. - Stabilization of water flooding
Type of Benefit/Cost/Risk	Actual and direct benefit
Change in Well-being	Youth are involved in all project activities, they have increased their knowledge in mangrove ecosystem, its function as well as they are able to develop different sustainable businesses associated to this ecosystem in the future.

4.1.2 Negative Community Impact Mitigation (CM2.2)

No critical ecosystem services are been negatively affected by the project activities to people who live in the project zone, no negative well-being impact have been identified for them.

4.1.3 Net Positive Community Well-Being (CM2.3, GL1.4)

The net positive well-being impacts of the project are showed in the table below, through the comparison between without-project and with project scenario, using the five key livelihood assets:

- Human capital: education, health, physical capability, knowledge and skills possession
- Social capital: Community cohesiveness, responsibility, affiliation and socio-political relations
- Financial capital: Access to financing support and financial assets including cash, loans, savings and cattle
- Physical capital: Access to infrastructure (roads, transport, electricity), production equipment, shelter and technology (communication system)
- Natural capital: Natural resource stocks (soil, water, air, genetic resources, etc.) and environmental service

Table 20. Summary of net positive community benefits

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
AREA BASED				
HCV4 Areas with an environmental function	The mangrove restoration and coastal green-belt protection plays a critical role in maintaining hydrological function, preventing erosion risk through maintaining forest cover	Under the BS mangrove cover would be lost, leading to increased erosion and increased risk of salt water intrusion	Under the PS the mangrove forest is protected and reforestation have been conducted reducing the risk of erosion and sedimentation	The main objective of the project is the mangrove restoration and coastal green-belt protection. Details of the project activities in Section 2.1.8
HCV5 Areas fundamental to meeting basic needs of local people	The fish ponds if the project area are traditionally used by project-zone communities as a silvo-fishery system for the provision of numerous products	Under the BS intensive aquaculture and bad practices in the management of mangroves will take place, leading to mangrove cover loss	Under the PS the mangrove forest has been restored in the fish ponds as a sustainable silvo-fishery system	Restoration of the fish ponds as silvo-fishery systems and capacity building to communities regarding mangrove management and good practices
WELL-BEING BASED				
Human capital	Traditional knowledge of local people is high, but knowledge is lacking in how to utilize this within a contemporary market-driven society and how important are the social and environmental services provided by mangrove ecosystem.	Under the BS traditional knowledge will be lost gradually, since deforestation and no protection of mangroves will continue. Sea fishery production depends on mangrove forest, that provide breeding sites for various fish species and other species.	Under the PS, local people have been assisted to be aware of the relevance of mangrove ecosystems, as well as to develop sustainability economies based on traditional knowledge.	Project activities are designed to understand the relevance of mangrove ecosystems. Small and Medium sized business development is a main objective of this approach, incorporating training and capacity building and involving local people in other project activities.

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
Social capital	Village communities are units that function through well-established institutions backed by Indonesian law.	Under the BS the social cohesion will be at risk because the unequally distribution of benefits and costs.	Under the PS local people is commit with the project and support it.	Project activities aim to get mutual support, such as village and district meeting to establish agreements
Financial capital	With the mangrove forest cleared, the villagers experienced a drastic decline in their fishery products, since these products depends directly on the mangrove conditions.	Under the BS the production of fishery farms will decrease, since the existing mangroves are degraded and it will be lost by illegal logging and other threats.	Under the PS local people have obtained sustainable economic development and land use. They have seen their incomes increased due to the develop of sustainable businesses associated to mangrove forests.	Small and Medium sized sustainable business development is a main objective of this approach, incorporating training and capacity building and involving local people in other project activities
Physical capital	Infrastructure in the project zone is poor, in order to enhance ecotourism in the project zone, road access and other facilities should be developed.	Under the BS, is unlikely these facilities will be developed, since these traditional businesses will decrease.	Under the PS mangrove project is working with communities and governments to ensure the sustainable development of infrastructure	Project ensure the development of ecotourism and village facilities, supported by different donors.
Natural capital	Natural capital within the project area is extremely high. Mangroves support critical ecosystem services as well as contain natural resources utilized by the project zone communities	Under the BS, the natural capital would be exploited for short-term, leading to short-term benefits whom effects would be short lived	Under the PS, the natural capital of the project is being restored and protected, and local people have been assisted to develop ways that exploit these resources in a sustainable way	The main objective of the project is the mangrove restoration and coastal green-belt protection. Details of the project activities in CCB PD Section 2.1.8
EXCEPTIONAL BENEFITS (Gold Standard)				
Improved land rights	Some people from outside the village take mangrove plants (illegal cutting), and catch fish, shrimp using fine nets.	Under the BS conflicts between people inside the villages and between villagers and people from outside wouldn't	Under the PS, local people have management rights to land in the project area. An agreement between land owners and Yagasu	Clear mechanism and regulation on utilization of resources and land-use, help to avoid conflict among villagers (perdes)

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
		be solved, at least in the short-term.	has being developed, as well as specific regulations regarding mangrove use, that guarantee the right on the land.	
Positive well-being	See WELL-BEING BASED of this table	See WELL-BEING BASED of this table	See WELL-BEING BASED of this table	See WELL-BEING BASED of this table
Risk reduction	Local people lack of social, financial and physical resilience so are at risk from economic or environmental shock and external forces beyond their control	Under the BS local people will continue lacking these aspects.	Under the PS, local people's resilience had increased while risk has been reduced.	Small and Medium sized sustainable business development is a main objective of this approach, incorporating training and capacity building and involving local people in other project activities
Women's well-being	Women typically have specific roles in households and society due to patriarchal culture	Under the BS there will be lack of participation in decision-making, leadership of business and economic independence.	Under the PS, women have developed Small and Medium sized enterprises that allow them integrate in society as active members	Small and Medium sized sustainable business development is a main objective of this approach, incorporating training and capacity building and involving local people in other project activities
Active involvement	Communities have been involved in the project design since the inception of it. According to Yagasu CCB study in 2009, around 80% of local villagers thought that local community should participate in the management and conservation of the mangrove forest.	Under the BS, there will not be any kind of program that involve local communities in restoration and protection of the mangrove ecosystem	Under the PS, the communities are part of the implementation of the project, taking part in the conservation and restoration of the mangrove ecosystem	Project activities aim to get mutual support, and establish agreements to involve local people in the project activities
Capacity building	Opportunities in training and capacity building regarding	Under the BS is unlikely to be a rise in training and	Under the PS, a program of training and capacity	Small and Medium sized sustainable business

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
	project topics rarely occurs in the local communities.	capacity building opportunities.	building has been implemented to give local people knowledge and resources to develop economic and environmentally sustainable business, at the same time that they restore and protect the mangrove ecosystem	development is a main objective of this approach, incorporating training and capacity building and involving local people in other project activities

4.1.4 Protection of High Conservation Values (CM2.4)

As mentioned before, in Section 4.1.3, none of the HCVs related to community well-being has been negatively affected by the project.

The project proponents have internalized environmental and social sustainability criteria into their projects, including but not limited to respect for native species, preservation and protection of natural ecosystems and implementation of productive systems under good practices.

4.2 Other Stakeholder Impacts

4.2.1 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

No negative impacts have been identified on other stakeholders, no measures or activities have been developed.

4.2.2 Net Impacts on Other Stakeholders (CM3.3)

The impacts on other stakeholders are shown in the table below.

Table 21 Other stakeholders' impacts

Stakeholder	Expected impacts	Expected benefits
Project developers	Transfer of knowledge and skills with the generation of knowledge to the economic, social and environmental development of the region	Exchange of experience, technological knowledge and skills between the actors involved. The benefits are indirect, positive and expected medium term
Government authorities	Sustainable co-benefits projects in the region	Increased sustainable forestry projects in the region according to land use planning Conservation of protected areas, restoration of natural areas, development of effective environmental management

Stakeholder	Expected impacts	Expected benefits
	Effective protection of natural resources in compliance with existing environmental regulations	plans, sustainable development of the region. The benefits are indirect, positive and are expected medium term

4.3 Community Impact Monitoring

4.3.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The community monitoring plan is included in the CCB Project Description.

Yagasu developed a study in 2009 about climate, community and biodiversity in four villages in Northern Sumatra. This study shows that mostly people in coastal villages were living as fishermen. They catch for fishes, breeding ponds fishes and crabs traditionally which meant that nutrient for fisheries are gathered from ocean water that came with tidal. To maintain these kinds of traditional fisheries, having a good mangrove habitat is essential that significantly supports the local livelihood. The constant mangrove conservation would set up leverage pro poor economic sustainably, improve the existence community-based institutions, and prospect of reduce conflict.

According to Yagasu study, average income in Tanjung Rejo and Percut village is about Rp 1,850,000 (117 €), per month. Fisheries are the most activities in generate income in this area. Otherwise, community in this area also plant rice field even only for they own consumption. Oilpalm start to grow in this area. Income in Belawan Sicanang village is about Rp 1,420,000 (90 €) per month. Many degraded mangrove occurred here. Most of the community here built fish pond through cutting mangrove. When fish pond productivity decreasing, they left it and looking for other place. Replanting mangrove trees have not seen here.

Yagasu carried out other research about community in 2014 (in Alur Cempedak, Beras Basah, Lubuk Kertang, Sei Meran), where income, outcome and respondent perception about mangrove were analysed. In these places, income from fishery sector is very dominant, representing around 80% (excepting Lubuk Kertang). Depending on the villages, the perception on mangrove is different, many people said that mangrove is very significant for community, but there were people who said never getting information about mangroves.

Human capital

There has been an increase in perception and recognition on mangrove restoration and protection, as well as an improvement in knowledge and skills in mangrove management, carried out through capacity building since the start of the project.

A total of 800 meetings for project socialization and establishment of agreements with village government and community groups have been carried out, where 38% of participants were women and 60% were young people. Since the start of the project, 206 meetings have taken place for training local people in preparing nursery and planting mangroves, where 60% were young people and 40% were women (see section 7.2).

Figure 7. Interviewing the respondents in Kelurrahan Sicanang

Yagasu, from 2011 to 2019, have organized 360 district meetings and 29 workshops for developing protection policy and supporting mangrove restoration with communities (48% of participants), decision makers (20%), legislatives (6%), private sectors (10%), universities (10%) and NGOs (6%).

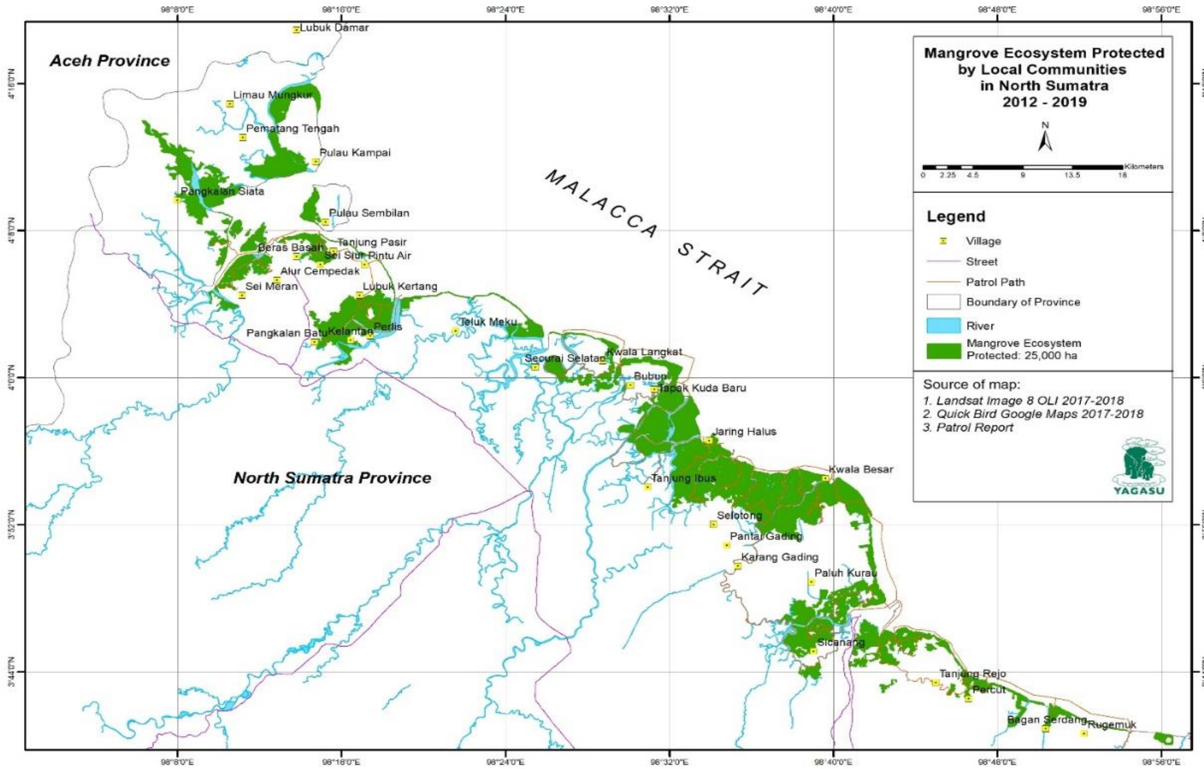
Awareness actions have been implemented by Yagasu to villagers (25% of participants), teachers (17%), students (45%) and other participants (13%), about climate change issues, mangrove conservation and benefits of mangroves for their life, where 60% of participants were young people.

Capacity building sessions aiming the training on preparing nursery & planting mangroves for local stakeholders were carried out, participating a total of 2,678 people, 40% women. 984 people are implementing the knowledge regarding: mangrove nursery, planting & replanting, mangrove patrolling, income generating activities and other long-term program activities.

There has been also an increase in the protection of the mangrove for interferences from outside parties. Protection actions on mangrove ecosystem through Community Mangrove Patrolling Unit (154 patrols), were carried out in 25,000 ha of protected area, covering 17,388.77 ha of villages and involving 1,224 people.

The patrols have reported illegal findings in these areas, where majority of them are illegal logging (77%), and poaching of fauna (10%) and land encroachment (10%) to a lesser extent.

Figure 8. Map of MPU operations (Source: Yagasu)



Social capital

Project has been able of improve technical and management knowledge on how to run village business through mangrove products and services. A total of 231 training sessions on Income Generation Activities have been carried out, where 60% were young people and 40% were women.

The table below shows the profitable and sustainable existing local communities' businesses (aquaculture, organic batik, mangrove eco-tourism, mangrove food processing and others) and people involved running the businesses.

Table 22. Profitable and sustainable of social businesses ran by local communities

District	Number of social business on					People involved		
	Aquaculture	Organic batik	Eco-tourism	Food processing	Others	Men	Women	Total
Banda Aceh	8	3	1	15	27	275	184	459
Aceh Besar	7	3	-	8	18	184	122	306
Pidie	5	3	-	9	17	173	116	289
Pidie Jaya	6	3	-	9	18	184	122	306
Bireun	5	2	-	11	18	184	122	306
Aceh Utara	6	2	-	12	20	204	136	340
Aceh Timur	6	3	1	17	27	324	216	540
Langsa	6	3	2	18	29	348	232	580
Aceh Tamiang	5	3	1	15	24	245	163	408

District	Number of social business on					People involved		
	Aquaculture	Organic batik	Eco-tourism	Food processing	Others	Men	Women	Total
Langkat	8	5	3	26	42	630	420	1.050
Deli Serdang	5	3	2	37	47	536	357	893
Medan	3	1	1	25	30	360	240	600
Total	**70**	**34**	**11**	**202**	**317**	**3,646**	**2,431**	**6,077**

These businesses employ 715 local people directly and indirectly, where 44% are women and 60% are young people. Women are employed in organic batik business and food business, as well as in eco-tourism business. Aquaculture business only employs men. The total number of people employed in project activities (expressed as number of full-time employees) is 144, being 86 women.

An effective community governance communication through village and district meetings and workshops occurs in the project, where a total of 8,833 people has been participated, being 35% women and 60% young people.

There is also an improvement of local commitment and relations of trust and mutual supports on project implementation and long-term mangrove carbon conservation, with 178 MoUs signed between Yagasu and local community groups, 126 MoUs with village governments and 6 MoUs signed at provincial level. This improvement can also be demonstrated by the existence of working contracts produced for nursery works (178 contracts) and planting mangroves (178 contracts).

Since the start of the project Yagasu has supported the development of different village policy initiatives that support carbon conservation and sustainable income generations long-term mangrove carbon conservation. These initiatives are VLP, MPA and VR.

Production of mangrove species processed for foods is shown in the table below, helps to increase, enhance and ensure food security, and is developed by 124 women.

Table 23 Production of mangrove species proceed for foods

Name of village	Production of mangrove species proceed for foods (kg)/month					
	Avicennia spp	Bruguiera gymnorhiza	Sonneratia spp	Nypa fructicans	Achanthus ilicifolius	Number of women
Alur Naga	20	-	-	-	-	13
Gampong Pande	-	-	-	30	-	11
Udeung	20	-	40	-	-	14
Neuheun	20	-	-	-	10	12
Kuala Langsa	25	-	35	-	10	18
Alur Nunang	-	-	10	-	-	8
Alur Cempedak	-	-	15	-	-	10
Lubuk Kertang	30	5	30	-	20	16
Sicanang	30	8	35	10	20	15
Tanjung Rejo	18	-	25	-	20	7
Total	**163**	**13**	**190**	**40**	**80**	**124**

Project implementation have increased and strengthen gender equity and youth participation. Around 60% of participants in capacity building were young people. A total number of 190 women were trained on innovate activities associated to mangroves ecosystems, such as organic batik, food processing (syrup, dodol, jam), shrimp paste production and dried fish/oyster processing, as shown in table below.

Table 24. Capacity building for women groups

Name of village	Number of women trained on innovative activities				
	Organic batik	Food processing (Syrup, dodol, jam)	Shrimp paste production	Dried fish/oester processing	Others
Alur Naga	-	6	-	6	
Lam Nga	-	6	-	5	
Lam Ujung	-	4	-	7	
Asoe Nangroe	-	5	-	-	
Lamlumpu	-	5	-	-	
Neuhun	-	5	8	-	
Udeung	-	6	8	-	
Lubuk Kertang	5	8	6	-	
Kebun Ubi	-	5	7	6	
Alur Cempedak	7	6	-	-	
Tanjung Rejo	12	6	-	-	
Sicanang	7	5	6	-	
Rugemuk	6	4	5	6	
Percut	6	6	-	-	
Total	**43**	**77**	**40**	**30**	

Physical capital

Regarding physical capital, project zone facilities are improved due to project intervention. Facilities supported by the projects have been supported by multi donors, communities, government or private companies. The facilities developed are road access, communal toilets, water pumps, sign boards and meeting room.

Table 25. Village facilities developed by the project, self-community participation and other supports related to project activities

Type of facilities	Physical facilities supported by					Locations
	the projects	community self-initiatives	central/district governments	village governments	private companies	
Road access	-	1	2	1	1	Lubuk Kertang, Tanjung Rejo, Sicanang and Aceh Timur
Communal toilet	2	-	2	-	-	Sicanang, Aceh Timur
Water pump	-	-	2	-	1	Aceh Timur, Lubuk Kertang
Sign-board	5	2	2	1	2	Lubuk Kertang, Tanjung Rejo, Sicanang and Aceh Timur

Type of facilities	Physical facilities supported by					Locations
	the projects	community self-initiatives	central/district governments	village governments	private companies	
Meeting room	5	2	3	2	1	Aceh Tamiang, Pidie Jaya, Aceh Utara, Sicanang, Lubuk Kertang
Total	**12**	**5**	**11**	**4**	**5**	

Project activities aimed to develop and enhance eco-tourism associated to mangrove ecosystem. Table below shows the mangrove eco-tourism facilities developed in the sites:

Table 26. Mangrove eco-tourism facilities developed

Type of facilities	Location	Number of physical facilities supported by				
		the projects	community self-initiatives	local governments	private companies	Yagasu investment
Mangrove trail	Lubuk Kertang, Jaring Halus, Sicanang, Tanjung Rejo	4	6	1	1	2
Canteen	Lubuk Kertang	1	3	1	2	-
Coffee shop	Lubuk Kertang	1	4	1	2	-
Shelter	Lubuk Kertang	1	-	-	-	-
Entrance gate	Lubuk Kertang, Sicanang, Tanjung Rejo	3	-	-	2	1
Toilet	Lubuk Kertang, Tanjung Rejo, Sicanang	3	1	1		-
Road access	Lubuk Kertang	-	-	-	1	-
Sign-board	Lubuk Kertang, Jaring Halus, Sicanang, Tanjung Rejo	4	-	-	2	-
Meeting room	Lubuk Kertang, Sicanang, Tanjung Rejo	3	-	-	-	-
Total		**20**	**14**	**4**	**10**	**3**

Financial capital

Without planting mangrove, decreasing income and environment condition would be occurred in the future. Decreasing income will happened because of losing shrimps, crabs and fish from this area due to loosing mangrove tree. According to Sudarmono (2005), around 30% of sea fishing production depends on the existence of mangrove forests, because this ecosystem provides breeding place for various marine biotas, including several fish species. Fallen mangrove leaves can be detritus for land fertility, thus attracting marine biota for laying eggs, breeding larvae and as hunting area for aquatic species especially penacidae shrimp and milkfish (“chanos chanos”).

After of mangrove was cleared, the villagers started to experience a drastic decline in their fishery products. Some villagers complained that the yield had been reduced by a much on 70%, they also suffered from severe winds and storms (Suthawan Sathirathai, 2003).

Local community living in coastal area much understand about mangrove tree and impact of losing mangrove to fisheries product. Planting mangrove which is offering by project is not something new for community. They actually need it, but they have financial problem for planting. Financial support and seedling are helpful for community on protecting mangrove.

There are almost 10,000 people employed due to project implementation and activities, were almost half of them (42%) are women.

Table 27. Number of people employed on nursery works, planting mangroves, policy development and IGAs

District	Number of people employed								
	Nursery works		Planting mangroves		Policy development		IGAs		Total
	Men	Women	Men	Women	Men	Women	Men	Women	
Banda Aceh	72	54	99	63	72	36	45	54	495
Aceh Besar	72	54	99	63	72	36	45	54	495
Pidie	56	42	77	49	56	28	35	42	385
Pidie Jaya	112	84	154	98	112	56	70	84	770
Bireun	152	114	209	133	152	76	95	114	1,045
Aceh Utara	56	42	77	49	56	28	35	42	385
Aceh Timur	272	204	374	238	272	136	170	204	1,870
Langsa	80	60	110	70	80	40	50	60	550
Aceh Tamiang	176	132	242	154	176	88	110	132	1,210
Langkat	320	240	440	280	320	160	200	240	2,200
Deli Serdang	48	36	66	42	48	24	30	36	330
Medan	8	6	11	7	8	4	5	6	55
Total	**1,424**	**1,068**	**1,958**	**1,246**	**1,424**	**712**	**890**	**1,068**	**9,790**

Looking at the long-term (20 years) ecosystem resilience, the program scenario will bring various benefits to the coastal communities. Based on the business feasibility study, the products and services created by this program are very feasible and profitable. Net Benefit Cost of each business is more than 1 point and the Pay Back Period of each product and service varies from 13 to 36 months. During 2011 – 2019, the Income Generating Activities (IGAs) has increased beneficiaries' income by 50,5% (126.21 to 190.03 USD per family per month) and reduce their vulnerability to climate events. Within additional supports on IGAs, the program will create at least 500 entrepreneurs on sustainable social business. Within next 6 years, village business will contribute 6,841,080 USD (1,140,180 USD per annum) income of coastal families that equal to 1,6 times of total investment. By helping to ensure that the mangroves and more generally the coastal green-belt are well maintained, the program will contribute to the Northern Sumatra’s future ecological, social and green-economic development.

Table 28. Increase in income per family and month due to project intervention on Income Generation Activities

Name of village	Amount of average income per family per month			
	Without project 2011	After project intervention 2019	Increase income (IDR)	% income increase
Asoe Nangroe	1,700,000	2,600,000	900,000	52.9
Lam Ujung	1,500,000	2,250,000	750,000	50.0

Name of village	Amount of average income per family per month			
	Without project 2011	After project intervention 2019	Increase income (IDR)	% income increase
Cut Njoeng	1,600,000	2,500,000	900,000	56.3
Rantau Panjang	1,800,000	2,700,000	900,000	50.0
Kuala Langsa	1,900,000	3,000,000	1,100,000	57.9
Alur Nunang	1.750.000	2,600,000	850,000	48.6
Pangkalan Siata	1,500,000	2,200,000	700,000	46.7
Lubuk Kertang	2,000,000	3,000,000	1,000,000	50.0
Jaring Halus	1,600,000	2,300,000	700,000	43.8
Percut	1,700,000	2,500,000	800,000	47.1
Tanjung Rejo	1,600,000	2,400,000	800,000	50.0
Pantai Labu	1,500,000	2,300,000	800,000	5.,3
Sicanang	2,000,000	3,000,000	1,000,000	50.0
Average IDR	**1,703,846**	**2,565,385**	**861,538**	**50.50**
Average USD	**126,21**	**190,03**	**63,82**	**50,50**

The program has also created multiplier effects:

- Supporting local government tax income from village business
- Providing profit sharing cash money for social missions (orphanage care)
- Creating new employment, and locally-identified production and infrastructure needs

The non-monetary benefits of the program include:

- The stable condition of mangrove ecosystem supports the presence of commercial biodiversity: 7 – 12 tons of crabs, 3 – 5 tons of shrimps and 500 – 700 tons of fishes per week for non-stop production through the year;
- Increased community skills on value-added processing, quality control system, export specification, environmentally-sound processing practices (zero waste system);
- Improved community resilience on the coastal disaster prevention and anticipation to sea-rise impacts;
- Improved community awareness and sharing knowledges through trainings

Within help from various donors, the long-term Coastal Carbon Corridor program proposed will scale-up sustainable practices on biodiversity conservation/management – including IGAs around mangrove ecosystem and ecological services. The aim is to bring back socio-economic benefits of mangrove ecosystems to local populations, promote fair/equitable development, and reconcile biodiversity use with economic growth. The program establishes a business model on processing and manufacturing of mangrove-based products. These businesses integrate the processing, packaging, branding, certificating and licensing, marketing, promoting products and developing potential exports. This program will answer the current gap that Sumatra has a big number of natural resource products and services but there is not enough processing unit to give added values of those products.

The business unit in the communities include organic silvo-fishery, crab production, fish processing, shrimp-paste, mangrove honey, mangrove foods and organic batik mangrove. The social business is intended to expand trade and export opportunities to directly assist the market value chains. Sustainability of added-value products to meet anticipated rising demand might occur, and creates distribution of community's product at displayed shops and on-line shop. In addition, the carbon credits may attract national and international investors.

Pond fisheries production average is 1,431 kg/month for crab production, 881 kg/month for shrimp production and 919 kg/month for fish production. The income from non-aquaculture farms and from mangrove-based products and services are shown in the tables below.

Table 29. Income from non-aquaculture farms

Name of village	Amount of income from non-aquaculture farms (IDR) per month				
	Traditional fishing	Collecting crabs/mussels/ oysters	Shrimp paste production	Dried fish production	Other fishermen works
Asoe Nangroe	-	2,500,000	-	-	-
Lam Ujung	-	1,800,000	-	-	-
Rantau Panjang	-	1,500,000	-	-	-
Kuala Langsa	2,000,000	1,000,000	-	-	500,000
Alur Nunang	2,000,000	-	-	-	-
Tanjung Rejo	3,500,000	-	-	-	-
Sicanang	1,500,000	-		2,000,000	-
Kebun Ubi	-	-	3,500,000	2,000,000	-
Pantai Labu	-	-	3,000,000	2,500,000	-
Average IDR	**9,000,000**	**6,800,000**	**6,500,000**	**6,500,000**	**500,000**
Average USD	**667**	**504**	**481**	**481**	**37**

Table 30. Income from mangrove-based products and services

Name of village	Amount of income from mangrove-based products and services (IDR) per month				
	Mangrove foods	Organic batik mangroves	Nypa products	Other mangrove products (oyster, etc)	Mangrove eco-tourism
Asoe Nangroe	-	-	-	1,500,000	-
Lam Ujung	-	-	-	1,500,000	-
Cut Njoeng	-	-	-	1,500,000	-
Rantau Panjang	-	-	-	-	-
Kuala Langsa	2,500,000	-	1,500,000	-	-
Alur Nunang	1,000,000	-	-	-	-
Pangkalan Siata	-	-	-	-	-
Lubuk Kertang	2,000,000	-	1,000,000	-	16,000,000
Jaring Halus	-	-	-	-	-
Percut	-	-	-	-	-
Tanjung Rejo	2,500,000	2,500,000	-		2,000,000
Pantai Labu	-	-	-	-	-

Name of village	Amount of income from mangrove-based products and services (IDR) per month				
	Mangrove foods	Organic batik mangroves	Nypa products	Other mangrove products (oyster, etc)	Mangrove eco-tourism
Sicanang	3,000,000	1,000,000	1,000,000	-	6,000,000
Average IDR	**11,000,000**	**3,500,000**	**3,500,000**	**4,500,000**	**24,000,000**
Average USD	**815**	**259**	**259**	**333**	**1.,78**

Natural capital

Enhancement of natural capital includes the improvement of coastal stabilization in preventing abrasion due to heavy wave as well as improvement of coastal stabilization in protecting coastal community settlements from strong wind, by protecting 25,000 ha and restoring more than 5,000 ha of mangrove ecosystem. One of the mangrove roles is to prevent erosion, sedimentation stabilizers, inhibiting the effects of sea storm attack/tsunamis and flooding, maintaining water quality and increases the limits on the existence of salt solution and sedimentation anaerobic or detoxify contaminants as well as the habitat or supporting of wildlife ranges.

In addition to environmental services, mangrove ecosystem increases and enhance the populations and presence of fish, shrimps, mussel, crabs and other commercial biodiversity, used as resource of the local communities' sustainable businesses. Mangrove ecosystem has a high biological productivity and is very dependent on the nutrient content of coastal waters. The high content of organic in the form of detritus provides a source of food that contains high nutrients for mangrove plants itself and for animals that are on the mangrove area to the adjacent ecosystems of the marine ecosystem. Besides as a producer of nutrients, mangrove also contributes in fisheries to its function as a magnification/breeding place for some commercially valuable species of fishes and shrimps. Therefore, there is a positive correlation between mangrove ecosystem and fishery productivity.

Other improvements in natural capital are the enhancement of wildlife population associated to mangrove ecosystem. See more information in Section 5.3.1 Biodiversity Monitoring Plan and Section 7.3. Biodiversity monitoring data.

4.3.2 Monitoring Plan Dissemination (CM4.3)

All documents and information about the results on the monitoring and verification of this project will be published in the platforms of the VCS and CCBS.

4.4 Optional Criterion: Exceptional Community Benefits

4.4.1 Short-term and Long-term Community Benefits (GL2.2)

The maintenance of the project activities through time derives in protection and restoration of mangrove ecosystem through the development of sustainable businesses associated to mangroves, that are designed to persist in time, even after project ends. For more information, see Section 4.1.3.

4.4.2 Marginalized and/or Vulnerable Community Groups (GL2.4)

The marginalized and/or vulnerable community groups have been identified in Table 1 from Section 4.1.3.

One of the aims of the project is promote women's participation. The planting has been done mostly by women, and also field monitoring training mangrove processing. Project activities involve women in the participation in the nursery stage, planting and maintaining mangroves. In nurseries involving women 60% and 40% male workers, at the stage of planting 50% of female workers and 50% of male workers while in the maintenance of 60% of female workers and 40% of male labor.

The project aims to develop a new business to sell processed mangrove products (food and batik) and support women. Capacity building in this topic have been made to train women in the processing of mangrove products such as syrup, fruits, leaves and organic batik, and various businesses are being developed ran by women.

Community Group	Women
Net positive impacts	Women are able to develop sustainable businesses associated to mangroves, having a mean of life that helps the familiar economy and being included the in the society as active members.
Benefit access	Project activities involve women in the participation of planting and maintaining mangroves and the development a new business related to mangrove products.
Negative impacts	No negative impacts have been identified

Community Group	Youth
Net positive impacts	Young people are involved in all trainings building up a knowledge associated to development of sustainable business related to mangrove ecosystem, and to the relevance of protecting mangrove ecosystem in the future.
Benefit access	Project is able to involve young people in participating in the project activities, in order to give them necessary tools to develop businesses associated to mangroves (fishery and products processing) to be used in the future, and teach them the benefits of mangrove ecosystem to society, and the relevance of protecting the mangrove.
Negative impacts	No negative impacts have been identified

4.4.3 Net Impacts on Women (GL2.5)

Net impacts on women are positive and have been described in Section 4.3.1 Community Monitoring Plan.

Figure 9. Working with a group of women in Desa Kuala Langsa

4.4.4 Benefit Sharing Mechanisms (GL2.6)

Livelihoods Fund owns the total emission reductions generated by the project.

Distribution of all benefits resulting from the mangrove restoration carried through the project for the beneficiary community. The communities have agreed that the plantation mangrove is carried out for the benefit of the local communities who have recognized the benefits generated out of such plantations, including mangroves as a bio-shield, increased biodiversity, to protect embankments against climate change consequences, increased productive activities like fishing, crab production, organic silvo-fishery, mangrove foods, mangrove inks for batik, eco-tourism, etc. and increased new income generating options, with the only exception being the carbon credits generated by project.

The communities have agreed that the property rights on the carbon credits generated by this restoration are exclusively allocated to Livelihoods Fund. Under this agreement, the beneficiary community is committed not to assert any property rights over the carbon credits generated and/or to be generated. The sharing of VCUs between project stakeholders is governed by a series of agreements that extend over all activities in Aceh and North Sumatra.

4.4.5 Governance and Implementation Structures (GL2.8)

Livelihoods Fund will act as the project proponents, supporting the project, managing investments for it and holds all rights to the emission reduction from the Mangrove project.

Yagasu will act as project proponent too, developing and establishing direct relation with communities, implementing the project activities according to the Project Implementation Workplan, and providing necessary data to support the carbon development activities for measurement, monitoring and registering, verifying and issuing emission reduction for the crediting period of the project.

Governance and Implementation Structures have enabled effective participation of community members in project implementation through different participatory village meetings. See Section 4.3.1 Community Monitoring Plan and Section 7.2. Community monitoring data.

4.4.6 Smallholders/Community Members Capacity Development (GL2.9)

The project will undertake feasibility for participatory social and economic development outside and within the project areas and its boundaries, which is designed to enhance conservation and mitigate the risk. Intensive capacity building and awareness will bring upon an understanding of the importance of coastal ecosystem and its integrity function for sustainable development. An effective collaborative management will integrate more to the mechanism to maximize potential participation and roles of local communities and the authorities to develop sustainable coastal and watershed management plan.

The project not require the relocation of people, because mangrove restoration would indeed need local community, which a total participation would carry a better chance to achieve the goals on conserving and maintaining the mangrove habitat in the more sustainable processes. Supporting the community needs by building up capacity and empower their economic would fundamentally resolve a part of tenurial problems.

Capacity building on policy is prioritized regarding rehabilitation and conservation of mangrove forest ecosystem and maintain marine ecosystem integrity and its function as the land-use scenario that would affect water and soil resources and the biodiversity within the project area.

The necessary structure to accommodate the needs of capacity building and community empowerment in many sectors related to mangrove are listed below:

- Establish and conserve mangrove ecosystem,
- Improve knowledge on added value on utilization of mangrove fruits as alternative foods.
- Improve technology on traditional production of ponds for shrimps, fishes and crabs.
- Capacity building on community to improve local economic value through market link corporation

The capacity building developed have consisted in:

- Village meetings for project socialization and establishing Memorandum of Agreements (MoA) with village governments
- Village meetings for project socialization and MoU with community groups
- Capacity building on preparing nursery and planting mangroves
- Village meetings on developing Village Policy Initiatives (VLP, MPA and VR)
- Village meetings for community development and Income Generation Activities (IGA) preparation
- Awareness actions about climate change issues, mangrove conservation and benefits of mangroves for their life
- Capacity building aimed at women on different topics: organic batik, food processing (syrup, dodol, jam), shrimp paste production, dried fish/oyster processing.

Capacity building sessions will be aimed at different stakeholders: villagers, villagers administrators, university students, teachers, government administrators and NGOs.

See Section 4.3.1 Community Monitoring Plan and Section 7.2. Community monitoring data.

5 BIODIVERSITY

5.1 Net Positive Biodiversity Impacts

5.1.1 Biodiversity Changes (B2.1)

The without-project scenario, "Continuation of the pre-project land use", predicted a high biodiversity loss in the project zone due to the continued deforestation of mangroves, conversion for intensive aquaculture, illegal logging for wood and charcoal, coastal construction and settlement, water pollution and tourism activities.

In contrast, the present project seeks to protect 25,000 ha of existing coastal mangrove forest and restore mangrove by planting and increasing 5,228 ha of mangrove cover, by no using invasive species nor genetically modified organism (GMOs). Resulting in avoided deforestation from the beginning of the project, with no negative biodiversity impacts estimated.

The project zone is categorized as a Key Biodiversity Area (See Section 5.5.1 of CCB Project Description) and is considered biologically very rich. Besides, the project generated net positive impacts while maintaining high conservation values from the beginning of the project.

The tables below describe the changes in biodiversity resulting from project activities under with-project scenario in the project zone during this monitoring period:

Change in Biodiversity	Increase in forest and native plants cover
Monitored Change	The increase of restored land is a positive and direct change as a result of project activities.
Justification of Change	The total number of native species planted and the number of hectares restored contribute to this change. The change is estimated by using GIS data and ground check and comparing with pre-project state.

Change in Biodiversity	Recovery of ecological niches
Monitored Change	The recovery of the ecosystem is a predicted, actual, positive and direct change.
Justification of Change	The recovery of ecological niches is a result of the protected mangrove cover as well as the restoration of abandoned and deforested lands, resulting in the permanence of mangrove ecosystems

Change in Biodiversity	Recovery of fauna habitats
Monitored Change	The recovery of the ecosystem is a predicted, actual, positive and direct change.
Justification of Change	The recovery of fauna habitats is a result of the protected mangrove cover as well as the restored lands, that allows protected species as water-birds and migratory birds to spread and live in their natural habitats.

Change in Biodiversity	Increase of biodiversity: fauna and flora
Monitored Change	This is a predicted, positive and direct change, resulting from the mangrove cover protection and restoration
Justification of Change	The increase of fauna in the project zone is monitored through observation of fauna. The increase of flora is a direct result of the restoration with native plants.

5.1.2 Mitigation Actions (B2.3)

The negative impact for biodiversity identified is the potential disturbance of natural balance of nutrients in the soil, changes of pH, salinization, nitrification due to fish feeding.

Yagasu team has taken some samples of the fishpond soil to analyze the composition, and they are explaining to fisherman how they should feed the fish, and what are the risks for their sustainable business.

5.1.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

The Coastal Carbon Corridor project is expected to generate significant net positive biodiversity benefits. The table below presents the expected net positive biodiversity benefits, including high conservation values.

The coastal mangrove protection and restoration will increase and enhance the area of cover for the mangrove forest dependent species. Besides, project will increase the population of threatened species by enhancing their natural habitat.

The project aims to improve biodiversity by increasing mangrove cover and protecting coastal mangrove, thereby, increasing and improving habitat for plant and animal life.

Tabla 1. Summary of net positive biodiversity benefits

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
AREA-BASED CRITERIA				
Globally, regionally or nationally significant concentrations of	The project site presents a Priority Site for Biodiversity in Deli Serdang District “East Coast of North	The BS predicts loss of biodiversity and ecosystems due to continued deforestation of	Under the PS the coastal mangrove is been protected and the abandoned fishponds are being	Project activities ensure the protection of existing coastal mangrove and

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
biodiversity values (HCV1)	Sumatra Beach", has been identified as Important Bird and Biodiversity Area and Key Biodiversity Area based on the presence of significant populations of globally threatened species and significant congregations of one or more birds species at certain times in their lifecycle or seasonal migration	mangroves, conversion for intensive aquaculture, illegal logging for wood and charcoal, coastal construction and settlement, water pollution and tourism activities.	restored, which ensures the recovery of ecosystems and the maintenance and enhancement of populations of species supported by them.	the restoration of abandoned fishponds. The activities are currently protecting and enhancing the high conservation values areas. Some of these activities are the identification of mangrove planting sites, training local communities on nursery, planting and monitoring mangroves, planting mangroves and doing regular monitoring on planted mangroves, develop policy and technical guideline for Community Patrolling Units (MPU) operations, training of MPU and actions development. Establishing sustainable business that ensures the local livelihoods without threat or alter the subsistence of the ecosystems.
Globally, regionally or nationally significant large landscape-level areas where viable populations of most if not all naturally occurring species exist in natural patterns of distribution and abundance (HCV2)	Indonesia is part of the Sundaland Hotspot, and is home to 10% of the world's known plant species, 12% of all mammals, 17% of all birds, 16% of all reptiles and amphibians, and 25% of all fish. Most of Sumatra's endemic plant species are found in lowland forests below 500 m			
Threatened or rare ecosystems (HCV3)	Indonesian mangrove forests are the largest area of this ecosystem in the world (23% of all mangrove ecosystems in the world, Giri *et al*., 2011), being among the most carbon rich forest and providing habitats for marine and terrestrial species. According to the FAO, over the past three decades, Indonesia has lost			

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
	40% of its mangroves, and It keeps decreasing due to land use change.			
SPECIES-BASED CRITERIA				
Critically endangered: Helmeted Hornbil (Rhinoplax vigil, *Buceros vigil*)	The population has not been quantified but, according to UICN, the species has almost disappeared on Sumatra (J. Eaton in litt. 2015).	Under the BS, hunting pressure, illegal trade and habitat loss are the main threats for this species, which population is predicted to decrease extremely rapid.	The protection and restoration of mangrove forest is expected to enhance the population of this species, that occurs mostly in tropical lowland.	The activities headed to protect coastal mangrove and are the main actions taken to maintain and enhance the population of Helmeted Hornbil. Another action taken is the conduction of surveys to locate colonies, seasonal movements, ecological requirements.
Critically endangered: Painted turtle (*Batagur borneoensis*)	Population not quantified for this species, but is listed in The World's Most 25 Endangered Freshwater Turtles and Tortoises 2011 (Turtle Conservation Coalition).	Without-project, this species is expected to decrease its population, the main threats for this species are habitat loss due to land use change, harvesting by fishermen, poaching to meet pet and food demand, and intensive fish and shrimp farming. This tortoise is a priority species to be conserved in Indonesia according Minister of Forestry Decree No. 57 Year 2008 about Strategic Direction	The protection and restoration of mangrove forest is expected to enhance the population of this species, that occurs in inland freshwaters.	The activities headed to protect coastal mangrove and are the main actions taken to maintain and enhance the population of Painted turtle. Another action taken is the conduction of surveys to locate colonies, seasonal movements, ecological requirements.

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
		of National Species Conservation 2008–18		
Endangered: Irrawaddy dolphin (*Orcaella brevirostris*)	A population of roughly 50 individuals has been documented in Balikpapan Bay, Kalimantan, Indonesia (D. Kreb, unpub. data).	Under the BS the population of this species will keep on decreasing. The main threat is incidental mortality from entanglement in fishing gear, particularly gillnets. Besides, habitat loss due to deforestation, dams in riverine populations, degradation from altered freshwater flows, pollutants (oil, pesticides, industrial wastes, coal, etc) and noise.	The Irrawaddy Dolphin occur in wetlands (inland), just as permanent rivers, streams or creeks and estuarines. The project activities leaded to protect the coastal mangrove ecosystems will help to maintain, enhance and recover the habitat for this species.	The activities headed to protect coastal mangrove and are the main actions taken to maintain and enhance the population of Irrawaddy dolphin. Another action taken is the conduction of surveys to locate colonies, seasonal movements, ecological requirements.
Endangered Bird: Milky Stork (*Ciconia stormi*)	According to IUCN, is suspected to be declining rapidly. A total population of 5,000 individuals was estimated in late 1980s in Sumatra, recent estimates (2009) put the population at 1,600 individuals in Aceh province, 500 in North Sumatra province.	Suspected to be declining rapidly due to intense hunting pressure at nesting colonies and the rapid loss and conversion of coastal habitat.	Milky Stork is a predominantly coastal resident in Indonesia, inhabiting mangroves, fishponds and swamps. It is necessary to conduct surveys and research to locate additional colonies, seasonal movements and clarify its ecological requirements.	The main action taken to maintain and enhance the population status of Milky Stork is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the conduction of surveys periodically, as well as researches to locate colonies, seasonal movements,

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
				ecological requirements.
Vulnerable Mammal: Indo-Pacific Humpback Dolphin (*Sousa chinensis*)	According to IUCN, total species population would have to number 13.333-16.666 individuals	Under the BS the population is predicted to keep on decrease, the greatest threat to this species is fishing gear followed by habitat loss and degradation	Occur in tropical to warm-temperate coastal waters, like mangrove swamps and estuarine areas, they appear to have their highest densities in and around estuaries. The protection and restoration of mangrove forest is expected to enhance the population of this species	The main action taken to maintain and enhance the population status of Indo-Pacific Humpback Dolphin is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the conduction of surveys to locate colonies, seasonal movements, ecological requirements
Vulnerable Bird: Lesser Adjutant (*Leptoptilos javanicus*)	According to IUCN, is suspected to be declining rapidly. A total population of 5,000 individuals was estimated in 1993, unknown number in Indonesia, previously estimated at 2,000 individuals in 1993 but it may have severely declined since.	Decreasing of population due to drainage and conversion wetland, agricultural intensification, pesticide use, disturbance and large-scale development in coastal areas, the persistent and unregulated harvesting of eggs and chicks from colonies.	Lesser Adjutant frequent mangroves and intertidal flats According to IUCN, is necessary to monitor key colonies, protecting nesting colonies outside protected areas, promote control pesticide use around feeding areas. The protection and restoration of mangrove forest is expected to enhance the population of this species	The main action taken to maintain and enhance the population status of Lesser Adjutant is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the conduction of surveys to locate colonies,

Criteria	Status	Baseline scenario (BS)	With-project scenario (PS)	Activities
				seasonal movements, ecological requirements
Vulnerable Bird: Kerak Kerbau (*Acridotheres javanicus*)	Total population size is estimated in 2,500-9,999 individuals. The population of Sumatra is considered to come from the introduced population of the species outside of the native range.	The population is expected to decrease, due to bird trade and maybe the effect of pesticide.	This bird occurs in grasslands and wetlands, so the project activities are expected to enhance the population.	The main action taken to maintain and enhance the population status of Kerak Kerbau is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the conduction of surveys to locate colonies, seasonal movements, ecological requirements

5.1.4 High Conservation Values Protected (B2.4)

As mentioned before, in Section 5.1.3, none of the HCVs related to biodiversity will be negatively affected by the project.

The project proponents have internalized environmental and social sustainability criteria into their projects, including but not limited to respect for native species, preservation and protection of natural ecosystems and implementation of productive systems under good practices.

5.1.5 Invasive Species (B2.5)

As mentioned before, the species used in the restoration activities were native species all of them, thereby, no known invasive species have been or will be introduced into any area affected by the project.

5.1.6 Impacts of Non-native Species (B2.6)

There is no non-native species used in the project.

Species	
Justification of Use	
Adverse Effect	

5.1.7 GMO Exclusion (B2.7)

No genetically modified organisms, fertilizers or chemical pesticides have been used in the project.

5.1.8 Inputs Justification (B2.8)

No genetically modified organism, fertilizers or chemical pesticides have been used in the project.

Name	
Justification of Use	
Adverse Effect	

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

According to the assessment described in Section B.2, net impacts on biodiversity in project zone will be positive (compared to the without-project scenario and considering that measures to mitigate possible adverse impacts are being implemented by the project owners).

As previously mentioned, it is expected that project activities generate more favorable conditions for natural regeneration than conditions linked to abandoned fishponds and lands, and that tree plantations act as connectivity habitat and as a "soft barrier" to bird dispersal between remnants of native forest, facilitating the movement of animals across the habitat matrix and providing suitable habitat for some forest-dependents at landscape level; it means even outside the project zone.

Therefore, the potential impacts on biodiversity that the project activities are likely to cause outside the Project Zone are positive; no negative impacts outside the Project Zone have been identified.

Negative Offsite Impact	Mitigation Measure(s)

5.2.2 Net Offsite Biodiversity Benefits (B3.3)

According to the assessment described in Section B.2, net impacts on biodiversity in project zone will be positive (compared to the without Project scenario and considering that measures to mitigate possible adverse impacts are being implemented by the project owners).

As previously mentioned, it is expected that project activities generate more favorable conditions for natural regeneration than conditions linked to abandoned fishponds and lands, and that tree plantations act as connectivity habitat and as a "soft barrier" to bird dispersal between remnants of native forest, facilitating the movement of animals across the habitat matrix and providing suitable habitat for some forest-dependents at landscape level; it means even outside the project zone.

Therefore, the potential impacts on biodiversity that the project activities are likely to cause outside the Project Zone are positive; no negative impacts outside the Project Zone have been identified.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

The biodiversity monitoring plan is included in the CCB Project Description.

Flora biodiversity

First CCB Yagasu biodiversity report (2009) shows the initial state of the project zone along 25 km from Sicanang to Percut, crossing two districts of Medan and Deli Serdang zone. Most parts of the whole area are home to more than 16 mangrove species such as *Rhizophora* spp, *Soneratia* spp, *Ceriops* spp and *Avicenia* spp. The outcomes of flora identification in Yagasu study (2009) can be seen in Section 7.4.

Yagasu data (2009) gathered from remote sensing indicated that both areas (Sicanang and Percut) had been degraded for decades.

The vegetation of Percut and Secanang consisted of low, medium and high density of natural stand mangrove, bare land and agriculture land. High density mangrove is not really dense, because only has basal area 20 m^2/ha. Basal area in medium density is 15 m^2/ha and low density 3 m^2/ha. Bare land consists in fish pond which is left by the owner because of productivity decreasing.

Table 31. Vegetation composition of mangroves in Sicanang and Percut at 2009 (Source: Yagasu CCB report 2009)

Family	Sicanang	Percut
Avicenniaceae	*Avicennia marina*	*Avicennia marina*
Avicenniaceae	*Avicennia officinalis*	*Avicennia officinalis*
Avicenniaceae	*Avicennia alba*	*Avicennia lanata*
Rhizophoraceae	*Bruguiera cylindrica*	*Bruguiera sexangula*
Rhizophoraceae	*Bruguiera gymnorrhiza*	*Bruguiera gymnorrhiza*
Rhizophoraceae		*Bruguiera parviflora*
Rhizophoraceae		*Ceriops tagal*
Rhizophoraceae	*Rhizophora apiculata*	*Rhizophora apiculata*
Rhizophoraceae	*Rhizophora mucronata*	*Rhizophora mucronata*
Rhizophoraceae	*Rhizophora stylosa*	*Kandelia kandel*
Sonneratiaceae	*Sonneratia alba*	*Sonneratia alba*
Maliaceae	*Xylocarpus granatum*	*Xylocarpus granatum*
Araceae	*Nypa fructicans*	*Nypa fructicans*
Pandanaceae	*Pandanus tectorius*	*Pandanus tectorius*
Acanthaceae	*Acanthus sp*	*Acanthus sp*
Euphorbiaceae	*Excoecaria agalocha*	*Excoecaria agallocha*
Combretaceae	*Lumnitzera littorea*	

According to Yagasu CCB report (2009), the biodiversity in the project areas represent species inhabiting coastal ecosystem with a specific mangrove habitat. From the total of 39 mangrove species existing in

Indonesia, more than 50% species of mangrove species are present in the project site, which specific 11 species in Percut and 12 species in Sicanang. The sites are dominated by *Avicennia* spp. that widely cover the outer coastal zone and then gradually rowed by *Rhizophora* spp. and *Lumnitzera* spp.

The Shannon-Wiener diversity Index (H') obtained by Yagasu in these sites is considerably low. H' might indicate the level of habitat condition (highly degraded and requiring rehabilitation for H' < 1.5; the condition is fair and good to have a rehabilitation for 1.5 < H' < 3.0, and mangrove is still in good conditions for H' > 3.0). Based on the level of condition above, Sicanang (H' = 1.70), though it is categorized as a light degrade, but rehabilitations are necessary. For Percut (H' = 1.17) indicated heavy degraded mangrove habitat.

Table 32. Shannon-Wiener diversity Index (H') for growth phase on mangrove vegetations (Source: Yagasu CCB report 2009)

No	Phase of growth	Sicanang	Percut
1	Seedlings	1.56	1.41
2	Trees	1.70	1.17

The Ministry of Environment (KepMen 201/ 2004) released the criteria for mangrove condition as bellow:

- Good (high density) more than 1,500 trees per hectare
- Degraded (medium density) between 1,000 to 1,500 trees per hectare
- Damaged (Low density) less than 1,000 trees per hectare

Frequency of seedling and trees in Percut and Sicanang showed in the following table, resulting in Percut having a high frequency of trees, that gives an opportunity for that area for self-recovery, as long as human activities within the areas are minimum.

Table 33. Frequency of mangrove stem per hectare in Percut and Secanang (Source: Yagasu CCB report 2009)

No	Growth phase	Secanang	Percut
1	Sedlings	6,608	22,500
2	Smaller trees	1,362	5,722
3	Trees		
	- frequency	529	250
	- Damage criteria	damage (rarely)	damage (rarely)

A total of 25,000 ha of coastal mangrove are being protected by local communities due to project activities in Medan, Deli Serdang and Langkat district. The existing mangrove it can be found in these sites are secondary mangrove forest, swamp forest and primary mangrove forest. The number of hectares protected in each village can be consulted in Section 7, subsection 7.3.

The total number of degraded mangrove ecosystem restored is 5,278.71 ha, using native species *Rhizophora mucronata*, *Rhizophora apiculata*, *Rhizophora stylosa* and *Avicennia* spp. And planting around 16,500,900 trees.

The seedlings come from the nursery carried out in the different villages involved in the project, where community groups are trained in good silvicultural practices to grow up the seedlings that will be used in the plantations. A total amount of 18,150,991 seedlings have been grew up in 100 different villages.

Natural regeneration in restored areas is given due to restoration of degraded mangrove, the total population of natural regeneration is around 10% from total population between 2 and 12 years of restored mangroves. The table below shows the species of natural regeneration in restored areas:

Table 34. Natural regeneration in restored areas

Family	Species	Local name	Sub-ecosystem	
			Pond	Riverine/coastline
Rhizophoraceae	*Rhizophora apiculata*	Bakau merah	+++	+++
	Rhizophora mucronata	Bakau besar	+++	+++
	Rhizophopra stylosa	Bakau hijau	+++	++
	Bruguiera sexangula	Mata buaya	+	+
	Bruguiera gymnorrhiza	Mata buaya merah	++	+
	Ceriops tagal	Tengar	++	++
Sonneratiaceae	*Sonneratia alba*	Berembang	++	+++
Avicenniaceae	*Avicennia marina*	Api-api putih	+++	+++
	Avicennia officinalis	Api-api oval	-	+
	A.vicennia alba	Api-api hitam	+++	+++
Comretaceae	*Lumnitzera littorea*	Terumtum putih	++	+
Arecaceae	*Nypa fructicans*	nipah	++	++
Euphorbiaceae	*Excoecaria agallocha*	Buta-buta	++	+
Acanthaceae	*Acanthus sp*	Jeruju	+++	+++
Meliaceae	*Xylocarpus granatum*	Nyirih	+	+
+low population ++ medium population +++high population				

The outcomes resulting from the flora inventory in both North Sumatra and Banda Aceh province are showed in the tables below. In North Sumatra all sites present a medium-high population of almost all the species, being Sei Meran the location that shows a richer biodiversity in species associated to mangrove ecosystems, and Beras Basah the site where these species are rarely presents. Regarding Aceh province all the locations present more or less the same population, which is medium high, being Alur Nunag the site presenting all species associated to mangrove ecosystems.

Table 35. Outcomes of flora inventory in North Sumatra province.

Family	Latin name	Location of Study					
		LK	SM	BB	PP	P	SC
MANGROVES							
Rhizophoraceae	*Ceriops decandra*	++	++		++		++
	Ceriops tagal		++	+	++		+
	Rhizophora stylosa	++	+	+	+	+	+
	Rhizophora apiculata	+++	+++	++	+++	+	+++
	Rhizophora mucronata	+++	+++	+	++	+	+++
	Bruguiera gymnorrhiza	++	++	+	+		++

Family	Latin name	Location of Study					
		LK	SM	BB	PP	P	SC
	Bruguiera sexangula	++	+	+	+	+	+
	Bruguiera parviflora		++		++		
Sonneratiaceae	*Sonneratia alba*	+++	++	++	+	++	+++
	Sonneratia caseolaris	+++	++	+		++	+++
	Sonneratia griffithii			+			
Aracaceae	*Nypa fruticans*	+++	+++	+++	+++	+++	+++
Combretaceae	*Lumnitzera racemosa*	+	+		+	+	
	Lumnitzera littorea	+		+	+		
Avicenniaceae	*Avicennia alba*	+++	+++	+++	+++	+++	+++
	Avicennia eucalyptifolia		+				+
	Avicennia lanata		+	+		+	+
	Avicennia marina	+++	+++	+++	+++	+++	+++
	Avicennia officinalis	+++	+++	+++	+++	+++	+++
Meliaceae	*Xylocarpus granatum*	++	++	+	++	+++	+
Polypodiaceae	*Acrostichum aureum*	++	++		++	+	++
Myrsinaceae	*Aegiceras floridum*	+		+	+	+	
Euphorbiaceae	*Excoecaria agallocha*		+++	++	+++	++	++
MANGROVE ASSOCIATION							
Asteraceae	*Pluchea indica*	+	+			+	
Myrtaceae	*Syzygium myrtifolium*		+				+
Myrtaceae	*Osbornia octodonta*	+	+		+		
Bignoniaceae	*Dolichandrone spathacea*		+		+		
Rubiaceae	*Morinda citrifolia*	++	++		++	+	++
Flagellariaceae	*Flagellaria indica*		+				
Meliaceae	*Aglaia cucullata*		+				
	Azadirachta indica					+	
Fabaceae	*Derris trifoliata*	+	++		++		
Melastomataceae	*Melastoma candidum*		+		+	+	+
Simaroubaceae	*Eurycoma apiculata*		+				
Anacardiaceae	*Campnosperma auriculatum*		+				
Verbenaceae	*Gmelina arborea*		+				
Clusiaceae	*Calophyllum inophyllum*					+	
Gentianaceae	*Fagraea crenulata*					+	
Malvaceae	*Hibiscus tiliaceus*			++	+	+	+
Acantahaceae	*Achanthus ilicifolius*	+++	+++	+++	+++	+++	+++

LK: Lubuk Kertang, SM: Sei Meran, BB: Beras Basah, P: Percut, PP Pulau Panjang, SC: Sicanang

+low population
++ medium population
+++high population

Family	Latin name	Location of Study					
		LK	SM	BB	PP	P	SC

Table 36. Outcomes of flora inventory in Aceh province.

Family	Latin name	Location of study				
		GP	LU	RP	KL	AN
MANGROVES						
Rhizophoraceae	*Rhizophora apiculata*	+++	++	+++	+++	+++
	Rhizophora mucronata	++	++	+++	+++	+++
	Rhizophopra stylosa	++	++	-	-	++
	Bruguiera cylindrica	++	++	-	-	++
	Bruguiera gymnorrhiza	++	++	++	++	++
	Bruguiera parviflora	-	++	+++	++	++
	Ceriops tagal	+	++	++	++	++
	Kandelia kandel	+	+	-	++	-
Sonneratiaceae	*Sonneratia alba*	+++	+++	+++	+++	+++
Avicenniaceae	*Avicennia marina*	+++	+++	+++	+++	+++
	Avicennia officinalis	-	-	++	+	+
	Avicennia alba	+++	+++	+++	+++	+++
Comretaceae	*Lumnitzera littorea*	++	+	++	++	++
Arecaceae	*Nypa fructicans*	+++	+++	+++	+++	+++
Plumbaginaceae	*Aegialitis annulata*	-	-	++	-	+
Euphorbiaceae	*Excoecaria agallocha*	++	++	++	++	++
Acanthaceae	*Acanthus sp*	+++	+++	+++	+++	+++
Meliaceae	*Xylocarpus granatum*	++	++	++	++	++
MANGROVE ASSOCIATION						
Apocynaceae	*Calotropis gigantea*	-	-	-	++	++
Araceae	*Cocos nucifera*	++	-	++	-	++
Fabaceae	*Derris trifoliata*	++	++	++	++	++
Pandanaceae	*Pandanus tectorius*	-	-	-	+++	+++
GP: Gampong pande, LU: Lam Ujong, RP: Rantau Panjang, KL: Kuala Langsa, AN: Alur Nunang +low population ++ medium population +++high population						

Figure 10. Flora identification in Desa Sei Meran

Fauna biodiversity

North Sumatra's remaining mangrove ecosystem is an ideal home for endemic and migratory birds as part of Malacca Straits biodiversity. A lot of fish species and bird species are living in mangrove swamps, where also often to find some threatened mammal species. Mangrove helps to structure and enrich the ground soil, filter sediment from upper stream, make cleaner water, and protect the corals, etc., only if these ecosystems are well managed.

Yagasu CCB 2009 study encountered at least 23 species of birds, 7 species of mammals, six species of reptiles in Sicananag, while in Percut were found 34 species of birds, 2 species of mammals and three species of reptiles. The biodiversity in the project area is considerably low because of most mangrove vegetation had already destroyed.

Yagasu developed another research in 2014 to assess the biodiversity conditions in the project sites (Langkat district of Biodiversity, Northern Sumatra, villages of Beras Basah, Lubuk Kertang and Sei Meran). The results showed wildlife found, which includes: 7 species of mammals, 74 species of birds, 15 species of herpetofauna and 22 species of fish. 18 avifauna species were wild animals protected by the laws in force in Indonesia. Lesser Adjutant (*Leptoptilus javanicus*) is a species of bird that is in danger of extinction worldwide, in the Vulnerable category, according to IUCN Red list. The outcomes of this study can be seen in Section 7.5.

Wildlife and its habitat threats identified in this Yagasu study were some activities including the felling of mangrove trees (in the Sei Meran area, registration is done to supply the raw material to make coal); and transfer of functions from mangrove forests to oil palm plantations (in Sei Meran) and aquaculture (in Beras Basarh, Sei Meran and Pulau Panjang).

Figure 11. Biodiversity survey preparation in the Lubuk Kertang area. (Source: Yagasu biodiversity study, 2014)

Figure 12. Lesser Adjutant (Leptoptilus javanicus) is observed in the mudflat section on the coast of Wet Rice. (Source: Yagasu biodiversity study, 2014)

Figure 13. Crab-eating macaque (Macaca fascicularis) is one of the mammals that can still be observed in the remaining part of the mangrove in Wet Rice, which borders settlements. (Source: Yagasu biodiversity study, 2014)

Figure 14. Several species of molluscs found in the survey area: (1) Telescopium; (2) Chicoreus sp .; (3) Terebralia sulcata; (4) Platyvindex sp. ; (5) Ellobium sp .; (6) Nerita sp .; (7) Ellobium aurisidae. (Source: Yagasu biodiversity study, 2014).

The outcomes resulting from the fauna inventory in North Sumatra and Aceh province are showed in the tables below. It may be seen that Percut is the location with the richest biodiversity and highest population of species.

Regarding bird species, Milky Stork (*Mycteria cinerea*), Endangered species according to IUCN Red List, is present in Pulau Panjang, Percut and Sicanang with a medium-low population, and is have not been seen in Aceh province. Lesser Adjutant (*Leptoptilos javanicus*) Vulnerable species according to IUCN Red List, have been found in the same places as Milky Stork, and also it is present in all locations of study in Aceh province, but presenting low population. Another Vulnerable bird species present in the project zone is White-vented Myna (*Acridotheres javanicus*), but only seen in Northern Sumatra and medium-low population. Hornbil (*Buceros vigil*) have been seen in low population in Rantau Panjang, this species is Critically Endangered according to IUCN Red List.

Figure 15. Bird watching in Desa Lubuk Kertang

Painted turtle (*Batagur borneoensis*) is considered as Critically Endangered in the IUCN Red List, and is present in Aceh province in medium population in Kuala Langsa and Alur Nunang.

Majority of the rest of fauna observed in the locations are considered in the IUCN Red List as Least Concern, excepting Monitor lizard (*Varanus salvator*), considered as Near Threatened. All these species are protected by law in Indonesia.

Table 37. Outcomes of fauna inventory in North Sumatra province.

Family	Latin name	English Name	Location of Study					
			LK	SM	BB	PP	P	SC
AVIFAUNA								
Ardeidae	*Egretta alba*	Great Egret	++	+++	+++	+++	+++	+++
	Egretta garzetta	Little Egret	++	+	++	++	++	+
	Egretta intermedia	Intermediate Egret	+	++	++	++	+++	+++
	Egretta sacra	Reef Egret	-	-	++	++	++	+
Ciconiidae	*Mycteria cinerea*	Milky Stork	-	-	-	+	++	+
	Leptoptilos javanicus	Lesser adjutant	-	-	-	+	++	+
Accipitridae	*Haliaeetus leucogaster*	White-bellied Sea-eagle	+	+	++	++	++	+
	Haliastur indus	Brahminy Kite	++	++	++	+	+++	+++
	Aviceda leuphotes	Black Baza	+	-	-	-	+	-
	Ichtyhophaga ichthyaetus	Grey-Headed Fish-Eagle	-	-	-	++	-	-
Laridae	*Chlidonias leucopterus*	White-winged Tern	-	+	++	+	+++	-
	Sterna albifrons	Little Tern	-	-	++	++	++	++
	Sterna hirundo	Common Tern	-	++	+	+	++	+
	Sterna anaethetus	Bridled Tern	-	-	-	-	++	-
	Chlidonias hybridus	Whiskered Tern	-	-	-	-	++	-
	Sterna sumatrana	Black-naped Tern	-	-	-	-	++	-
Alcedinidae	*Halcyon smyrnensis*	White-throated Kingfisher	+	+	+	-	++	++
	Halcyon sancta	Sacred Kingfisher	-	-	+	-	+	-
	Halcyon chloris	Collared kingfisher	++	++	++	++	++	++
Monarchidae	*Rhipidura javanica*	Pied Fantail	++	++	++	-	+	+
Nectariniidae	*Nectarinia calcostetha*	Copper-throated Sunbird	+++	+++	+++	+	+++	+++
	Nectarinia jugularis	Olive-backed Sunbird	++	++	-	-	+	-
	Anthereptes malacensis	Brown-throated Sunbird	++	-	-	+	-	-
	Anthereptes simplex	Plain Sunbird	+	+	-	-	+	+
Sturnidae	*Acridotheres javanicus*	White-vented Myna	++	+	-	-	++	-
Tytonidae	*Tyto Alba*	Barn owl	-	-	-	+	+	+
HERPETOFAUNA								
Colubridae	*Python reticulatus*	Reticulated Python	++	+		++		
Viperidae	*Trimeresurus albolabris*	White-Lipped Tree Viper	++	++	++	++	++	++
Varanidae	*Varanus salvator*	Monitor lizard	+++	+++	+++	+	+++	+++
LK: Lubuk Kertang, SM: Sei Meran, BB: Beras Basah, P: Percut, PP Pulau Panjang, SC: Sicanang								

Family	Latin name	English Name	Location of Study					
			LK	SM	BB	PP	P	SC
+low population ++ medium population +++high population								

Table 38. Outcomes of fauna inventory in Aceh province.

Family	Latin name	English Name	Location of study				
			GP	LU	RP	KL	AN
AVIFAUNA							
Ciconiidae	*Leptoptilos javanicus*	Lesser Adjutant	++	+	+	+	+
Ardeidae	*Egretta garzetta*	Little Egret	+++	+++	+++	++	+++
Cisticolidae	*Prinia flaviventris*	Yellow-bellied Prinia	++	++	++	-	++
Bucerotidae	Buceros vigil	Hornbill	-	-	+	-	+
Muscicapidae	*Copsychus saularis*	Magpie	++	++	++	+	++
HERPETOFAUNA							
Crocodylidae	*Crocodylus porosus*	Crocodylidae	-	-	-	++	++
Geoemydidae	*Batagur borneoensis*	Painted turtle	-	-	-	++	++
GP: Gampong pande, LU: Lam Ujong, RP: Rantau Panjang, KL: Kuala Langsa, AN: Alur Nunang +low population ++ medium population +++high population							

Regarding both species of dolphin, flag species in stable mangrove ecosystem, Irrawaddy Dolphin (*Orcaella brevirostris*), considered Endangered according to IUCN Red List, and Indo-Pacific Humpback Dolphin (*Sousa chinensis*), Vulnerable species, occurs in Langkat and Deli Serdang, with a low estimated population.

Other species whose presence is indicator of good conditions of mangrove ecosystems are the bird species Lesser Adjutant (*Leptoptilos javanicus*), Milky Stork (*Mycteria cinerea*), Brahmini kite eagle (*Haliastur indus*); the Vulnerable Estuarine Crocodile (*Crocodylus porosus*) occurring in Aceh and Langkat; and the Mangrove primate (*Presbytis cristata*).

Table below shows the presence of migratory birds in stable mangrove ecosystem, and the location of occurrence in the project zone.

Table 39. Location of occurrence regarding migratory birds in stable mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence
Scolopacidae	*Numenius phaeopus*	Whimbrel	LC	Deli Serdang
	Tringa totanus	Common Redshank	LC	Langkat, Deli Serdang
	Actitis hypoleucos	Common Sandpiper	LC	Langkat, Deli Serdang
Laridae	*Chlidonias leucopterus*	White-winged Tern	LC	Aceh, Langkat, Deli Serdang
	Sterna albifrons	Little Tern	LC	Aceh, Langkat, Deli Serdang
	Sterna hirundo	Common Tern	LC	Aceh, Langkat, Deli Serdang
	Sterna anaethetus	Bridled Tern	LC	Aceh, Langkat, Deli Serdang
	Chlidonias hybridus	Whiskered Tern	LC	Aceh, Langkat, Deli Serdang
Ciconiidae	*Mycteria cinerea*	Milky Stork	EN	Langkat, Deli Serdang
	Leptoptilus javanicus	Lesser adjutant	VU	Aceh, Langkat, Deli Serdang
+low population ++ medium population +++high population				

Figure 16. The colony of Lesser Adjutant (Leptoptilos javanicus) in Percut

The presence of reptiles is estimated in the following table, where it can be seen that Endangered Painted Turtle (*Batagur borneoensis*) is present in low population in Langkat and Aceh.

Table 40. Location of occurrence regarding reptile species in mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Bufonidae	*Duttaphrynus melanostictus*	Asian toad	LC	Aceh, Langkat, Deli Serdang	++
Microhylidae	*Microhyla sp.*	Frog	LC	Aceh, Langkat, Deli Serdang	++
Ranidae	*Fajervarya cancrivora*	Crab-eating Frog	LC	Aceh, Langkat, Deli Serdang	++
Colubridae	*Cerberus rhyncops*	Dog-faced watersnake	LC	Aceh, Langkat, Deli Serdang	++
	Python reticulatus	Reticulated Python	LC	Aceh, Langkat, Deli Serdang	++
Viperidae	*Trimeresurus albolabris*	White-Lipped Tree Viper	LC	Aceh, Langkat, Deli Serdang	++
Elapidae	*Bungarus candidus*	Malayan krait	LC	Aceh, Langkat, Deli Serdang	++
Geckonidae	*Gehyra mutilata*	Four-Clamed House Gecko	LC	Aceh, Langkat, Deli Serdang	++
	Hemidactylus frenatus	Common House Gecko	LC	Aceh, Langkat, Deli Serdang	++
Scincidae	*Eutropis multifasciata*	Common Sun Skink	LC	Aceh, Langkat, Deli Serdang	++
Agamidae	*Bronchocela cristatella*	Green-crested lizard	LC	Aceh, Langkat, Deli Serdang	++
Varanidae	*Varanus salvator*	Monitor lizard	LC	Aceh, Langkat, Deli Serdang	++
Rhacophoridae	*Fajervarya leucomystax*	Striped tree-frog	LC	Aceh, Langkat, Deli Serdang	++
Dicroglossidae	*Fajervarya limnocharis*	Frog	LC	Aceh, Langkat, Deli Serdang	++
Elapidae	*Bungarus laticep*	Snake	LC	Aceh, Langkat, Deli Serdang	++
	Laticauda colubrina	Yellow-lipped Sea Krait	LC	Aceh, Langkat, Deli Serdang	++
	Naja sumatrana	Cobra snake	LC	Aceh, Langkat, Deli Serdang	++
Colubridae	*Boiga dendrophila*	gold-ringed cat snake	LC	Aceh, Langkat, Deli Serdang	++
Homalopsidae	*Cerberus rynchops*	Dog-faced Water Snake	LC	Aceh, Langkat, Deli Serdang	++
	Enhydris	rainbow water snake	LC	Aceh, Langkat, Deli Serdang	++
	Fordonia leucobalia	White bellied mangrove snake	LC	Aceh, Langkat, Deli Serdang	++
Crocodylidae	*Crocodylus porosus*	Crocodylidae	LC	Aceh, Langkat, Deli Serdang	++
Gekkonidae	*Gekko gecko*	Tucktoo	LC	Aceh, Langkat, Deli Serdang	++
Scincidae	*Mabouya multifasciata*	Lizard	LC	Aceh, Langkat, Deli Serdang	++
Geoemydidae	*Batagur borneoensis*	Painted turtle	EN	Langkat, Aceh	+
+low population ++ medium population					

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
+++high population					

The monitoring of non-commercial macrozoobenthos in the project zone is shown in the table below, where it can be seen that all species are considered as Least Concern according to the IUCN Red List.

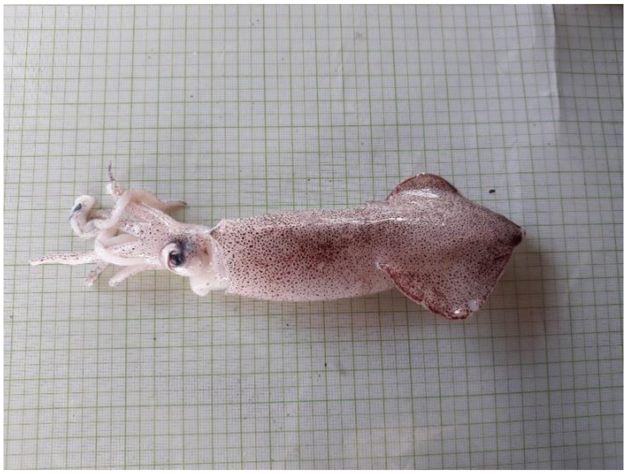
Table 41. Location of occurrence regarding non-commercial macrozoobenthos species in mangrove ecosystem

Family	Latin name	English	Location of occurrence	Estimated population
Ulmaridae	*Aurelia sp*	Jelly fish	Aceh, Langkat, Deli Serdang	+++
	Aurelia aurita	Moon Jelly fish	Aceh, Langkat, Deli Serdang	+++
Thiaridae	*Balanocochlis glans*	Trumpet snail	Aceh, Langkat, Deli Serdang	+++
Buccinidae	*Babylonia spirata*	Tiger snail	Aceh, Langkat, Deli Serdang	+++
Ellobiidae	*Cassidula aurisfelis*	Mangrove ear snail	Aceh, Langkat, Deli Serdang	+++
	Ellobium aurisjudae	Judas ear cassidula	Aceh, Langkat, Deli Serdang	+++
	Ellobium sp.	Judas ear	Aceh, Langkat, Deli Serdang	+++
Littorinidae	*Littoraria melanostoma*	Black mangrove snail	Aceh, Langkat, Deli Serdang	+++
Neritiidae	*Nerita polita*	Sea snail	Aceh, Langkat, Deli Serdang	+++
Strombidae	*Strombus labiatus*	Samar conch	Aceh, Langkat, Deli Serdang	+++
Neritidae	*Septaria lineata*	Brackish snail	Aceh, Langkat, Deli Serdang	+++
Potamididae	*Terebralia palustris*	Giant mangrove whelk	Aceh, Langkat, Deli Serdang	+++
	Terebralia sulcata	Sulcate swamp cerith	Aceh, Langkat, Deli Serdang	+++
Discodorididae	*Discodoris sp.*	Sea slug	Aceh, Langkat, Deli Serdang	+++
Mytilidae	*Mytilus sp.*	Blue mussel	Aceh, Langkat, Deli Serdang	++
Nereididae	*Namalycastis sp.*	Sea worm	Aceh, Langkat, Deli Serdang	++
	Nereis sp.	Sea worm	Aceh, Langkat, Deli Serdang	++
Asteroidea	*Asteroidea sp.*	Sea Star	Aceh, Langkat, Deli Serdang	++
Cerithiidae	*Cerithium stigmosum*		Aceh, Langkat, Deli Serdang	++
+low population ++ medium population +++high population				

Commercial fauna biodiversity

The biodiversity inventory has also monitored the presence of ichtyofauna, meaning fish, crabs and shrimps, and other macrozoobenthos in mangrove ecosystem that are related to economic development in the project zone due to commercial character of these species. Most of them are used for consumption and trade by local communities. Results of these inventories can be seen in Section 7, subsection 7.3, indicators B.C.1 and B.C.2.

Figure 17. The presence of squid in Desa Percut



5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

All documents and information about the results on the monitoring and verification of this project will be published in the platforms of the VCS and CCBS.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

The project activities have generated and are generating exceptional biodiversity benefits.

5.4.1 Trigger Species Population Trends (GL3.3)

The tables below show the trigger population trends (CR, EN and VU).

Trigger Species	Helmeted Hornbil (*Rhinoplax vigil*, *Buceros vigil*), CRITICALLY ENDANGERED (IUCN; 2018)
With-project Scenario	The protection and restoration of mangrove forest is expected to enhance the population of this species, that occurs mostly in tropical lowland.

Trigger Species	Painted Terrapin (Batagur borneoensis), CRITICALLY ENDANGERED (IUCN, 2000)
With-project Scenario	The protection and restoration of mangrove forest is expected to enhance the population of this species, that occurs in inland freshwaters.

Trigger Species	Irrawaddy Dolphin *(Orcaella brevirostris),* ENDANGERED (IUCN; 2017)
With-project Scenario	The activities headed to protect coastal mangrove and are the main actions taken to maintain and enhance the population of Irrawaddy dolphin.

Trigger Species	Milky Stork (*Mycteria cinerea*), ENDANGERED (IUCN, 2016)
With-project Scenario	The main action taken to maintain and enhance the population status of Milky Stork is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the conduction of surveys periodically, as well as researches to locate colonies, seasonal movements, ecological requirements.

Trigger Species	Indo-Pacific Humpback Dolphin (*Sousa chinensis*), VULNERABLE (IUCN; 2017)
With-project Scenario	The protection of coastal mangrove and restoration of fishponds is the main action taken to maintain and enhance the population of Indo-Pacific Humpback Dolphin, since it likes mangrove swamps and estuarine areas, like those that are presents in the project zone.

Trigger Species	Lesser Adjutant (*Leptoptilos javanicus*), VULNERABLE (IUCN, 2017)
With-project Scenario	The main action taken to maintain and enhance the population status of Lesser Adjutant is protect the coastal mangrove, natural habitat for this species, and restore abandoned fishponds. Another action taken is the monitoring of key colonies.

Trigger Species	Kerak Kerbau (Acridotheres javanicus), VULNERABLE (IUCN, 2016)
With-project Scenario	This bird occurs in grasslands and wetlands, so the project activities are expected to enhance the population.

6 ADDITIONAL PROJECT IMPLEMENTATION INFORMATION

No additional project implementation information is necessary.

7 ADDITONAL PROJECT IMPACT INFORMATION

7.1 Climate monitoring data

7.2 Community monitoring data

C.HC 1. A) Village meeting for project socialization and village government MoUs, B) Village meetings for project socialization and community group MoUs, C) Village meetings for preparing nursery and planting mangroves, D) Village meetings for Village Policy Initiatives (VLP, MPA and VR), E) Village meetings for community development and IGA preparation

A) Village meeting for project socialization and village government MoUs

District	Number of meetings	Participants		
		Men	Women	Total
Banda Aceh	16	156	84	240
Aceh Besar	26	253	137	390
Pidie	8	78	42	120
Pidie Jaya	19	185	100	285
Bireun	35	341	184	525
Aceh Utara	15	146	79	225
Aceh Timur	80	780	420	1,200
Langsa	28	273	147	420
Aceh Tamiang	21	205	110	315
Langkat	64	624	336	960
Deli Serdang	20	195	105	300
Medan	2	19	11	30
Total	**334**	**3,255**	**1,755**	**5,010**

B) Village meetings for project socialization and community group MoUs

District	Number of meetings	Participants		
		Men	Women	Total
Banda Aceh	12	166	110	276
Aceh Besar	25	345	230	575
Pidie	9	124	83	207
Pidie Jaya	40	552	368	920
Bireun	52	718	478	1,196

District	Number of meetings	Participants		
		Men	Women	Total
Aceh Utara	20	276	184	460
Aceh Timur	95	1,311	874	2,185
Langsa	27	372	249	621
Aceh Tamiang	55	759	506	1,265
Langkat	111	1,532	1,021	2,553
Deli Serdang	18	249	165	414
Medan	2	28	18	46
Total	**466**	**6,432**	**4,286**	**10,718**

C) Village meetings for preparing nursery and planting mangroves

District	Number of meetings	Participants		
		Men	Women	Total
Banda Aceh	7	54	37	91
Aceh Besar	16	124	84	208
Pidie	8	62	42	104
Pidie Jaya	13	102	67	169
Bireun	10	78	52	130
Aceh Utara	17	132	89	221
Aceh Timur	62	483	323	806
Langsa	23	180	119	299
Aceh Tamiang	14	109	73	182
Langkat	28	218	146	364
Deli Serdang	5	39	26	65
Medan	3	24	15	39
Total	**206**	**1,605**	**1,073**	**2,678**

D) Village meetings for Village Policy Initiatives (VLP, MPA and VR)

District	Number of meetings	Participants		
		Men	Women	Total
Aceh Tamiang	14	168	112	280
Langkat	54	648	432	1080
Deli Serdang	17	204	136	340
Medan	37	444	296	740
Total	**122**	**1,464**	**976**	**2,440**

E) Village meetings for community development and IGA preparation

District	Number of meetings	Participants		
		Men	Women	Total
Banda Aceh	10	88	92	180
Aceh Besar	10	104	104	208
Pidie	9	54	50	104
Pidie Jaya	10	80	89	169
Bireun	12	65	65	130
Aceh Utara	15	110	111	221
Aceh Timur	42	396	410	806
Langsa	26	155	144	299
Aceh Tamiang	35	80	102	182
Langkat	51	145	219	364
Deli Serdang	47	30	35	65
Medan	22	15	24	39
	289	**1,322**	**1,445**	**2,767**

C.HC 2. A) District meetings supporting mangrove restoration and protection through policy development and activity participation (2011 - 2019), B) Provincial workshop supporting mangrove restoration and protection through policy development and activity participation

A) District meetings supporting mangrove restoration and protection through policy development and activity participation (2011 - 2019)

District	Number of meetings	Participants						
		communities	decision makers	leglislatives	private sectors	universities	NGOs	Total
Banda Aceh	29	696	290	87	145	145	87	1,450
Aceh Besar	24	576	240	72	120	120	72	1,200
Pidie	21	504	210	63	105	105	63	1,050
Pidie Jaya	21	504	210	63	105	105	63	1,050
Bireun	19	456	190	57	95	95	57	950
Aceh Utara	30	720	300	90	150	150	90	1,500
Aceh Timur	37	888	370	111	185	185	111	1,850
Langsa	41	984	410	123	205	205	123	2,050
Aceh Tamiang	35	840	350	105	175	175	105	1,750
Langkat	46	1,104	460	138	230	230	138	2,300
Deli Serdang	33	792	330	99	165	165	99	1,650
Medan	24	576	240	72	120	120	72	1,200
Total	**360**	**8,640**	**3,600**	**1,080**	**1,800**	**1,800**	**1,080**	**18,000**

B) Provincial workshop supporting mangrove restoration and protection through policy development and activity participation

District	Number of workshops	Participants						
		communities	decision makers	leglislatives	private sectors	universities	NGOs	Total
Banda Aceh	8	288	128	80	112	88	72	768
Aceh Besar	2	72	32	20	28	22	18	192
Aceh Timur	3	108	48	30	42	33	27	288
Langsa	2	72	32	20	28	22	18	192
Aceh Tamiang	1	36	16	10	14	11	9	96
Medan	13	468	208	130	182	143	117	1,248
Total	**29**	**1,044**	**464**	**290**	**406**	**319**	**261**	**2,784**

C.HC 3. Awareness actions about climate change issues, mangrove conservation and benefits of mangroves for their life

District	Number of awareness	Participants				
		villagers	teachers	students	others	Total
Banda Aceh	32	384	256	640	192	1,472
Aceh Besar	17	204	136	340	102	782
Pidie	14	168	112	280	84	644
Pidie Jaya	15	180	120	300	90	690
Bireun	13	156	104	260	78	598
Aceh Utara	16	192	128	320	96	736
Aceh Timur	27	324	216	540	162	1,242
Langsa	30	360	240	600	180	1,380
Aceh Tamiang	25	300	200	500	150	1,150
Langkat	42	504	336	840	252	1,932
Deli Serdang	33	396	264	660	198	1,518
Medan	35	420	280	1050	210	1,960
Total	**299**	**3,588**	**2,392**	**6,330**	**1,794**	**14,104**

C.HC 4. Capacity building sessions for local stakeholders

District	Number of trainings	Participants						
		villagers	village administrators	university students	teachers	government administrators	NGOs	Total
Banda Aceh	26	364	182	390	156	156	130	1378
Aceh Besar	18	252	126	270	108	108	90	954
Pidie	15	210	105	225	90	90	75	795
Pidie Jaya	16	224	112	240	96	96	80	848
Bireun	13	182	91	195	78	78	65	689
Aceh Utara	17	238	119	255	102	102	85	901
Aceh Timur	20	280	140	300	120	120	100	1,060
Langsa	25	350	175	375	150	150	125	1,325

District	Number of trainings	Participants						
		villagers	village administrators	university students	teachers	government administrators	NGOs	Total
Aceh Tamiang	19	266	133	285	114	114	95	1,007
Langkat	33	462	231	495	198	198	165	1,749
Deli Serdang	26	364	182	390	156	156	130	1,378
Medan	21	294	147	315	126	126	105	1,113
Total	**249**	**3,486**	**1,743**	**3,735**	**1,494**	**1,494**	**1,245**	**13,197**

C.HC 5. People implementing the knowledge for long-term program activities

District	Number persons implementing on the actions on				
	Mangrove nursery	Planting mangroves	Replanting dead tress	Mangrove patrolling	Total
Banda Aceh	21	25	15	7	68
Aceh Besar	17	22	11	6	56
Pidie	18	24	12	6	60
Pidie Jaya	15	20	9	5	49
Bireun	17	21	11	5	54
Aceh Utara	20	24	14	7	65
Aceh Timur	33	36	27	13	109
Langsa	36	40	30	15	121
Aceh Tamiang	30	35	24	12	101
Langkat	42	46	36	18	142
Deli Serdang	25	30	19	9	83
Medan	23	28	17	8	76
Total	**297**	**351**	**225**	**111**	**984**

C.HC 6. Protection actions on mangrove ecosystem through Community Mangrove Patrolling Unit (CMPU)

Zone	CMPU actions			
	Number of patrols	Areas of village covered	Areas protected (ha)	People involved
1	18	2,461.28	3,516.12	144
2	20	2,702.48	3,860.69	162
3	33	3,742.93	5,347.05	255
4	26	2,916.45	4,166.36	207
5	25	1,355.05	1,935.79	200
6	32	4,210.56	6,015.09	256
Total	**154**	**17,388.77**	**24,841.10**	**1,224**

C.HC 7. Illegal findings reported and legally enforced by government rangers and local police

Zone		Number of illegal findings			
	Areas of village covered	Illegal logging	Poaching of fauna	Land encroachment	Others
1	2,461.28	35	2	7	
2	2,702.48	55	4	6	
3	3,742.93	34	6	3	
4	2,916.45	20	5	4	
5	1,355.05	17	3	5	
6	4,210.56	25	4	6	
Total	**17,388.77**	**186**	**24**	**31**	

C.SC. 1. Capacity building sessions on Income Generation Activities (IGAs)

District	Number of trainings	Participants		
		men	women	Total
Banda Aceh	15	153	102	255
Aceh Besar	14	143	95	238
Pidie	10	102	68	170
Pidie Jaya	12	122	82	204
Bireun	13	132	88	220
Aceh Utara	17	173	115	288
Aceh Timur	24	244	163	407
Langsa	27	275	183	458
Aceh Tamiang	22	224	149	373
Langkat	35	357	238	595
Deli Serdang	26	265	177	442
Medan	16	163	109	272
Total	**231**	**2,353**	**1,569**	**3,922**

C.SC. 2 Profitable and sustainable of social business running by local communities

District	Number of social business on					People involved		
	Aquaculture	Organic batik	Eco-tourism	Food processing	Others	men	women	Total
Banda Aceh	8	3	1	15	27	275	184	459
Aceh Besar	7	3	-	8	18	184	122	306
Pidie	5	3	-	9	17	173	116	289
Pidie Jaya	6	3	-	9	18	184	122	306
Bireun	5	2	-	11	18	184	122	306
Aceh Utara	6	2	-	12	20	204	136	340
Aceh Timur	6	3	1	17	27	324	216	540

District	Number of social business on					People involved		
	Aquaculture	Organic batik	Eco-tourism	Food processing	Others	men	women	Total
Langsa	6	3	2	18	29	348	232	580
Aceh Tamiang	5	3	1	15	24	245	163	408
Langkat	8	5	3	26	42	630	420	1,050
Deli Serdang	5	3	2	37	47	536	357	893
Medan	3	1	1	25	30	360	240	600
Total	**70**	**34**	**11**	**202**	**317**	**3,646**	**2,431**	**6,077**

C.SC. 3. Number of people (directly and indirectly) employed in the village business

District	Number of people employed										Total
	Aquaculture		Organic batik		Eco-tourism		Food processing		Others		
	men	women	men	women	men	women	men	women	men	women	
Banda Aceh	14	-	1	6	7	2	2	8			40
Aceh Besar	10	-	-	7	8	2	1	7			35
Pidie	7	-	-	6	7	-	1	8			29
Pidie Jaya	9	-	-	6	7	-	1	7			30
Bireun	8	-	-	6	5	-	3	7			29
Aceh Utara	8	-	-	7	5	-	3	6			29
Aceh Timur	12	-	2	9	7	4	4	7			45
Langsa	23	-	-	9	7	6	3	12			60
Aceh Tamiang	23	-	3	12	8	3	2	10			61
Langkat	42	-	-	20	24	15	5	26			132
Deli Serdang	33	-	3	25	37	10	7	24			139
Medan	18	-	2	17	28	8		13			86
	207	**-**	**11**	**130**	**150**	**50**	**32**	**135**	**-**	**-**	**715**

C.SC. 4. Effective community governance communication through village and district meetings, and workshops

District	Number of governance communication			Participants		
	Village meetings	District meetings	Workshops	men	women	Total
Banda Aceh	35	18	6	758	408	1,167
Aceh Besar	20	10	3	433	233	667
Pidie	18	9	3	390	210	600
Pidie Jaya	18	9	3	390	210	600
Bireun	15	8	3	325	175	500
Aceh Utara	13	7	2	282	152	433
Aceh Timur	17	9	3	368	198	567
Langsa	23	12	4	498	268	767
Aceh Tamiang	17	9	3	368	198	567
Langkat	34	17	6	737	397	1,133
Deli Serdang	26	13	4	563	303	867
Medan	29	15	5	628	338	967
Total	**265**	**133**	**44**	**5,742**	**3,092**	**8,833**

C.SC. 5. Commitments of local stakeholders in supporting project implementation

District	Number of MoUs with		
	Village governments	Local community groups	Provincial level
Banda Aceh	8	9	-
Aceh Besar	9	9	-
Pidie	10	7	-
Pidie Jaya	6	14	-
Bireun	10	19	-
Aceh Utara	8	7	-
Aceh Timur	30	34	-
Langsa	11	10	-
Aceh Tamiang	10	22	-
Langkat	21	40	-

District	Number of MoUs with		
	Village governments	Local community groups	Provincial level
Deli Serdang	2	6	-
Medan	1	1	-
North Sumatra province	-	-	3
Aceh province	-	-	3
Total	**126**	**178**	**6**

C.SC. 6 Working contracts produced for nursery works and planting mangroves

District	Number of contracts	
	Nursery works	Planting mangroves
Banda Aceh	9	9
Aceh Besar	9	9
Pidie	7	7
Pidie Jaya	14	14
Bireun	19	19
Aceh Utara	7	7
Aceh Timur	34	34
Langsa	10	10
Aceh Tamiang	22	22
Langkat	40	40
Deli Serdang	6	6
Medan	1	1
Total	**178**	**178**

C.SC. 7 Village policy models established to support long-term mangrove carbon conservation

Name of village	Number of village policy initiatives		
	Village Land-use Plan	Mangrove Protected Area	Village Regulation
Kuala Langsa	-	1	1
Sungai Leung	1	-	1
Alur Nunang	1	1	1
Lubuk Kertang	1	1	1
Pintu Air	1	-	1
Sei Meran	1	-	1
Jaring Halus	-	-	1
Bubun	1	1	1
Tanjung Rejo	1	1	1
Sicanang	1	1	1
Total	**8**	**6**	**10**

C.SC. 8 Mangrove species used for food security

Name of village	Production of mangrove species proceed for foods (kg)/month					Number of women
	Avicennia spp	Bruguiera gymnorhiza	Sonneratia spp	Nypa fructicans	Achanthus ilicifolius	
Alur Naga	20	-	-	-	-	13
Gampong Pande	-	-	-	30	-	11
Udeung	20	-	40	-	-	14
Neuheun	20	-	-	-	10	12
Kuala Langsa	25	-	35	-	10	18
Alur Nunang	-	-	10	-	-	8
Alur Cempedak	-	-	15	-	-	10
Lubuk Kertang	30	5	30	-	20	16

Name of village	Production of mangrove species proceed for foods (kg)/month					Number of women
	Avicennia spp	Bruguiera gymnorhiza	Sonneratia spp	Nypa fructicans	Achanthus ilicifolius	
Sicanang	30	8	35	10	20	15
Tanjung Rejo	18	-	25	-	20	7
Total	163	13	190	40	80	124

C.SC. 9. Capacity building for women groups

Name of village	Number of women trained on innovative activities				
	Organic batik	Food processing (Syrup, dodol, jam)	Shrimp paste production	Dried fish/oester processing	Others
Alur Naga	-	6	-	6	
Lam Nga	-	6	-	5	
Lam Ujung	-	4	-	7	
Asoe Nangroe	-	5	-	-	
Lamlumpu	-	5	-	-	
Neuhun	-	5	8	-	
Udeung	-	6	8	-	
Lubuk Kertang	5	8	6	-	
Kebun Ubi	-	5	7	6	
Alur Cempedak	7	6	-	-	
Tanjung Rejo	12	6	-	-	
Sicanang	7	5	6	-	
Rugemuk	6	4	5	6	
Percut	6	6	-	-	
Total	**43**	**77**	**40**	**30**	

C.PC 1 Mangrove eco-tourism facilities developed in the sites

Type of facilities	location	Number of physical facilities supported by				
		the projects*	community self-initiatives	local governments	private companies	yagasu investment
Mangrove trail	Lubuk Kertang, Jaring Halus, Sicanang, Tanjung Rejo	4	6	1	1	2
Canteen	Lubuk Kertang	1	3	1	2	-
Coffee shop	Lubuk Kertang	1	4	1	2	-
Shelter	Lubuk Kertang	1	-	-	-	-
Entrance gate	Lubuk Kertang, Sicanang, Tanjung Rejo	3	-	-	2	1
Toilet	Lubuk Kertang, Tanjung Rejo, Sicanang	3	1	1		-
Road access	Lubuk Kertang	-	-	-	1	-
Sign-board	Lubuk Kertang, Jaring Halus, Sicanang, Tanjung Rejo	4	-	-	2	-
Meeting room	Lubuk Kertang, Sicanang, Tanjung Rejo	3	-	-	-	-
Total		**20**	**14**	**4**	**10**	**3**

C.PC 2 Village facilities developed by the project, self-community participation and other supports related to project activities

Type of facilities	Physical facilities supported by					Locations
	the projects (multidonors)	community self-initiatives	central/district governments	village governments	private companies	
Road access	-	1	2	1	1	Lubuk Kertang, Tanjung Rejo, Sicanang and Aceh Timur
Communal toilet	2	-	2	-	-	Sicanang, Aceh Timur
Water pump	-	-	2	-	1	Aceh Timur, Lubuk Kertang

Type of facilities	Physical facilities supported by					Locations
	the projects (multidonors)	community self-initiatives	central/district governments	village governments	private companies	
Sign-board	5	2	2	1	2	Lubuk Kertang, Tanjung Rejo, Sicanang and Aceh Timur
Meeting room	5	2	3	2	1	Aceh Tamiang, Pidie Jaya, Aceh Utara, Sicanang, Lubuk Kertang
Total	12	5	11	4	5	

C.FC 1 Number of people employed on nursery works, planting mangroves, policy development and IGAs

District	Number of people employed								
	Nursery works		Planting mangroves		Policy development		IGAs		
	Men	Women	Men	Women	Men	Women	Men	Women	Total
Banda Aceh	72	54	99	63	72	36	45	54	495
Aceh Besar	72	54	99	63	72	36	45	54	495
Pidie	56	42	77	49	56	28	35	42	385
Pidie Jaya	112	84	154	98	112	56	70	84	770
Bireun	152	114	209	133	152	76	95	114	1,045
Aceh Utara	56	42	77	49	56	28	35	42	385
Aceh Timur	272	204	374	238	272	136	170	204	1,870
Langsa	80	60	110	70	80	40	50	60	550
Aceh Tamiang	176	132	242	154	176	88	110	132	1,210
Langkat	320	240	440	280	320	160	200	240	2,200
Deli Serdang	48	36	66	42	48	24	30	36	330
Medan	8	6	11	7	8	4	5	6	55
	1,424	**1,068**	**1,958**	**1,246**	**1,424**	**712**	**890**	**1,068**	**9,790**

C.FC 2 Profitable and sustainable business developed in the villages

Name of village	Number Group of social business in the villages					
	Aquaculture	Organic batik	Food processing	Shrimp paste production	Mangrove eco-tourism	Others
Asoe Nangroe	1		2			
Lam Ujung	1	1	2	1		
Cut Njoeng			1			
Rantau Panjang	2		1	1		
Kuala Langsa	4	1	3	1	1	
Alur Nunang	2	1	2			
Pangkalan Siata			2	1		
Lubuk Kertang	1	1	1		1	
Jaring Halus			1	1	1	
Percut			1	1		
Tanjung Rejo	22	2	3	1	1	
Pantai Labu			2	1		
Sicanang	5	1	3		1	
Total	**38**	**7**	**24**	**8**	**5**	

C.FC 3. Increase income per family and month due to project intervention on Income Generation Activities

Name of village	Amount of average income per family per month			
	Without project 2011	After project intervention 2019	Increase income (IDR)	% income increase
Asoe Nangroe	1,700,000	2,600,000	900,000	52.9
Lam Ujung	1,500,000	2,250,000	750,000	50.0
Cut Njoeng	1,600,000	2,500,000	900,000	56,3
Rantau Panjang	1,800,000	2,700,000	900,000	50.0
Kuala Langsa	1,900,000	3,000,000	1.100,000	57.9

Name of village	Amount of average income per family per month			
	Without project 2011	After project intervention 2019	Increase income (IDR)	% income increase
Alur Nunang	1,750,000	2,600,000	850,000	48.6
Pangkalan Siata	1,500,000	2,200,000	700,000	46.7
Lubuk Kertang	2,000,000	3,000,000	1,000,000	50.0
Jaring Halus	1,600,000	2,300,000	700,000	43.8
Percut	1,700,000	2,500,000	800,000	47.1
Tanjung Rejo	1,600,000	2,400,000	800,000	50.0
Pantai Labu	1,500,000	2,300,000	800,000	53.3
Sicanang	2,000,000	3,000,000	1,000,000	50.0
Average IDR	**1,703,846**	**2,565,385**	**861,538**	**50.50**
Average USD	**126.21**	**190.03**	**63.82**	

C.FC 4 Pond fisheries production

Name of village	Amount of income from pond fisheries (Kg)/month		
	crab production	shrimp production	fish production
Asoe Nangroe	800	500	400
Lam Ujung	700	600	400
Rantau Panjang	1,500	1,000	850
Kuala Langsa	1,250	1,250	1,100
Alur Nunang	1,000	600	600
Lubuk Kertang	1,700	800	1,000
Tanjung Rejo	2,500	1,500	2,000
Sicanang	2,000	800	1,000
Average	**1,431**	**881**	**919**

C.FC 5. Income from non-aquaculture farms

Name of village	Amount of income from non-aquaculture farms (IDR) per month				
	Traditional fishing	Collecting crabs/mussels/ oysters	Shrimp paste production	Dried fish production	Other fishermen works
Asoe Nangroe	-	2,500,000	-	-	-
Lam Ujung	-	1,800,000	-	-	-
Rantau Panjang	-	1,500,000	-	-	-
Kuala Langsa	2,000,000	1,000,000	-	-	500,000
Alur Nunang	2,000,000	-	-	-	-
Tanjung Rejo	3,500,000	-	-	-	-
Sicanang	1,500,000	-		2,000,000	-
Kebun Ubi	-	-	3,500,000	2,000,000	-
Pantai Labu	-	-	3,000,000	2,500,000	-
Average IDR	**9,000,000**	**6,800,000**	**6,500,000**	**6,500,000**	**500,000**
Average USD	**667**	**504**	**481**	**481**	**37**

C.FC 6. Income from mangrove-based products and services

Name of village	Amount of income from mangrove-based products and services (IDR)				
	Mangrove foods	Organic batik mangroves	Nypa products	Other mangrove products (oyster, etc)	Mangrove eco-tourism
Asoe Nangroe	-	-	-	1,500,000	-
Lam Ujung	-	-	-	1,500,000	-
Cut Njoeng	-	-	-	1,500,000	-
Rantau Panjang	-	-	-	-	-
Kuala Langsa	2,500,000	-	1,500,000	-	-
Alur Nunang	1,000,000	-	-	-	-

Name of village	Amount of income from mangrove-based products and services (IDR)				
	Mangrove foods	Organic batik mangroves	Nypa products	Other mangrove products (oyster, etc)	Mangrove eco-tourism
Pangkalan Siata	-	-	-	-	-
Lubuk Kertang	2,000,000	-	1,000,000	-	16,000,000
Jaring Halus	-	-	-	-	-
Percut	-	-	-	-	-
Tanjung Rejo	2,500,000	2,500,000	-		2,000,000
Pantai Labu	-	-	-	-	-
Sicanang	3,000,000	1,000,000	1,000,000	-	6,000,000
Average IDR	**11,000,000**	**3,500,000**	**3,500,000**	**4,500,000**	**24,000,000**
Average USD	**815**	**259**	**259**	**333**	**1,778**

7.3 Biodiversity monitoring data

B.A. 1. Outcomes if flora inventory in North Sumatra province and Aceh province

Family	Latin name	Local name	Location of Study					
			LK	SM	BB	PP	P	SC
MANGROVES								
Rhizophoraceae	*Ceriops decandra*	Tengar	++	++		++		++
	Ceriops tagal	Tengar		++	+	++		+
	Rhizophora stylosa	Bakau hijau	++	+	+	+	+	+
	Rhizophora apiculata	Bakau merah	+++	+++	++	+++	+	+++
	Rhizophora mucronata	Bakau besar	+++	+++	+	++	+	+++
	Bruguiera gymnorrhiza	Mata Buaya Merah	++	++	+	+		++
	Bruguiera sexangula	Mata buaya	++	+	+	+	+	+
	Bruguiera parviflora			++		++		
Sonneratiaceae	*Sonneratia alba*	Perepat	+++	++	++	+	++	+++
	Sonneratia caseolaris	Pedada merah	+++	++	+		++	+++
	Sonneratia griffithii	Pedada			+			
Aracaceae	*Nypa fruticans*	Nipah	+++	+++	+++	+++	+++	+++
Combretaceae	*Lumnitzera racemosa*	teruntum putih	+	+		+	+	
	Lumnitzera littorea	Teruntum merah	+		+	+		
Avicenniaceae	*Avicennia alba*	Api-api hitam	+++	+++	+++	+++	+++	+++
	Avicennia eucalyptifolia	Api-api		+				+
	Avicennia lanata	Api-api bulu		+	+		+	+
	Avicennia marina	Api-api putih	+++	+++	+++	+++	+++	+++
	Avicennia officinalis	Api-api daun lebar	+++	+++	+++	+++	+++	+++
Meliaceae	*Xylocarpus granatum*	Nyirih	++	++	+	++	+++	+
Polypodiaceae	*Acrostichum aureum*	Paku Laut	++	++		++	+	++
Myrsinaceae	*Aegiceras floridum*	Kacangan	+		+	+	+	
Euphorbiaceae	*Excoecaria agallocha*	Buta-buta		+++	++	+++	++	++
MANGROVE ASSOCIATION								

Family	Latin name	Local name	Location of Study					
			LK	SM	BB	PP	P	SC
Asteraceae	*Pluchea indica*	Beluntas	+	+			+	
Myrtaceae	*Syzygium myrtifolium*	Kayu merah		+				+
Myrtaceae	*Osbornia octodonta*	Baru-baru	+	+		+		
Bignoniaceae	*Dolichandrone spathacea*	Kayu Jaran		+		+		
Rubiaceae	*Morinda citrifolia*	Mengkudu	++	++		++	+	++
Flagellariaceae	*Flagellaria indica*	Rotan Laki		+				
Meliaceae	*Aglaia cucullata*	Karet laut		+				
	Azadirachta indica	Mimba					+	
Fabaceae	*Derris trifoliata*	Tuba laut	+	++		++		
Melastomataceae	*Melastoma candidum*	Senduduk		+		+	+	+
Simaroubaceae	*Eurycoma apiculata*	Pasak Bumi		+				
Anacardiaceae	*Campnosperma auriculatum*	Terentang		+				
Verbenaceae	*Gmelina arborea*	Jati Putih		+				
Clusiaceae	*Calophyllum inophyllum*	Bintangur					+	
Gentianaceae	*Fagraea crenulata*	Bira-bira					+	
Malvaceae	*Hibiscus tiliaceus*	Waru			++	+	+	+
Acantahaceae	*Achanthus ilicifolius*	Jeruju	+++	+++	+++	+++	+++	+++

LK: Lubuk Kertang, SM: Sei Meran, BB: Beras Basah, P: Percut, PP Pulau Panjang, SC: Sicanang

+low population
++ medium population
+++high population

Family	Latin name	Local name	Location of study				
			GP	LU	RP	KL	AN
MANGROVES							
Rhizophoraceae	*Rhizophora apiculata*	Bakau merah	+++	++	+++	+++	+++
	Rhizophora mucronata	Bakau besar	++	++	+++	+++	+++
	Rhizophopra stylosa	Bakau hijau	++	++	-	-	++
	Bruguiera cylindrica	Burus	++	++	-	-	++
	Bruguiera gymnorrhiza	Mata buaya merah	++	++	++	++	++
	Bruguiera parviflora	Lenggadai	-	++	+++	++	++
	Ceriops tagal	Tengar	+	++	++	++	++
	Kandelia kandel	kandelia	+	+	-	++	-
Sonneratiaceae	*Sonneratia alba*	Berembang	+++	+++	+++	+++	+++
Avicenniaceae	*Avicennia marina*	Api-api putih	+++	+++	+++	+++	+++
	Avicennia officinalis	Api-api oval	-	-	++	+	+
	Avicennia alba	Api-api hitam	+++	+++	+++	+++	+++
Comretaceae	*Lumnitzera littorea*	Terumtum putih	++	+	++	++	++
Arecaceae	*Nypa fructicans*	Nipah	+++	+++	+++	+++	+++
Plumbaginaceae	*Aegialitis annulata*	Bonsai alam	-	-	++	-	+
Euphorbiaceae	*Excoecaria agallocha*	Buta-buta	++	++	++	++	++
Acanthaceae	*Acanthus sp*	Jeruju	+++	+++	+++	+++	+++
Meliaceae	*Xylocarpus granatum*	Nyirih	++	++	++	++	++
MANGROVE ASSOCIATION							
Apocynaceae	*Calotropis gigantea*	Biduri	-	-	-	++	++
Araceae	*Cocos nucifera*	Kelapa	++	-	++	-	++
Fabaceae	*Derris trifoliata*	Tuba laut	++	++	++	++	++
Pandanaceae	*Pandanus tectorius*	Pandan mangrove	-	-	-	+++	+++

GP: Gampong pande, LU: Lam Ujong, RP: Rantau Panjang, KL: Kuala Langsa, AN: Alur Nunang

+low population
++ medium population

Family	Latin name	Local name	Location of study				
			GP	LU	RP	KL	AN
+++high population							

B.A. 2. Number of degraded mangrove ecosystem hectares restored

District	Number of				Species
	Villages	ha	Plots	Trees	
Banda Aceh	4	66.00	23	217,210	*R. mucronata, R.apiculata*
Aceh Besar	9	110.87	69	365,864	*R. mucronata, R.apiculata, R.stylosa*
Pidie	4	74.00	49	243,932	*R. mucronata, R.apiculata*
Pidie Jaya	7	386.00	327	1,273,412	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Bireun	5	17.00	17	56,393	*R. mucronata, R.apiculata*
Aceh Utara	8	200.00	205	660,735	*R. mucronata, R.apiculata, R.stylosa*
Aceh Timur	30	2,131.00	1,040	7,030,714	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Langsa	11	551.95	301	1,823,150	*R. mucronata, R.apiculata, R.stylosa*
Aceh Tamiang	7	933.00	437	3,078,363	*R. mucronata, R.apiculata, R.stylosa*
Langkat	13	436.45	150	1,440,557	*R. mucronata, R.apiculata, R.stylosa*
Deli Serdang	2	94.00	15	310,571	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Total	**100**	**5,000.27**	**2,633**	**16,500,901**	

B.A. 3. Improvement in good silvicultural practices (nursery)

District	Number of		Species
	Villages	Seedlings	
Banda Aceh	4	238,931	*R. mucronata, R.apiculata*
Aceh Besar	9	402,450	*R. mucronata, R.apiculata, R.stylosa*
Pidie	4	268,325	*R. mucronata, R.apiculata*
Pidie Jaya	7	1,400,753	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Bireun	5	62,032	*R. mucronata, R.apiculata*
Aceh Utara	8	726,809	*R. mucronata, R.apiculata, R.stylosa*

District	Number of		Species
	Villages	Seedlings	
Aceh Timur	30	7,733,785	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Langsa	11	2,005,465	*R. mucronata, R.apiculata, R.stylosa*
Aceh Tamiang	7	3,386,200	*R. mucronata, R.apiculata, R.stylosa*
Langkat	13	1,584,613	*R. mucronata, R.apiculata, R.stylosa*
Deli Serdang	2	341,629	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Total	**100**	**18,150,991**	

B.A. 4. Species of natural regeneration in restored areas

Family	Species	Local name	Sub-ecosystem	
			Pond	Riverine/ coastline
Rhizophoraceae	*Rhizophora apiculata*	Bakau merah	+++	+++
	Rhizophora mucronata	Bakau besar	+++	+++
	Rhizophopra stylosa	Bakau hijau	+++	++
	Bruguiera sexangula	Mata buaya	+	+
	Bruguiera gymnorrhiza	Mata buaya merah	++	+
	Ceriops tagal	Tengar	++	++
Sonneratiaceae	*Sonneratia alba*	Berembang	++	+++
Avicenniaceae	*Avicennia marina*	Api-api putih	+++	+++
	Avicennia officinalis	Api-api oval	-	+
	A.vicennia alba	Api-api hitam	+++	+++
Comretaceae	*Lumnitzera littorea*	Terumtum putih	++	+
Arecaceae	*Nypa fructicans*	Nipah	++	++
Euphorbiaceae	*Excoecaria agallocha*	Buta-buta	++	+
Acanthaceae	*Acanthus sp*	Jeruju	+++	+++
Meliaceae	*Xylocarpus granatum*	Nyirih	+	+

Family	Species	Local name	Sub-ecosystem	
			Pond	Riverine/ coastline
+low population ++ medium population +++high population				

B.A. 5. Number of hectares where have been carried out replanting actions

District	Number of			Species
	Villages	ha	Seedlings	
Banda Aceh	5	14.48	47,784	*R. mucronata, R.apiculata*
Aceh Timur	12	147.72	487,476	*R. mucronata, R.apiculata, R.stylosa, Avicennia spp*
Langsa	8	78.02	257,466	*R. mucronata, R.apiculata, R.stylosa*
Aceh Tamiang	3	73.57	242,781	*R. mucronata, R.apiculata, R.stylosa*
Langkat	3	11.46	37,818	*R. mucronata, R.apiculata, R.stylosa*
Total	**31**	**325.25**	**1,073,325**	

B.A. 6. Number of protected mangrove ecosystem hectares by local communities

Zone	District	Villages	Areas protected (ha)	Remarks
I	Medan	Belawan Bahari	232.70	*Secondary Mangrove Forest*
I	Medan	Nelayan Indah	331.27	*Secondary Mangrove Forest*
I	Medan	Sicanang	1,015.39	*Secondary Mangrove Forest, Swamp Forest*
I	Deli Serdang	Percut	513.48	*Swamp Forest*
I	Deli Serdang	Bagan Serdang	241,04	*Secondary Mangrove Forest, Swamp Forest*
I	Deli Serdang	Pematang Johar	373.69	*Secondary Mangrove Forest*

Zone	District	Villages	Areas protected (ha)	Remarks
I	Deli Serdang	Tanjung Rejo	655.52	*Secondary Mangrove Forest, Swamp Forest*
I	Deli Serdang	Denai Kuala	120.69	*Secondary Mangrove Forest*
I	Deli Serdang	Rugemuk	32.34	*Secondary Mangrove Forest*
II	Deli Serdang	Paluh Kurau	846.30	*Secondary Mangrove Forest, Swamp Forest*
II	Deli Serdang	Karang Gading	1,837.48	*Primary Mangrove Forest*
II	Deli Serdang	Hamparan Perak	1,176.91	*Secondary Mangrove Forest, Swamp Forest*
III	Langkat	Jaring Halus	78.36	*Primary Mangrove Forest*
III	Langkat	Kwala Besar	1,410.01	*Primary Mangrove Forest*
III	Langkat	Pantai Gading	706.89	*Primary Mangrove Forest*
III	Langkat	Selotong	1,579.55	*Primary Mangrove Forest*
III	Langkat	Tanjung Ibus	1.572,24	*Primary Mangrove Forest*
IV	Langkat	Bubun	1,096.44	*Dominant Secondary Mangrove Forest*
IV	Langkat	Kwala Gebang	9.68	*Secondary Mangrove Forest*
IV	Langkat	Kwala Langkat	416.38	*Secondary Mangrove Forest*
IV	Langkat	Kwala Serapuh	431.11	*Secondary Mangrove Forest*
IV	Langkat	Pasar Rawa	351.21	*Secondary Mangrove Forest*
IV	Langkat	Securai Selatan	48,26	*Secondary Mangrove Forest*
IV	Langkat	Tapak Kuda Baru	1,813.28	*Secondary Mangrove Forest, Swamp Forest*
V	Langkat	Teluk Meku	138.27	*Dominant Secondary Mangrove Forest*
V	Langkat	Kelantan	123.41	*Secondary Mangrove Forest*
V	Langkat	Lubuk Kertang	1,503.72	*Secondary Mangrove Forest*

Zone	District	Villages	Areas protected (ha)	Remarks
V	Langkat	Perlis	361.88	*Secondary Mangrove Forest*
V	Langkat	Pintu Air	31.70	*Secondary Mangrove Forest*
V	Langkat	Tanjung Pasir	38.50	*Secondary Mangrove Forest*
VI	Langkat	Alur Cempedak	281.16	*Dominant Secondary Mangrove Forest*
VI	Langkat	Beras Basah	241.13	*Swamp Forest*
VI	Langkat	Pangkalan Siata	1,659.73	*Swamp Forest*
VI	Langkat	Pematang Tengah	744.59	*Secondary Mangrove Forest*
VI	Langkat	Pulau Kampai	1.790.48	*Secondary Mangrove Forest*
VI	Langkat	Pulau Sembilan	300.16	*Secondary Mangrove Forest*
VI	Langkat	Sei Meran	774.30	*Secondary Mangrove Forest, Swamp Forest*
VI	Langkat	Sei Siur	223.55	*Secondary Mangrove Forest*
		Total	**25,102.77**	

B.B 1. Outcomes of protected fauna inventory in North Sumatra and Aceh province

NO	Family	Latin name	Local name	English Name	Location of Study					
					LK	SM	BB	PP	P	SC
A. AVIFAUNA										
1	Ardeidae	*Egretta alba*	Kuntul Besar	Great Egret	++	+++	+++	+++	+++	+++
2		*Egretta garzetta*	Kuntul Kecil	Little Egret	++	+	++	++	++	+
3		*Egretta intermedia*	Kuntul Sedang	Intermediate Egret	+	++	++	++	+++	+++
4		*Egretta sacra*	Kuntul Karang	Reef Egret	-	-	++	++	++	+
5	Ciconiidae	*Mycteria cinerea*	Bangau Bluwok	Milky Stork	-	-	-	+	++	+
6		*Leptoptilos javanicus*	Bangau Tongtong	Lesser adjutant	-	-	-	+	++	+

NO	Family	Latin name	Local name	English Name	Location of Study					
					LK	SM	BB	PP	P	SC
7	Accipitridae	*Haliaeetus leucogaster*	Elang-laut Perut-putih	White-bellied Sea-eagle	+	+	++	++	++	+
8		*Haliastur indus*	Elang Bondol	Brahminy Kite	++	++	++	+	+++	+++
9		*Aviceda leuphotes*	Baza Hitam	Black Baza	+	-	-	-	+	-
10		*Ichtyhophaga ichthyaetus*	Elang Ikan Kepala Kelabu	Grey-Headed Fish-Eagle	-	-	-	++	-	-
11	Laridae	*Chlidonias leucopterus*	Dara-laut Sayap Putih	White-winged Tern	-	+	++	+	+++	-
12		*Sterna albifrons*	Dara-laut Kecil	Little Tern	-	-	++	++	++	++
13		*Sterna hirundo*	Dara Laut Biasa	Common Tern	-	++	+	+	++	+
14		*Sterna anaethetus*	Dara-laut Batu	Bridled Tern	-	-	-	-	++	-
15		*Chlidonias hybridus*	Dara-laut Kumis	Whiskered Tern	-	-	-	-	++	-
16		*Sterna sumatrana*	Dara-laut Tengkuk-hitam	Black-naped Tern	-	-	-	-	++	-
17	Alcedinidae	*Halcyon smyrnensis*	Cekakak Belukar	White-throated Kingfisher	+	+	+	-	++	++
18		*Halcyon sancta*	Cekakak Suci	Sacred Kingfisher	-	-	+	-	+	-
19		*Halcyon chloris*	Cekakak Sungai	Collared kingfisher	++	++	++	++	++	++
20	Monarchidae	*Rhipidura javanica*	Kipasan Belang	Pied Fantail	++	++	++	-	+	+
21	Nectariniidae	*Nectarinia calcostetha*	Burung-madu Bakau	Copper-throated Sunbird	+++	+++	+++	+	+++	+++
22		*Nectarinia jugularis*	Burung-madu Sriganti	Olive-backed Sunbird	++	++	-	-	+	-
23		*Anthereptes malacensis*	Burung Madu Kelapa	Brown-throated Sunbird	++	-	-	+	-	-
24		*Anthereptes simplex*	Burung Madu Polos	Plain Sunbird	+	+	-	-	+	+
25	Sturnidae	*Acridotheres javanicus*	Kerak Kerbau	White-vented Myna	++	+	-	-	++	-
26	Tytonidae	*Tyto Alba*	Serak Jawa	Barn owl	-	-	-	+	+	+
B. HERPETOFAUNA										
27	Colubridae	*Python reticulatus*	Ular Sawah	Reticulated Python	++	+		++		
28	Viperidae	*Trimeresurus albolabris*	Ular Hijau	White-Lipped Tree Viper	++	++	++	++	++	++
29	Varanidae	*Varanus salvator*	Biawak	Monitor lizard	+++	+++	+++	+	+++	+++

LK: Lubuk Kertang, SM: Sei Meran, BB: Beras Basah, P: Percut, PP Pulau Panjang, SC: Sicanang

+low population
++ medium population
+++high population

NO	Family	Latin name	Local name	English Name	Location of Study					
					LK	SM	BB	PP	P	SC

NO	Family	Latin name	Local name	English Name	Location of study				
					GP	LU	RP	KL	AN
A. AVIFAUNA									
1	Ciconiidae	*Leptoptilos javanicus*	Bangau Tong	Lesser Adjutant	++	+	+	+	+
2	Ardeidae	*Egretta garzetta*	Kuntul kecil	Little Egret	+++	+++	+++	++	+++
3	Cisticolidae	*Prinia flaviventris*	Perenjak Rawa	Yellow-bellied Prinia	++	++	++	-	++
4	Bucerotidae	Buceros vigil	Enggang	Hornbill	-	-	+	-	+
5	Muscicapidae	*Copsychus saularis*	Kacer	Magpie	++	++	++	+	++
B. HERPETOFAUNA									
6	Crocodylidae	*Crocodylus porosus*	Buaya muara	Crocodylidae	-	-	-	++	++
7	Geoemydidae	*Batagur borneoensis*	Tuntong	Painted turtle	-	-	-	++	++

GP: Gampong pande, LU: Lam Ujong, RP: Rantau Panjang, KL: Kuala Langsa, AN: Alur Nunang

+low population
++ medium population
+++high population

B.B 2. The return of flagship species in stable mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Delphinidae	*Orcaella brevirotris*	Dolphin	EN	Langkat, Deli Serdang	+
	Sousa chinensis		VU	Langkat, Deli Serdang	+
Crocodylidae	*Crocodilus porosus*	Crocodile	VU	Aceh, Langkat	++
Cercopithecidae	*Presbytis cristata*	Mangrove primate	NT	Aceh, Langkat, Deli Serdang	+++
Ciconiidae	*Leptoptilus javanicus*	Lesser Adjutant	VU	Aceh, Langkat, Deli Serdang	+
	Mycteria cinerea	Milky Stork	EN	Aceh, Langkat, Deli Serdang	+
Accipitridae	*Haliastur indus*	Brahminy kite eagle	LC	Aceh, Langkat, Deli Serdang	+++
+low population ++ medium population +++high population					

B.B 3. Presence of migratory birds in stable mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Scolopacidae	*Numenius phaeopus*	Whimbrel	LC	Deli Serdang	+++
	Tringa totanus	Common Redshank	LC	Langkat, Deli Serdang	+++
	Actitis hypoleucos	Common Sandpiper	LC	Langkat, Deli Serdang	+++
Laridae	*Chlidonias leucopterus*	White-winged Tern	LC	Aceh, Langkat, Deli Serdang	+++
	Sterna albifrons	Little Tern	LC	Aceh, Langkat, Deli Serdang	+++
	Sterna hirundo	Common Tern	LC	Aceh, Langkat, Deli Serdang	+++
	Sterna anaethetus	Bridled Tern	LC	Aceh, Langkat, Deli Serdang	+++
	Chlidonias hybridus	Whiskered Tern	LC	Aceh, Langkat, Deli Serdang	+++
Ciconiidae	*Mycteria cinerea*	Milky Stork	EN	Langkat, Deli Serdang	+
	Leptoptilus javanicus	Lesser adjutant	VU	Aceh, Langkat, Deli Serdang	++
+low population					

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
++ medium population +++high population					

B.B 4. Presence of non-commercial macrozoobenthos in stable mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Ulmaridae	*Aurelia sp*	Jelly fish	LC	Aceh, Langkat, Deli Serdang	+++
	Aurelia aurita	Moon Jelly fish	LC	Aceh, Langkat, Deli Serdang	+++
Thiaridae	*Balanocochlis glans*	Trumpet snail	LC	Aceh, Langkat, Deli Serdang	+++
Buccinidae	*Babylonia spirata*	Tiger snail	LC	Aceh, Langkat, Deli Serdang	+++
Ellobiidae	*Cassidula aurisfelis*	Mangrove ear snail	LC	Aceh, Langkat, Deli Serdang	+++
	Ellobium aurisjudae	Judas ear cassidula	LC	Aceh, Langkat, Deli Serdang	+++
	Ellobium sp.	Judas ear	LC	Aceh, Langkat, Deli Serdang	+++
Littorinidae	*Littoraria melanostoma*	Black mangrove snail	LC	Aceh, Langkat, Deli Serdang	+++
Neritiidae	*Nerita polita polita*	Sea snail	LC	Aceh, Langkat, Deli Serdang	+++
Strombidae	*Strombus labiatus labiatus*	Samar conch	LC	Aceh, Langkat, Deli Serdang	+++
Neritidae	*Septaria lineata*	Brackish snail	LC	Aceh, Langkat, Deli Serdang	+++
Potamididae	*Terebralia palustris*	Giant mangrove whelk	LC	Aceh, Langkat, Deli Serdang	+++
	Terebralia sulcata		LC	Aceh, Langkat, Deli Serdang	+++
Discodorididae	*Discodoris sp.*	Sea slug	LC	Aceh, Langkat, Deli Serdang	+++
Mytilidae	*Mytilus sp.*	Blue mussel	LC	Aceh, Langkat, Deli Serdang	++

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Nereididae	*Namalycastis sp.*	Sea worm	LC	Aceh, Langkat, Deli Serdang	++
	Nereis sp.	Sea worm	LC	Aceh, Langkat, Deli Serdang	++
Asteroidea	*Asteroidea sp.*	Sea Star	LC	Aceh, Langkat, Deli Serdang	++
Cerithiidae	*Cerithium stigmosum*		LC	Aceh, Langkat, Deli Serdang	++
+low population ++ medium population +++high population					

B.B 5. Presence of herpetofauna (reptile) in stable mangrove ecosystem

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Bufonidae	*Duttaphrynus melanostictus*	Asian toad	LC	Aceh, Langkat, Deli Serdang	++
Microhylidae	*Microhyla sp.*	Frog	LC	Aceh, Langkat, Deli Serdang	++
Ranidae	*Fajervarya cancrivora*	Crab-eating Frog	LC	Aceh, Langkat, Deli Serdang	++
Colubridae	*Cerberus rhyncops*	Dog-faced watersnake	LC	Aceh, Langkat, Deli Serdang	++
	Python reticulatus	Reticulated Python	LC	Aceh, Langkat, Deli Serdang	++
Viperidae	*Trimeresurus albolabris*	White-Lipped Tree Viper	LC	Aceh, Langkat, Deli Serdang	++
Elapidae	*Bungarus candidus*	Malayan krait	LC	Aceh, Langkat, Deli Serdang	++
Geckonidae	*Gehyra mutilata*	Four-Clamed House Gecko	LC	Aceh, Langkat, Deli Serdang	++
	Hemidactylus frenatus	Common House Gecko	LC	Aceh, Langkat, Deli Serdang	++
Scincidae	*Eutropis multifasciata*	Common Sun Skink	LC	Aceh, Langkat, Deli Serdang	++

Family	Latin name	English	IUCN status	Location of occurrence	Estimated population
Agamidae	*Bronchocela cristatella*	Green-crested lizard	LC	Aceh, Langkat, Deli Serdang	++
Varanidae	*Varanus salvator*	Monitor lizard	LC	Aceh, Langkat, Deli Serdang	++
Rhacophoridae	*Fajervarya leucomystax*	Striped tree-frog	LC	Aceh, Langkat, Deli Serdang	++
Dicroglossidae	*Fajervarya limnocharis*	Frog	LC	Aceh, Langkat, Deli Serdang	++
Elapidae	*Bungarus laticep*	Snake	LC	Aceh, Langkat, Deli Serdang	++
	Laticauda colubrina	Yellow-lipped Sea Krait	LC	Aceh, Langkat, Deli Serdang	++
	Naja sumatrana	Cobra snake	LC	Aceh, Langkat, Deli Serdang	++
Colubridae	*Boiga dendrophila*	gold-ringed cat snake	LC	Aceh, Langkat, Deli Serdang	++
Homalopsidae	*Cerberus rynchops*	Dog-faced Water Snake	LC	Aceh, Langkat, Deli Serdang	++
	Enhydris enhydris	rainbow water snake	LC	Aceh, Langkat, Deli Serdang	++
	Fordonia leucobalia	White bellied mangrove snake	LC	Aceh, Langkat, Deli Serdang	++
Crocodylidae	*Crocodylus porosus*	Crocodylidae	LC	Aceh, Langkat, Deli Serdang	++
Gekkonidae	*Gekko gecko*	Tucktoo	LC	Aceh, Langkat, Deli Serdang	++
Scincidae	*Mabouya multifasciata*	Lizard	LC	Aceh, Langkat, Deli Serdang	++
Geoemydidae	*Batagur borneoensis*	Painted turtle	EN	Langkat, Aceh	+
+low population ++ medium population +++high population					

B.C 1. Presence of ichtyofauna (fish, crab and shrimp) in mangrove ecosystem

Family	Latin name	English	Location of occurrence	Estimated population	Remarks
Serranidae	*Ephinephelus coioides*	Groupers	Aceh, Langkat, Deli Serdang	+++	Economical
Leiognathidae	*Leiognathus blochii*	Snicker	Langkat, Deli Serdang	++	Economical
Gerreidae	*Gerres sp.*	Silverbiddies	Langkat, Deli Serdang	++	Economical
Sciaenidae	*Johnius belengenii*	Croaker	Langkat, Deli Serdang	+++	Economical
Latidae	*Lates calcarifer*	Barramundi	Langkat, Deli Serdang	+++	Economical
Chandidae	*Ambasis sp.*	Glassfishes	Aceh, Langkat, Deli Serdang	++	Economical
Mugilidae	*Mugil sp.*	Mullert	Langkat, Deli Serdang	+++	Economical
	Valamugil sp.	Mullert	Langkat, Deli Serdang	+	Economical
Eleotrididae	*Ophiocara porocephala*	Flathead Sleeper	Langkat, Deli Serdang	+	Economical
Gobiidae	*Gobie sp 1*	Gobies	Aceh, Langkat, Deli Serdang	+++	Economical
	Acentrogobius sp.	Gobies	Aceh, Langkat, Deli Serdang	+++	Economical
	Glossogobius	Gobies	Aceh, Langkat, Deli Serdang	+++	Economical
	Periophthalmodon sp.	Mudskipper	Aceh, Langkat, Deli Serdang	+++	-
	Taenioides cirratus	Bearded Worm Goby	Aceh, Langkat, Deli Serdang	+	-
	Brachyogobius xanthozona	Gobies	Aceh, Langkat, Deli Serdang	++	Economical
Siganidae	*Siganus rivulatus*	Marbled Spinefoot	Langkat, Deli Serdang	+	Economical
Mullidae	*Upeneus sulphureus*	Sunrise Goatfish	Aceh, Langkat, Deli Serdang	+	Economical
Carangidae	*Selar crumenophthalmus*	Big Eye Scad	Aceh, Langkat, Deli Serdang	++	Economical
Drepanidae	*Drepane punctata*	Spotted Sicklefish	Langkat, Deli Serdang	+	Economical
Polynemidae	*Eleutheronema tetradactylum*	Fourfinger Threadfin	Langkat, Deli Serdang	+	Economical
Channidae	*Channa striata*	Snakehead Fish	Aceh, Langkat, Deli Serdang	+	Economical
Cichlidae	*Oreochromis niloticus*	Nile Tilapia	Aceh, Langkat, Deli Serdang	+++	Economical
Lutjanidae	*Lutjanus russellii*	Russell's Snapper	Aceh, Langkat, Deli Serdang	++	Economical

Family	Latin name	English	Location of occurrence	Estimated population	Remarks
	Lutjanus bitaeniatus	Indonesian Snapper	Aceh, Langkat, Deli Serdang	++	Economical
	Lutjanus chrenbergii	Ehrenberg's Snapper	Aceh, Langkat, Deli Serdang	++	-
Nemipteridae	*Scolopsis sp.*	Monocle Bream	Langkat, Deli Serdang	+	-
Stromateidae	*Pampus argenteus*	Silver Pomfret	Langkat, Deli Serdang	+	Economical
Poecillidae	*Gambusia affinis*	Mosquitofish	Langkat, Deli Serdang	++	-
Cyprinidae	*Osteochilus sp*		Langkat, Deli Serdang	++	Economical
Hemiramphidae	*Hyporhampus sp.*	Halfbeak	Langkat, Deli Serdang	++	-
	Zenarchopterus sp.	Halfbeak	Langkat, Deli Serdang	+	-
Plotosidae	*Plotosus canius*	Black-eel Catfish	Aceh, Langkat, Deli Serdang	+++	Economical
Ariidae	*Arius thalassinus*	Giant Salmon Catfish	Aceh, Langkat, Deli Serdang	++	Economical
	Hexanematichthys sagor	Sagor Catfish	Aceh, Langkat, Deli Serdang	++	Economical
Chanidae	*Chanos chanos*	Milky Fish	Aceh, Langkat, Deli Serdang	+++	Economical
Platycephalidae	*Onigocia spinosa*	Spinny Flathead	Aceh, Langkat, Deli Serdang	+++	Economical
Scorpaenidae	*Scorpaenopsis sp.*	Stone Fish	Langkat, Deli Serdang	++	-
Cynoglossidae	*Cynoglossus lingua*		Langkat, Deli Serdang	++	Economical
Soleidae	*Dexillichthys muelleri*	Tufted Sole	Langkat, Deli Serdang	++	Economical
Clupeidae	*Sardinella fimbriata*	Herring	Langkat, Deli Serdang	+	Economical
Characidae	*Bryconamericus sp.*		Langkat, Deli Serdang	++	Economical
Tetraodontidae	*Tetraodon nigroviridis*	Green Spotted Puffer	Langkat, Deli Serdang	++	-
Rajidae	*Raja sp.*	Eyed Skate	Langkat, Deli Serdang	+	Economical
Dasyatidae	*Himantura toshi*	Black Spotted Stingray	Langkat, Deli Serdang	+	Economical
	Dasyatis kuhlii	Blue Spotted Stingray	Langkat, Deli Serdang	+	Economical
Thalassinidae	*Thalassina anomala*	Mud Lobster	Aceh, Langkat, Deli Serdang	+++	-

Family	Latin name	English	Location of occurrence	Estimated population	Remarks
Coenobitidae	*Coenobita cavipes*	Land Hermit Crab	Aceh, Langkat, Deli Serdang	+++	-
Ocypodidae	*Sesarma sp.*		Aceh, Langkat, Deli Serdang	+++	-
	Uca rosea	Fiddler Crab	Aceh, Langkat, Deli Serdang	+++	-
	Uca tetragonan	Fiddler Crab	Aceh, Langkat, Deli Serdang	+++	-
	Uca princeps	Princely Fiddler Crab	Aceh, Langkat, Deli Serdang	+++	-
	Uca mayor	Fiddler Crab	Aceh, Langkat, Deli Serdang	+++	-
	Uca heteopleura	American Red Fiddler	Aceh, Langkat, Deli Serdang	+++	-
Penaeidae	*Metapenaeus monoceros*	Pink Prawan	Aceh, Langkat, Deli Serdang	+++	Economical (consumption)
	Litopenaeus vannamei	White Prawn	Aceh, Langkat, Deli Serdang	+++	Economical (consumption)
	Penaeus indicus	White Shrimp	Aceh, Langkat, Deli Serdang	+++	Economical (consumption)
	Penaeus monodon	Giant Tiger	Aceh, Langkat, Deli Serdang	+++	Economical (consumption)
Portunidae	*Scylla serrata*	Mud/mangrove crab	Aceh, Langkat, Deli Serdang	+++	Economical (consumption)
	Portunus pelagicus	Swimming Crab	Aceh, Langkat, Deli Serdang	+++	-
	Penaeus semisulcatus	Green Tiger Prawn	Aceh, Langkat, Deli Serdang	+++	-
	Metapenaeus lysianassa	Small White Shrimp	Aceh, Langkat, Deli Serdang	+++	-
	Portunus armatus	Crab	Aceh, Langkat, Deli Serdang	+++	-
Varunidae	*Eriocheir sp.*	Three Mitten Crabs	Aceh, Langkat, Deli Serdang	+++	-
Palaemonidae	*Macrobrachium dacqueti*	Giant River Prawn	Aceh, Langkat, Deli Serdang	+++	-
Limulidae	*Carcinoscorpius rotundicauda*	Mangrove horseshoe crab	Aceh, Langkat, Deli Serdang	+++	-
Squilidae	*Harpiosquilla raphidea*	Shrimp	Aceh, Langkat, Deli Serdang	+++	-

Family	Latin name	English	Location of occurrence	Estimated population	Remarks
Parastacidae	*Cherax sp*	Crayfish	Aceh, Langkat, Deli Serdang	++	Economical (consumption)
+low population ++ medium population +++high population					

B.C 2. Commercial macrozoobenthos collected from mangrove ecosystem

Family	Latin name	English	Location of occurrence	Estimated population
Potamididae	*Cerithidea obtusa*	Mud creeper	Aceh, Langkat, Deli Serdang	+++
	Cerithidea quadrata	Black chut snail	Aceh, Langkat, Deli Serdang	+++
Potamididae	*Telescopium telescopium*	Horn snail	Aceh, Langkat, Deli Serdang	+++
Arcidae	*Anadara sp.*	Blood shell	Aceh, Langkat, Deli Serdang	+++
Veneridae	*Paphia sp.*	Surf clam	Aceh, Langkat, Deli Serdang	+++
Pteriidae	*Pinctada sp.*	Mother of pearl	Aceh, Langkat, Deli Serdang	+
Cyrenidae	*Polymesoda expansa*	Marsh Clam	Aceh, Langkat, Deli Serdang	+++
	Polymesoda erosa	Sea clam	Aceh, Langkat, Deli Serdang	+++
Sepiidae	*Sepia sp.*	cuttlefish	Aceh, Langkat, Deli Serdang	+++
Loliginidae	*Loligo sp.*	squid	Aceh, Langkat, Deli Serdang	+++
Octopodidae	*Octopus sp.*	octopuse	Aceh, Langkat, Deli Serdang	+++
Muricidae	*Muricopsis necocheana*	sea shells	Aceh, Langkat, Deli Serdang	+++
Cyrenidae	*Polymesoda erosa*	Common geloina	Aceh, Langkat	++
Arcidae	*Anadara granosa*	Ark shell	Aceh, Langkat, Deli Serdang	+++
	Anadara antiquata	Antique ark	Aceh, Langkat, Deli Serdang	+++
Assimineidae	*Assiminea sp*	Red belly snail	Aceh, Langkat, Deli Serdang	+++
Potamididae	*Telescopium telescopium*	Horn snail	Aceh, Langkat, Deli Serdang	+++
Tellinidae	*Tellina sp.*	Alternate tellin	Aceh, Langkat, Deli Serdang	+++

Family	Latin name	English	Location of occurrence	Estimated population
Neritidae	*Nerita polita*	Vernacular	Aceh, Langkat, Deli Serdang	+++
+low population ++ medium population +++high population				

7.4 Yagasu CCB 2009 biodiversity study results

Tabla 2. Outcomes of flora inventory in North Sumatra province Yagasu CCB study (2009)

Family	Latin Name	Location of Study	
		SC	P
MANGROVES			
Avicenniaceae	*Avicennia marina*	+++	+++
	Avicennia officinalis	+++	+++
	Avicennia alba	+++	+++
	Avicennia lanata	++	+
Rhizophoraceae	*Bruguiera cylindrica*	+++	+
	Bruguiera gymnorrhiza	+++	++
	Bruguiera sexangula	+++	+
	Bruguiera parviflora	+++	++
	Rhizophora apiculata	+++	+
	Rhizophora mucronata	+++	+
	Rhizophora stylosa	+++	+
	Ceriops tagal	+	+
	Kandelia kandel	+	+
Sonneratiaceae	*Sonneratia alba*	+++	+++
Maliaceae	*Xylocarpus granatum*	+++	+++
Araceae	*Nypa fructicans*	+++	+++
Euphorbiaceae	*Excoecaria agalocha*	+++	+++
	Lumnitzera littorea	+++	+++
MANGROVE ASSOCIATION			
Pandanaceae	*Pandanus tectorius*		
Acanthaceae	*Jeruju*		
P: Percut, SC: Sicanang +low population			

Family	Latin Name	Location of Study	
		SC	P
++ medium population +++high population			

7.5 Yagasu 2014 biodiversity study results

Annex 2. The presence of mamalia in stable mangrove ecosystem

No	Familia, Scientific name	Local name	English name	Location of study					Status
				LK	SM	BB		PP	
	I. Cercopithecidae								
1	*Presbytis cristata*	Lutung kelabu	Silvered langur	++	++	++		++	App II
2	*Macaca fascicularis*	Monyet kra	Long-tailed Macaque	+++	+++	+++		+++	
	II. Muridae								
3	*Rattus sp.*	Tikus	Rat	+++	-	-		-	
	V. Mustelidae			-	-	-		-	
4	*Aonix cinerea*	Sero ambrang	Oriental small-clawed tter	+	+	+	+	+	App II
5	*Lutrogale perspicillata*	Berang-berang wregul	Smooth otter	+	++	-		++	
	I. Pteropodidae								
6	*Pteropus hypomelanus*	Kalong kecil	Island Flying Fox	-	-	++		++	
	III. Sciuridae								

Keterangan:
LK = Lubuk Kertang; **SM1 =** Sei Meran;**SM2**= Sei Meran-Bukit Jengkol; **BB** = Beras Basah;**PP** = Pulau Panjang.
o = ditemukan langsung; **t** = temuan faeses, jejak, dan atau cakaran di pohon; **i** = informasi
P = Dilindungi, menurut Peraturan Pemerintah RI No. 7 Tahun 1999;
EN = Endangered/genting; **VU** = *Vurnerable*/rentan, **nt** = *near threatened*/mendekati terancam punah; kategori keterancaman menurut *IUCN Red List of Threatened Species. Version 2013.2*;
App I (*Appendix* I) & App II (*Appendix* II) = kriteria perdagangan jenis satwa yang diatur dalam CITES (*Convention on International Trade in Endangered Species of Wild Flora and Fauna*)

Annex 3. The presence of avivauna in stable mangrove ecosystem

No	Familia, Scientific name	Local name	English name	Location of study				Status
				LK	SM	BB	PP	
	Phalacrocoracidae							
1	*Anhinga melanogaster*	Pecuk-ular	Asia Oriental Darter	++	++	++	++	P, NT
2	*Phalacrocorax sulcirostris*	Pecuk-padi Hitam	Little-black Cormorant	++	-	++	++	
	Ardeidae							
3	*Ardea cinerea*	Cangak Abu	Grey Heron	-	-	++	+	
4	*Ardea purpurea*	Cangak Merah	Purple Heron	+++	+++	+++	+++	
5	*Ardeola speciosa*	Blekok Sawah	Javan Pond Heron	+++	++	+++	+++	
6	*Bubulcus ibis*	Kuntul Kerbau	Cattle Egret	+++	+++	+++	+	
7	*Butorides striatus*	Kokokan Laut	Striated Heron	+++	+++	+++	+++	
8	*Egretta alba*	Kuntul Besar	Great Egret	+++	+++	+++	+++	P
9	*Egretta garzetta*	Kuntul Kecil	Little Egret	+++	+++	+++	+++	P
10	*Egretta intermedia*	Kuntul Sedang	Intermediate Egret	++	+	++	+++	P
11	*Egretta sacra*	Kuntul Karang	Reef Egret	-	-	++	++	P
12	*Nycticorax nycticorax*	Kowak-malam Kelabu	Black-crowned Night Heron	++	-	-	-	
13	*Ixobrychus cinnamomeus*	Bambangan Merah	Cinnamon Bittern	++	++	-	++	
	Ciconiidae							
14	*Leptoptilos javanicus*	Bangau Tongtong	Lesser adjutant	+	-	+	+	P, VU
	Accipitriidae							
15	*Elanus caeruleus*	Elang Tikus	Black-winged kite	++	+	++	++	P
16	*Haliaetus leucogaster*	Elang-laut Perut-putih	White-belli Sea-eagle	++	+	++	++	P, App II
17	*Haliastur indus*	Elang Bondol	Brahminy Kite	++	+	++	++	P, App II
18	*Accipiter badius*	Alap-alap Shikra	Shikra	+	-	-	+	
	Ralidae							

No	Familia, Scientific name	Local name	English name	Location of study				Status
				LK	SM	BB	PP	
19	*Amaurornis phoenicurus*	Kareo Padi	White-breated Water Hen	++	-	++	++	
	Charadriidae							
20	*Charadrius mongolus*	Cerek-pasir Mongolus	Lesser Sand-plover	-	-	++	++	
21	*Charadrius leschenaultii*	Cerek-pasir Besar	Greater Sand-plover	-	-	++	++	
	Scolopacidae							
22	*Numenius phaeopus*	Gajahan Pengala	Whimbrel	++	++	++	++	P
23	*Tringa totanus*	Trinil kaki-Merah	Common Redshank	++	++	++	++	
24	*Tringa stagnatilis*	Trinil Rawa	Marsh Sandpiper	-	-	-	++	
25	*Xenus cinereus*	Trinil Bedaran	Terek Sandpiper	-	-	-	++	
26	*Actitis hypoleucos*	Trinil Pantai	Common Sandpiper	++	++	++	++	
	Recurvirostridae							
cf	*Himantopus leucocephalus*	Gagang-bayam	White-headed Stilt	-	-	++	++	-
	Laridae							
27	*Chlidonias leucopterus*	Dara-laut	White-winged Tern	+++	+++	+++	+++	P
28	*Sterna albifrons*	Dara-laut Kecil	Little Tern	+++	+++	+++	+++	P
29	*Sterna bergii*	Dara-laut	Great-crested Tern	+++	+++	+++	+++	P
30	*Sterna bengalensis*	Dara-laut	Lesser Crested Tern	+++	+++	+++	+++	P
	Columbidae							
31	*Treron vernans*	Punai Gading	Pink-necked Green Pigeon	++	++	+	++	
32	*Streptopelia chinensis*	Tekukur Biasa	Spotted-dove	++	++	+	++	
33	*Geopelia striata*	Perkutut	Zebra Dove	++	++	+	++	
	Cuculidae							

No	Familia, Scientific name	Local name	English name	Location of study				Status
				LK	SM	BB	PP	
34	*Chrysococcyx minutillus*	Kedasi Laut	Little Bronze Cuckoo	+	+	+	+	
35	*Centropus sinensis*	Bubut Alang-alang	Greater Coucal	++	++	-	-	
36	*Centropus bengalensis*	Bubut Besar	Lesser Coucal	++	-	-	-	
	Strigidae							
37	*Ninox scutulata*	Celepuk	Brown Boobook	-	++	-	++	
	Caprimulgidae							
38	*Caprimulgus macrurus*	Cabak	Large-tailed Nightjar	++	-	-	-	
39	*Caprimulgus affinis*	Cabak Kota	Savannah Nightjar	++	++	-	++	
	Apopidae							
40	*Collocalia esculenta*	Walet Sapi	Glossy swiftlet	++	++	++	++	
	Alcedinidae							
41	*Alcedo atthis*	Raja-udang Erasia	Common Kingfisher	++	++	++	++	P
42	*Pelargopsis capensis*	Pekaka Emas	Stork-billed Kingfisher	-	+	+	-	P
43	*Halcyon smyrnensis*	Cekakak Belukar	White-throated Kingfisher	++	++	++	++	P
44	*Halcyon sancta*	Cekakak Suci	Sacred Kingfisher	++	++	++	++	P
45	*Halcyon chloris*	Cekakak Sungai	Collared kingfisher	++	++	++	++	P
	Meropidae							
46	*Merops philippinus*	Kirik-kirik Laut	Blue-tailed Bee-eater	-	++	-	-	
47	*Merops viridis*	Kirik-kirik Biru	Blue-throated Bee-eater	-	++	++	++	
	Capitonidae							
48	*Megalaima haemacephala*	Takur Ungkut-ungkut	Coppersmith Barbet	-	++	-	-	
	Picidae							
49	*Dendrocopus moluccensis*	Caladi Tilik	Sunda woodpecker	-	++	++	++	
	Hirundinidae							
50	*Hirundo rustica*	Layang-layang Api	Barn Swallow	++	++	-	++	

No	Familia, Scientific name	Local name	English name	Location of study				Status
				LK	SM	BB	PP	
51	*Hirundo tahitica*	Layang-layang Batu	Pacific Swallow	-	++	++	++	
	Motacilidae							
52	*Anthus novaeseelandiae*	Apung Tanah	Richard's Pipit	++	-	-	-	
	Campephagidae							
53	*Lalage nigra*	Kapasan Kemiri	Pied Triller	-	-	-	+	
	Pycnonotidae							
54	*Pycnonotus goiavier*	Merbah Cerukcuk	Yellow-vented Bulbul	++	++	++	-	
	Irenidae							
55	*Aegithina tiphia*	Cipoh Kacat	Common Iora	+++	+++	+++	+++	
	Laniidae							
56	*Lanius schach*	Bentet Kelabu	Long-tailed Shrike	-	++	++	-	
	Timaliidae							
57	*Macronus ptilosus*	Ciung-air Pongpong	Fluffy-backed Tit-babbler	-	++	-	++	
	Sylviidae							
58	*Cisticola juncidis*	Cici-padi	Zitting Cisticola	-	+++	+++	-	
59	*Orthotomus ruficeps*	Cinenen Kelabu	Ashy Tailordbird	+++	+++	+++	+++	
60	*Prinia flaviventris*	Perenjak Rawa	Yellow-bellied Prinia	+++	+++	+++	+++	
	Acanthizidae							
61	*Gerigone sulphurea*	Remetuk Laut	Flyeater	-	-	+++	+++	
	Monarchidae							
62	*Rhipidura javanica*	Kipasan Belang	Pied Fantail	++	++	++	-	
	Dicaeidae							
63	*Dicaeum trochileum*	Cabai Jawa	Scarlet-headed Flowerpecker	-	++	-	-	
	Nectariniidae							
64	*Nectarinia calcostheta*	Burung-madu Bakau	Copper-throated Sunbird	+++	+++	+++	+++	

No	Familia, Scientific name	Local name	English name	Location of study				Status
				LK	SM	BB	PP	
65	*Nectarinia jugularis*	Burung-madu Sriganti	Olive-backed Sunbird	+++	+++	+++	+++	
	Estrildidae							
66	*Lonchura malacca*	Bondol Rawa	Chestnut Munia	+++	+++	+++	+	
67	*Lonchura maja*	Bondol Haji	White-headed Munia	+++	-	+	+++	
	Ploceidae							
68	*Passer montanus*	Burung-gereja Eurasia	Eurasian tree-sparrow	+++	+++	+++	+++	
69	*Ploceus philippinus*	Manyar Tempua	Baya weaver	+	-	-	+	
	Sturnidae							
70	*Aplonis panayensis*	Perling Kumbang	Asian Glossy Starling	+	-	-	++	
71	*Acridotheres javanicus*	Kerak Kerbau	White-vented Myna	-	++	-	-	
	Oriolidae							
72	*Oriolus chinensis*	Kepudang Kuduk-hitam	Black-naped Oriole	-	-	-	++	
	Artamidae							
73	*Arthamus leucorhynchus*	Kekep Babi	White-breasted Wood-swallows	++	-	-	-	
	Corvidae							
74	*Corvus enca*	Gagak Hutan	Slender-billed Crow	++	-	-	++	

Keterangan:
LK = Lubuk Kertang; **SM1** = Sei Meran; **SM2** = Sei Meran-Bukit Jengkol; **BB** = Beras Basah; **PP** = Pulau Panjang.
o = ditemukan langsung; **s** = temuan suara; **i** = informasi, cf = perlu konfirmasi
P = Dilindungi, menurut Peraturan Pemerintah RI No. 7 Tahun 1999.;
EN = Endangered/genting; **VU** = *Vurnerable*/rentan, **nt** = *near threatened*/mendekati terancam punah; kategori keterancaman menurut *IUCN Red List of Threatened Species. Version 2013.2*
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Annex 4. The presence of herpetofauna (reptil)

No	ORDO - Familia, Scientific Name	Local Name	English Name	Lokasi Temuan				Status
				LK	SM	BB	PP	
	ANURA - Bufonidae							
1	*Bufo melanostictus*	Kodok Puru	Asian toad	++	++	-	-	

No	ORDO - Familia, Scientific Name	Local Name	English Name	Lokasi Temuan				Status
				LK	SM	BB	PP	
	ANURA - Microhylidae							
2	*Microhyla sp.*	Percil		-	-	-	-	
	ANURA - Ranidae							
3	*Fajervarya cancrivora*	Katak Hijau	Crab-eating Frog	++	-	-	-	
4	*Fajervarya limnocharis*	Katak	Frog	++	-	++	-	
5	*Polypedates leucomystax*	Katak Pohon	Striped tree-frog	++	-	++	-	
	SQUAMATA - Colubridae							
6	*Dendrelaphis pictus*	Ular Tali	Painted bronze-back	++	++	+	++	
7	*Cerberus rhyncops*	Ular Air	Dog-faced watersnake	+++	+++	+++	+++	
8	*Python reticulatus*	Ular Sawah	Reticulated Python	+	-	-	+	
	SQUAMATA - Viperidae							
9	*Cryptelytrops* (*Trimeresurus albolabris*)	Ular Hijau	White-Lipped Tree Viper	++	++	++	++	
	SQUAMATA - Elapidae							
10	*Bungarus candidus*	Weling	Malayan krait	++	-	-	-	
	SQUAMATA - Geckonidae							
11	*Gehyra mutilata*	Cicak Gula	Four-Clamed House Gecko	++	-	-	++	
12	*Hemidactylus frenatus*	Cicak Kayu	Common House Gecko	++	-	-	++	
	SQUAMATA - Scincidae							
13	*Eutropis (Mabuya) multifasciata*	Kadal Kebun	Common Sun Skink	++	++	++	++	
				++	++	++	++	
	SQUAMATA - Agamidae							
14	*Bronchocela cristatella*	Bunglon	Green-crested lizard	++	++	++	++	
	SQUAMATA - Varanidae							
15	*Varanus salvator*	Biawak	Monitor lizard	++	++	++	++	

Keterangan:
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o = ditemukan langsung; **t** = temuan telur, dan atau sisa sisik; **i** = informasi
P = Dilindungi, menurut Peraturan Pemerintah RI No. 7 Tahun 1999.;
EN = Endangered/genting; **VU** = *Vurnerable*/rentan, **nt** = *near threatened*/mendekati terancam punah; kategori keterancaman menurut *IUCN Red List of Threatened Species. Version 2013.2*
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Annex 5. The presence of fish

No	ORDO - Familia, Nama Ilmiah	Nama Indonesia	Nama Inggris	Lokasi Temuan				Keterangan
				LK	**SM**	**BB**	**PP**	
	PERCIFORMES – Serranidae							
1	*Ephinephelus coioides*	Kerapu	Groupers	+++	+++	+++	+++	
	PERCIFORMES – Apogonidae							
2	*Apogon hyalosoma*	Muntah Telur	Mangrove Cardinalfish	+++	-	-	+	
	PERCIFORMES – Leiognathidae							
3	*Leiognathus blochii*		Ponyfishes	++	-	-	++	
	PERCIFORMES – Gerreidae							
4	*Gerres sp.*	Kapas-kapas	Silverbiddies	++	-	-	++	
	PERCIFORMES – Sciaenidae							
5	*Johnius belengenii*	Gulamah	Croaker	+++	+++	+++	+++	
	PERCIFORMES – Latidae							
6	*Lates calcarifer*	Jenahat	Golden Snapper	++	-	-	++	
	SILURIFORMES - Bagridae							
7	*Mystus sp.*	Baung; Lunduh	Estuarine Catfish	++	-	++	-	
	PERCIFORMES – Haemulidae							
8	*Pomadasys unimaculatus*	Grut-grut	Red Patched Grunter	++	-	-	++	
	CYPRINODONTIFORMES - Hemiramphidae							
9	*Hyporhampus sp.*	Cucut; Julung-julung	Halfbeak	++	-	++	++	
10	*Zenarchopterus sp.*	Julung -julung	Halfbeak	++	-	++	-	
	SYNBRANCHIFORMES - Synbranchidae							
11	*Monopterus albus*	Belut		+++	+++	+++	+++	
	SILURIFORMES - Plotosidae							
12	*Plotosus canius*	Sembilang	Black-eel Catfish	+++	+++	+++	+++	
	Chanidae							
13	*Chanos chanos*	Bandeng	Milky Fish	+++	+++	+++	+++	
	PERCIFORMES – Chandidae							

14	*Ambasis sp*	Sriding	Glassfishes	+++	-	+++	+++
	PERCIFORMES – Mugilidae						
15	*Mugil sp.*	Belanak	Mullert	+++	+++	+++	+++
16	*Valamugil sp.*	Kombak; Ngongok		++	-	-	++
	PERCIFORMES – Eleotrididae						
17	*Butis butis*	Kedera	Sleepers	++	-	-	++
	PERCIFORMES – Gobiidae						
20	*Acentrogobius sp.*	Lontok Lubang	Gobies	++	++	++	++
21	*Glossogobius*	Gabus Pasir	Gobies	+++	+++	+++	+++
22	*Periophthalmodon sp.*	Slodok	Mudskipper	+++	+++	+++	+++
	PERCIFORMES – Siganidae						
23	*Siganus javus*	Kertang Bulat	Rabbit Fish	++	-	++	++
24	*Siganus guttatus*	Kertang Bintik	Goldlined spinefoot	++	-	++	-

Keterangan:LK = Lubuk Kertang; **SM1 =** Sei Meran; **SM2** = Sei Meran-Bukit Jengkol; **BB** = Beras Basah; **PP** = Pulau Panjang.
(**o**) = ditemukan langsung; (-) = tidak ditemukan, **P** = Dilindungi, menurut Peraturan Pemerintah RI No. 7 Tahun 1999.;

Annex 6. The presence of Crustaceae

No	Familia, Nama Ilmiah	Nama Indonesia	Nama Inggris	Lokasi Temuan				Status
				LK	SM	BB	PP	
1	*Carcinoscorpius rotundicauda*	Mimi/Belangkas	Horseshoe crab	+++	++	++	++	dikonsumsi
2	*Myomenippe hardwicki*	kepiting Batu	Stone crab	+	+	+	+	
3	*Scylla serrata*	kepiting Bakau	Mud/mangrove crab	+++	+++	+++	+++	dikonsumsi
4	*Episesarma sp.*	kepiting panjat/Wideng	Tree-climbing crab	+++	+++	+++	+++	
5		Kepiting Angin		+++	-	-	+++	
6	Penaeus indicus	Udang Kelong	Indian White Prawn	+++	-	-	+++	dikonsumsi
7		Udang Gantung		+++	-	-	+++	dikonsumsi
8	Penaeus merguensis	Udang putih/Jerbung	White Prawn	+++	-	+++	+++	

9	Metapenaeus monoceros	Udang Swallow	Pink Prawan	+++	-	-	+++	dikonsumsi
10	*Uca spp.*	Kepiting Biola	Fiddler Crab	+++	+++	+++	+++	
11	*Alpheus sp.*		Mangrove snapping prawn	+++	+++	+++	+++	dikonsumsi
12	*Haptosquilla sp.*	Udang mantis				+++	+++	

Keterangan:
LK = Lubuk Kertang; **SM1 =** Sei Meran; **SM2** = Sei Meran-Bukit Jengkol; **BB** = Beras Basah; **PP** = Pulau Panjang.
(**o**) = ditemukan langsung; (-) = tidak ditemukan

Annex 7. The presence of invertebrate

No	Filum, Class, Nama Ilmiah	Nama Indonesia	Nama Inggris	Lokasi Temuan				Status
				LK	SM	BB	PP	
	Cnidaria, Scyphozoa							
1	*Aurelia aurita*	Ubur-ubur	Jelly fish	+++	+++	+++	+++	
	Moluska, Gastropoda							
5	*Cerithidea cingulata*		a. Girdled horn snail	++	++	++	++	
6	*Cerithidea obtusa*			++	-	-	-	
7	*Cerithidea quadrata*		Quadrate horn shell	++	++	++	++	
8	*Chicoreus sp*		Rock shell	++	-	-	++	
9	*Ellobium* sp.	Siput Sedot		++	++	++	++	
10	*Littorina conica*			++	-	++	-	
11	*Nassarius sp.*			-	++	-	-	
12	*Nerita lineata*	Siput Timba	Common nerite	++	-	-	-	
13	*Neverita sp.*			-	-	++	-	
14	*Nucula sp.*			++	-	-	-	
18	*Stenothyra sp.*			++	-	-	-	
19	*Telescopium telescopium*	Belincong		++	++	++	++	
20	*Terebralia sulcata*	Belitong	Belitong	++	-	++	++	
21	*Turritella sp.*			++	-	-	-	
	Moluska, Bivalvia							

No	Filum, Class, Nama Ilmiah	Nama Indonesia	Nama Inggris	Lokasi Temuan					Status
				LK	SM	BB	PP		
23	*Anadara sp*	Kerang darah		++	-	-	-		
25	*Paphia sp.*	Kepah	Surf Clam	-	-	++	-	++	
27	*Polymesoda expansa*	Lokan	Marsh Clam	++	-	++	-	-	

Keterangan:
LK = Lubuk Kertang; **SM1 =** Sei Meran; **SM2** = Sei Meran-Bukit Jengkol; **BB** = Beras Basah; **PP** = Pulau Panjang.
(**o**) = ditemukan langsung; (-) = tidak ditemukan

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